

**AI-Based Short-Term Generation Forecasting of Cascaded Hydroelectric Power
Plants**

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CERTIFICATE OF PANEL APPROVAL

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ABSTRACT

The Agus 1–7 hydropower cascade is a hydraulically connected system where generation is affected by upstream–downstream interactions, unit availability, gate operations, reservoir conditions, rainfall, and other hydrologic factors. This study developed an AI-based short-term generation forecasting framework using a Radial Basis Function Neural Network (RBFNN) to support day-ahead capacity nomination and operational planning for the Agus cascaded hydroelectric power plants.

Historical operational and hydrologic data were cleaned, standardized into hourly time-series form, and used to construct temporal, lagged, rolling, outage-aware, hydrologic, and cascade-aware features. The RBFNN used a residual forecasting approach, where the model predicted the next-hour change in generation instead of the direct generation level to improve forecast continuity during recursive day-ahead prediction. The model was evaluated using a chronological training, validation, and testing procedure under a rolling 24-hour day-ahead backtest, where each forecast day was predicted using only information available before that day to reduce data leakage.

Performance was assessed using operational MAPE, MAE, RMSE, and R^2 . Results showed that the RBFNN achieved an average testing operational MAPE of 5.7949%, MAE of 3.4687 MW, RMSE of 8.5306 MW, and R^2 of 0.6705. Compared with Random Forest and XGBoost benchmark models, the RBFNN obtained the best overall testing performance, suggesting its potential for modeling nonlinear and time-dependent generation behavior in the Agus cascade. Since the evaluation was conducted using historical offline data, the proposed framework should be interpreted as a potential operational decision-support tool rather than a fully deployed forecasting solution.

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CHAPTER 1

INTRODUCTION

1.1. Background of the Study

Hydropower remains a cornerstone of the Philippine energy portfolio, providing a significant share of total generation and critical support for grid reliability (Polestico, 2019). Central to the national strategy is the Agus–Pulangi Hydroelectric Power Plant complex in Mindanao, a major government-owned facility operated by the National Power Corporation's (NPC) Mindanao Generation (MinGen) Group. The complex, with a total installed capacity of 1,001.1 MW, is sustained primarily by the hydraulically coupled Agus cascade along the Agus River, supplied by the vast storage volume of Lake Lanao (Polestico, 2019).

The Agus cascade, comprising Agus 1, 2, 4, 5, 6, and 7 plants, is a highly interdependent system, not a collection of stand-alone units, with a combined capacity of about 746.1 MW (Polestico, 2019). Water released by the upstream plants immediately dictates the inflow and hydraulic head available for all downstream plants as the river flows 36 km down to Iligan Bay (Polestico, 2019). Consequently, any operational decision, such as changes in gate opening or unit loading at one plant, directly and non-linearly influences the water available and the potential generation capacity of every plant below it. This cascading arrangement necessitates that operation, optimization, and power generation must be planned and modeled simultaneously for the entire system, making accurate short-term forecasting essential for scheduling and market participation (Polestico, 2019).

1.2. Statement of the Problem

Capacity nomination for the Agus hydropower cascade (Agus 1 to Agus 7) is frequently unstable. In practice, day-ahead nominations often need to be revised because the actual generation depends on a combination of hydrologic and operational factors such as forebay elevation, reservoir inflow and outflow, spillway and governor gate positions, riparian flow, and rainfall, which are not always measured consistently or fully reflected in

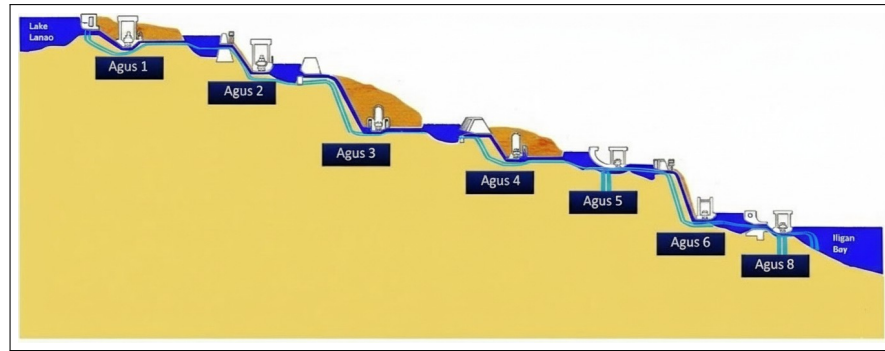


Figure 1.1. Schematic Representation of the Agus River Hydroelectric Power Plant (HEPP) Complex (National Power Corporation - Mindanao Generation & Mindanao State University - Iligan Institute of Technology, 2025).

existing planning tools. When these inputs are incomplete, outdated, or not systematically used, the current simulation models and rule-based practices struggle to represent the true behavior of the cascade, leading to conservative nominations, frequent re-nominations, and potential underutilization of available water and capacity.

1.3. Objectives of the Study

The main objective is to use an AI-based short-term (one-day-ahead) generation forecasting model for the cascaded Agus 1–7 hydropower plants using an RBFNN and to investigate how these methods improve the accuracy of hydropower generation forecasting driven by data on plant generation, forebay elevation, inflow and outflow, spillway gate positions, outage conditions, and rainfall. Specifically, the study aims:

1. To design and develop an RBFNN-based short-term (one-day-ahead) generation forecasting model for the Agus 1–7 hydropower plants.
2. To evaluate the forecasting performance of the developed RBFNN model using statistical accuracy measurements using MAPE by comparing the results of the developed forecasting system with the actual data.
3. To compare the performance of the RBFNN model against benchmark methods to determine its improvement in prediction stability.

1.4. Significance of the Study

The hydraulic coupling of the Agus cascade makes day-ahead capacity nomination sensitive to hydrologic variability and operational uncertainty (Polestico, 2019). Accurate short-term generation forecasting may therefore support scheduling, water-management assessment, and short-term operational planning (Polestico, 2019). The findings of this study may benefit the following stakeholders.

The study may benefit the National Power Corporation–Mindanao Generation (NPC–MinGen) by providing a prototype forecasting framework with potential application in day-ahead generation planning and capacity nomination review. By producing one-day-ahead generation estimates under historical testing conditions, the framework may assist users in examining expected generation trends and reviewing nomination plans. However, its use in actual plant operations would require further validation using live operational data. In the context of the Wholesale Electricity Spot Market, the developed RBFNN framework may also serve as a reference for future generation-bidding and day-ahead planning studies, although direct application in market operations would require additional technical validation and operator review (Polestico, 2019).

For the Mindanao power system and its operators, the study may contribute to future efforts to improve short-term generation visibility and operational planning. By incorporating historical generation, lagged upstream generation, elevation, gate-related, outage-related, rainfall, and outflow variables, the model may provide additional information regarding the expected generation patterns of the Agus plants over the next 24 hours. Such information may support dispatch review, reserve-planning studies, and the assessment of expected hydropower contribution, subject to further validation under actual operating conditions (Polestico, 2019).

The study may also benefit the academic community and future researchers by contributing to the theoretical and practical knowledge concerning artificial intelligence applications in renewable-energy forecasting. It addresses the limited availability of dedicated data-driven studies for coupled generation forecasting in cascaded hydropower systems

(Barzola-Monteses et al., 2019; Bernardes Jr et al., 2022). The documented preprocessing procedure and the inclusion of multi-plant operational, hydrologic, and cascade-related features in the RBFNN framework may serve as a reference for future studies involving nonlinear hydropower forecasting problems.

Finally, the study is relevant to the objectives of United Nations Sustainable Development Goal 7, which promotes access to affordable, reliable, sustainable, and modern energy (World Bank, 2020). By investigating a data-driven forecasting approach for the Agus cascade, the study contributes to ongoing research on decision-support methods intended to improve the planning and utilization of existing renewable hydropower resources (Lee et al., 2016; Zhou et al., 2020).

1.5. Scope and Limitations

This study focuses on the Agus River hydropower cascade, specifically Agus 1 to Agus 7, as operated by NPC–MinGen. Other power plants within the Mindanao grid are not included in the modeling and analysis. The forecasting task is limited to short-term, one-day-ahead (24-hour) generation forecasting using historical operational and environmental data, including plant generation, forebay elevation, inflow and outflow, spillway and governor gate positions, riparian flow, and rainfall. No additional sensors or field measurements are introduced in this study.

The modeling approach utilizes a Radial Basis Function Neural Network (RBFNN) as the primary forecasting method, while other methods, if considered, serve only as simple benchmarks. The study does not develop a full physical or hydrodynamic model of the Agus–Lake Lanao system. Furthermore, evaluation and implementation are confined to offline analysis using historical data and statistical accuracy measures, including operational Mean Absolute Percentage Error (MAPE), Mean Absolute Error (MAE), Root Mean Square Error (RMSE), and the coefficient of determination (R^2). Real-time system integration, detailed economic or market analysis, and Wholesale Electricity Spot Market (WESM) bidding simulations are beyond the scope of this thesis.

The forecasting framework was evaluated using historical data and offline testing

only. It was not validated under actual live plant operation or integrated into the operational decision-making workflow of the Agus plants. Therefore, the results should be interpreted as prototype-level evidence of forecasting potential rather than proof of full operational readiness or real-world deployment capability.

1.6. Theoretical Framework

The existing methods used in hydropower generation forecasting have distinct strengths and limitations. Some approaches are simple and interpretable but may not fully capture the nonlinear and interconnected behavior of cascaded hydropower plants. Other methods provide greater modeling flexibility but may require extensive tuning and may be more difficult to interpret (Bernardes Jr et al., 2022; Mediavilla et al., 2022). In this study, an artificial intelligence-based approach was used to learn the relationship between hydrologic–operational variables and plant generation directly from historical data and to generate one-day-ahead generation forecasts (Avesani et al., 2022; Cohen, 2022; Elsheikh et al., 2021; Slater et al., 2023).

The framework was based on the concept of data-driven learning. Rather than relying exclusively on explicitly defined physical equations and assumptions, the model learned patterns from historical operating conditions, including forebay elevation, inflow and outflow, spillway and governor gate positions, riparian flow, rainfall, and plant generation. These historical observations were used to establish a mapping between the predictor variables and subsequent plant generation (Amitaba et al., 2024; Balan et al., 2024; Bernardes Jr et al., 2022). The objective was to develop a forecasting model that approximated the historical generation behavior of Agus 1, Agus 2, Agus 4, Agus 5, Agus 6, and Agus 7 and that could potentially support operational decision-making after further validation (Ji et al., 2018; Zhou et al., 2020).

The data-driven learning approach was implemented using a Radial Basis Function Neural Network. The model inputs consisted of operational, hydrologic, temporal, and cascade-related predictor variables obtained from the Agus plants and the surrounding watershed. These variables were processed by the RBFNN to learn an internal representation

of the relationships between historical system conditions and plant generation. After training, the network was used as a prototype forecasting model capable of receiving recent input conditions and generating one-day-ahead generation estimates for each plant. Accordingly, the theoretical framework was organized around four main concepts: cascaded hydropower generation, forecasting, artificial neural networks, and the RBFNN-based forecasting framework (Schwenker et al., 2001; Sideratos & Hatziaargyriou, 2020; Soper, 2023; Zhang et al., 2024).

1.6.1. Cascaded Hydropower Generation

Cascaded hydropower plants share the same river flow. Water released by an upstream plant becomes the inflow of the next downstream plant. For Agus 1–7, water comes from Lake Lanao, passes through each plant in sequence, and finally discharges to Iligan Bay. The generation of each plant depends on how much water passes through its turbines and the effective head between forebay and tailwater (Bernardes Jr et al., 2022).

Because of this arrangement, the plants cannot be treated as independent units. A change in forebay elevation, gate opening, or spillage at one plant can change the inflow and head of the next plant. In addition, rainfall, riparian releases, and natural inflows affect lake level and river discharge. The theoretical view of this study is that the Agus cascade behaves as one dynamic, interconnected system, where generation at any time is the result of combined hydrologic conditions and operating decisions along the river (Ji et al., 2018; Moita & Monte, 2020; Rougé et al., 2019).

1.6.2. Forecasting

Forecasting is the process of making statements about future events whose actual outcomes have not yet been observed. In this study, the event of interest is the next-day hydropower generation of Agus 1–7. The forecast is expressed as a time series, such as hourly generation values for the next 24 hours (Mir et al., 2020; Prema et al., 2021).

Forecasting always involves uncertainty. The goal is not to predict the future perfectly but to provide an estimate that is as close as possible to what will actually happen,

given the available data. For hydropower generation, this requires using up-to-date records of plant operation and hydrologic conditions. The more accurately the model can relate past and present variables to future generations, the more useful it becomes for day-ahead capacity nomination and scheduling (Akhtar et al., 2023; Essenfelder et al., 2020; Sapitang et al., 2020; Tsai et al., 2023).

1.6.3. Artificial Neural Networks and RBFNN

Artificial Neural Networks (ANNs) are computational models inspired by the way biological neurons process information. They consist of layers of interconnected nodes that transform input data into output predictions. When used for forecasting, an ANN is trained by repeatedly adjusting its internal parameters so that its outputs match the historical data as closely as possible (Baliyan et al., 2015; Bucci, 2020; SV et al., 2020).

An RBFNN is a particular type of ANN. It has three layers: an input layer, a hidden layer with radial basis functions (usually Gaussian), and an output layer. Each hidden neuron is “sensitive” to a certain region of the input space; it responds strongly when the current inputs are close to its center and weakly when they are far away. By combining the responses of many such neurons, an RBFNN can approximate complex, nonlinear relationships (Sideratos & Hatziargyriou, 2020; Zhang et al., 2024).

In this study, the RBFNN was selected because it is suitable for approximating nonlinear relationships within a relatively compact neural-network structure. Its localized Gaussian basis functions allowed the model to represent different operating regions in the historical generation data. The RBFNN was therefore applied to learn plant-specific relationships among historical generation, temporal indicators, unit availability, hydrologic context, and upstream cascade conditions (Bernardes Jr et al., 2022; Schwenker et al., 2001; Soper, 2023).

1.6.4. The RBFNN-Based Forecasting System

The RBFNN-based forecasting framework used historical operational and hydrologic data from the Agus cascade, including plant generation, forebay elevation, Lake

Lanao outflow, gate-related variables, unit availability, rainfall, temporal indicators, and upstream cascade conditions. The collected time-series data were cleaned, temporally aligned, and organized into an hourly dataset suitable for model development.

Lagged generation values, rolling statistics, temporal indicators, outage-related variables, hydrologic context, and upstream cascade variables were constructed as forecasting inputs. Separate plant-specific RBFNN models were then trained to predict the next-hour change in generation. The predicted change was added to the most recent generation value to reconstruct the generation-level forecast.

For validation and testing, each forecast day was evaluated using a rolling 24-hour day-ahead backtest. Only information available before the beginning of the forecast day was used, while actual forecast-day generation values were retained solely for performance evaluation. Model performance was assessed using operational MAPE, MAE, RMSE, and the coefficient of determination (R^2).

The resulting forecasts were intended as prototype decision-support information for day-ahead generation planning and capacity nomination review. Their direct use in actual plant operation would require further validation using live operational data.

1.6.5. Model Behavior

The RBFNN was developed to approximate the nonlinear relationships between historical generation and the operational, temporal, hydrologic, outage-related, and upstream cascade variables included in the forecasting dataset. Its radial basis functions represented localized operating regions within the input space, allowing the model to respond differently to varying generation levels and system conditions.

Under the residual forecasting formulation, the RBFNN predicted the next-hour change in generation relative to the most recent available generation value. Positive predicted changes indicated an expected increase in generation, whereas negative predicted changes indicated an expected reduction. During recursive day-ahead forecasting, each predicted hourly generation value was appended to the temporary history and was used to construct the lagged and rolling features required for the succeeding forecast hour.

The framework was designed to represent recurring generation patterns associated with normal operation, varying generation levels, unit availability, hydrologic context, and upstream cascade behavior. Nevertheless, its forecasts remained dependent on the operating patterns represented in the historical dataset. Conditions not known before the forecast day, including forced outages, unexpected dispatch instructions, abrupt operational interventions, and unobserved hydrologic changes, could therefore result in larger forecasting errors.

The degree to which the predicted generation followed the observed generation was determined through the validation and testing procedures presented in Chapters 3 and 4 rather than assumed theoretically.

Definition of Terms

Agus Cascade A series of interconnected hydroelectric power plants (Agus 1, 2, 4, 5, 6, and 7) where the outflow of upstream plants serves as inflow to downstream plants.

Radial Basis Function Neural Network (RBFNN) A type of artificial neural network that uses radial basis functions as activation functions, effective for modeling nonlinear relationships in time-series data.

Day-Ahead Forecasting The process of predicting system output, such as power generation, for the next 24-hour period based on current and historical data.

Operational MAPE A modified Mean Absolute Percentage Error that excludes low-generation values where percentage error becomes unstable or misleading.

Mean Absolute Error (MAE) A performance metric that measures the average magnitude of prediction errors without considering their direction.

Root Mean Square Error (RMSE) A metric that measures the square root of the average squared differences between predicted and actual values, emphasizing larger errors.

Coefficient of Determination (R^2) A statistical measure indicating how well the model explains the variability of the observed data.

Residual (Delta) Modeling A forecasting approach where the model predicts the change in output rather than the absolute value, improving continuity in time-series predictions.

Feature Engineering The process of creating input variables from raw data, including lag features, rolling statistics, and temporal indicators, to improve model performance.

Lag Features Past values of a variable used as inputs to capture temporal dependencies in time-series forecasting.

Rolling Statistics Statistical measures such as mean, standard deviation, minimum, and maximum computed over a moving window of time.

Planned Outage A scheduled shutdown of a generating unit, represented as a binary input (1 = available, 0 = unavailable) in the forecasting model.

Cascade Dependency The relationship between upstream and downstream plants in a hydropower system, where upstream discharge affects downstream generation.

Capacity Nomination The process of declaring the expected power generation output for a given period in power system operations.

Re-nomination The adjustment of previously declared generation schedules due to forecasting errors or operational changes.

Chronological Data Split A dataset division method where training, validation, and testing sets are separated based on time order to preserve temporal integrity.

Benchmark Models Alternative machine learning models (e.g., Random Forest and XGBoost) used for performance comparison against the proposed model.

Time-Series Data Sequential data points recorded over time, commonly used in forecasting applications.

Hydropower Generation The production of electricity using the energy of flowing or falling water.

CHAPTER 2

REVIEW OF RELATED LITERATURE

2.1. Short-Term Generation Forecasting in Cascaded Hydropower Systems

Avesani et al. (2022) explained that the liberalization of electricity markets has shifted the operational role of storage hydropower producers from maximizing regional energy production toward active market participation and revenue-oriented operation. Hydropower remains an important energy source, accounting for approximately 15% of global electricity generation. Its operational flexibility also makes storage hydropower valuable for maintaining grid balance and supporting the integration of intermittent renewable energy sources such as wind and solar power.

The transition to a market-driven operational objective makes accurate short-term generation forecasting indispensable for two primary reasons: revenue maximization and grid stability. It enables producers to align water discharge schedules with high expected electricity price periods, optimizing revenue through a complex, non-linear process that is constrained by persistent hydrological uncertainties and the uncertain dynamics of the competitive electricity market (Avesani et al., 2022). At the same time, reliable forecasts are crucial for informed bidding in the day-ahead and intra-day markets, ensuring the timely commitment of flexible hydropower assets to support grid stability and maintain energy balance during periods of high demand or low intermittent RES output.

Cascaded hydropower systems significantly intensify the forecasting challenge. Unlike a single-plant system, a cascaded arrangement consists of multiple power plants situated sequentially along a river (Avesani et al., 2022). The power generation and water release decisions of an upstream plant directly influence the water inflow, and thus the potential generation of all subsequent downstream plants. This creates a highly complex, coupled, non-linear system where the optimal operational schedule must account for the interdependencies and constraints of the entire chain, rather than just isolated units. Accurate short-term forecasting must therefore model the dynamic hydraulic and economic relationships across the cascade.

Given the complexities and non-linear characteristics inherent in short-term generation forecasting, particularly in coupled hydro-econometric and cascaded environments, traditional statistical methods often lack the robust modeling capabilities required to maintain high prediction precision (Mediavilla et al., 2022).

As a result, there is substantial and growing research interest in the adoption of Artificial Intelligence (AI) and Machine Learning (ML) methodologies. AI models excel at identifying non-obvious correlations and complex, non-linear patterns within large, noisy datasets, frequently demonstrating superior accuracy over conventional models (Cohen, 2022; Elsheikh et al., 2021). Furthermore, the development of sophisticated hybrid forecasting models, which blend physics-based forecasts (such as those from numerical weather prediction) with data-driven AI/ML systems, is increasingly recognized as a vital approach for enhancing prediction skill and minimizing inherent biases (Slater et al., 2023).

The Radial Basis Function Neural Network (RBFNN) is a highly suitable Artificial Neural Network (ANN) architecture for this domain. It is distinguished by its capacity for effective generalization and simplicity in approximating nonlinear system behavior, even with a single hidden layer (Sideratos & Hatziaargyriou, 2020; Zhang et al., 2024). RBFNNs are efficiently utilized in hybrid schemes for tasks like input data clustering and localized regression, and their performance in short-term forecasting can be further optimized by combining them with meta-heuristic algorithms (Sideratos & Hatziaargyriou, 2020; Zhang et al., 2024).

2.2. Physical and Operational Characteristics of Cascaded Hydroelectric Power Plants

The foundation of hydropower optimization lies in understanding the conversion process: hydroelectric plants convert the kinetic and potential energy of flowing water into electrical energy (Bernardes Jr et al., 2022). Optimal power dispatch requires efficiently leveraging the available hydraulic head and river flows to maximize the resultant power, which is primarily determined by the flow of water through the turbine and the effective height of the waterfall (Balan et al., 2024). Accurate short-term generation forecasting must

therefore account for the gross head available, efficiency losses in the turbine–generator set, and hydraulic losses, often by integrating hydrological and climatic data to predict future water availability (Bernardes Jr et al., 2022).

The complexity of management is significantly increased in a cascaded hydro system, which features multiple power plants situated sequentially along the same river (Rougé et al., 2019). This setup creates a critical spatial coupling because the outflow (released water) from any upstream plant inherently becomes the inflow for its downstream counterpart (Moita & Monte, 2020). This interdependency means that upstream operational decisions, particularly release schedules, exhibit a progressive dominance over the entire chain, creating a highly non-linear system where localized downstream rules have limited influence (Rougé et al., 2019). Consequently, the core management challenge is the coordination of these interconnected facilities, which is governed by the mass balance principle (inflow minus outflow equals change in volume) and various operational thresholds.

Effective short-term forecasting and operation rely on monitoring and controlling several critical, coupled variables: reservoir volume and level, which must be maintained within predefined minimum and maximum storage limits and directly determine the available hydraulic head (Amitaba et al., 2024); inflow and outflow, where total outflow discharge consists of turbined flow that generates power and any intentional spillage (Balan et al., 2024); and turbine discharge and gate opening, which serve as key hourly decision variables that directly regulate power production and, in the context of flood management, dictate spillway operations (Amitaba et al., 2024).

These systems face practical coordination deficits, including managing uncertainty in future inflows, overcoming structural model deficits that fail to adequately represent inter-plant coordination, and mitigating the risk of strategic market behavior by upstream firms that might restrict water flow to limit downstream rivals (Moita & Monte, 2020; Rougé et al., 2019).

2.3. Hydropower Generation Forecasting and Time Horizons

The shift toward sustainable energy necessitates the accurate prediction of electricity generation across all resource types. Hydropower Generation Forecasting (HGF) involves estimating the electrical output of hydroelectric facilities over future time intervals, a process crucial for maintaining the operational reliability and economic viability of modern power systems (Akhtar et al., 2023). While HGF involves fewer direct meteorological uncertainties compared to highly intermittent sources like wind and solar, accurate short-term prediction remains critical for complex systems such as cascaded power plants to optimize water resource management and system integration. Forecasting techniques, whether applied to electricity load or generation, are conventionally stratified into distinct categories based on the temporal length of the predictive horizon, as these intervals correspond to specific planning and operational requirements (Mir et al., 2020). Long-Term Forecasting (LTLF) typically spans five years up to decades ahead, guiding strategic policy and power system expansion. Medium-Term Forecasting (MTLF) covers months to years ahead, supporting financial applications like energy trading and unit maintenance scheduling (Aquila et al., 2023; Prema et al., 2021). The most operationally relevant horizon is Short-Term Forecasting (STLF), characterized by timeframes ranging from a few hours up to one or two days ahead (Prema et al., 2021). This category is highly valuable for immediate system operation planning and coordination (Aquila et al., 2023).

The core focus of short-term generation forecasting in hydropower is essential for operational security, load-sharing decisions, and maximizing economic returns in the electricity market (Prema et al., 2021). Accurate STLF is vital for effective scheduling and dispatch, allowing operators to optimize system operations, minimize production costs, and avoid potential system overloads (Akhtar et al., 2023). Furthermore, STLF supports the integration of intermittent resources by providing the necessary predictive input for hydropower to maintain grid stability and compensate for fluctuations in wind and solar output. HGF is complicated because it involves many nonlinear physical and operational variables. These include hydrological inputs like rainfall and river inflow, reservoir metrics

like forebay elevation (water level), and operational controls like turbine discharge/outflow and gate openings (Balan et al., 2024; Bernardes Jr et al., 2022).

While hydropower is a conventional dispatchable source, the STLF challenge is significantly different from wind or solar, which struggle primarily with meteorological intermittency. For a cascaded hydroelectric system, the difficulties are rooted in hydraulic and spatial coupling. Specifically, the model must account for water routing time delays, which represent the highly variable and non-linear time it takes for released water to travel from an upstream dam to the reservoir of a downstream dam. It must also capture cascade effects, the precise interdependency where the outflow of one plant immediately and sequentially impacts the inflow and available head of the next plant in the chain (Ji et al., 2018).

2.4. Conventional Approaches and the Necessity of AI

Accurate forecasting of hydropower generation (HPG) is a fundamental prerequisite for the efficient management of water resources, capacity planning, and successful participation in dynamic energy markets (Hanoon et al., 2023). Historically, hydropower and related hydrological variables were forecasted using conventional statistical and physical models (Hanoon et al., 2023; Sapitang et al., 2020).

2.4.1. Traditional Forecasting Methods and Their Limitations

Among the classical techniques, the AutoRegressive Integrated Moving Average (ARIMA) model is a complex linear framework widely employed in time series analysis (Ghimire, 2017; Hanoon et al., 2023). Its seasonal counterpart, the Seasonal ARIMA (SARIMA) model, is applied when the time series exhibits clear periodic or seasonal patterns (Hanoon et al., 2023). Additionally, Multi-Linear Regression (MLR) and traditional hydrological/physical models represented the standard methodology used to forecast key variables like reservoir water levels (Sapitang et al., 2020).

Despite their mathematical simplicity and long-standing flexibility, these conventional methods face considerable limitations when applied to modern, complex hydroelec-

tric systems (Hanoon et al., 2023; Kontopoulou et al., 2023). A core difficulty is the restricted capability of linear models (such as MLR and ARIMA) to adequately model the significant non-linearity and non-stationarity found in hydrological and environmental time series (Kontopoulou et al., 2023). Complex, non-linear functions govern real-world systems, especially those related to coupled energy and water dynamics (Kontopoulou et al., 2023). Traditional techniques primarily focus on forecasting linear and static processes and struggle to identify these complex non-linear structures. Consequently, they exhibit poor performance, difficulty in accommodating multiple exogenous variables, and often fail to accurately generalize patterns, resulting in substantially higher prediction errors (Barzola-Monteses et al., 2019; Hanoon et al., 2023).

2.4.2. Application of AI in Energy Forecasting

The complexity arising from uncertain, dynamic, non-linear, and non-parametric variables in hydropower optimization necessitated a strategic shift toward AI and ML techniques (Bernardes Jr et al., 2022; Kontopoulou et al., 2023). Machine-learning models provide considerable adaptability in prediction because they can represent nonlinear processes without requiring a predefined mathematical relationship. Their applications in hydropower include river-flow forecasting, reservoir operation, and generation prediction (Sapitang et al., 2020).

Key AI and ML models applied in energy and hydropower forecasting include:

Artificial Neural Networks (ANNs). The most widely utilized ML technique, deployed to predict river flow, reservoir inflow, and short- to medium-term production due to their robustness and superior capability to derive complex non-linear relationships (Bernardes Jr et al., 2022; Hanoon et al., 2023). The Radial Basis Function Neural Network (RBFNN) is a specific ANN architecture frequently investigated in forecasting literature and is the focus of this study (Hanoon et al., 2023).

Support Vector Machines (SVMs) and Regression (SVR). Widely applied due to their demonstrated efficacy in modeling non-linear processes and their superior performance over simpler linear methods for river discharge forecasting (Hanoon et al., 2023).

Tree-based Models. Algorithms like Random Forest (RF) and Boosted Decision Tree Regression (BDTR) constitute a significant category, successfully applied to forecasting and noted for their ability to handle smaller datasets and multiple exogenous inputs (Sapitang et al., 2020).

2.4.3. Random Forest and XGBoost as Tree-Based Benchmark Methods

Tree-based ensemble learning methods have also become important in short-term energy forecasting because they can model nonlinear relationships without requiring strict assumptions about the form of the input-output function. Random Forest is an ensemble regression method that combines multiple decision trees to improve prediction accuracy and reduce variance. Random Forest is widely used for both classification and regression problems because of its high predictive accuracy, low variance, ease of learning, and relatively simple optimization (Dudek, 2022). Dudek (2022) showed that Random Forest could represent nonlinear trends, multiple seasonal patterns, random fluctuations, and changing daily profiles in short-term load forecasting. The study also found that a properly configured Random Forest model outperformed several statistical and machine-learning approaches in the evaluated cases, including ARIMA, ETS, MLP, SVM, LSTM, XGBoost, and LightGBM.

XGBoost, or Extreme Gradient Boosting, is another tree-based ensemble method that builds decision trees sequentially, where each new tree attempts to correct the errors of the previous trees. XGBoost achieved greater accuracy and shorter training time than SVM and LSTM in a short-term power-load forecasting application (Zheng et al., 2020). Their results showed that XGBoost obtained a MAPE of 2.01%, compared with 2.80% for SVM and 2.30% for LSTM, while also requiring less training time. The study also showed that when wavelet decomposition was added before XGBoost prediction, the forecasting error was further reduced, demonstrating the ability of XGBoost to perform well when combined with proper preprocessing (Zheng et al., 2020).

In this study, Random Forest and XGBoost were included as benchmark models because they represent strong state-of-the-art tree-based machine learning methods for short-term forecasting. Their inclusion allows the proposed RBFNN model to be evalu-

ated against established nonlinear regression models. However, they were not treated as the primary proposed model because the main contribution of this study focuses on the RBFNN-based forecasting framework, which is designed to learn localized nonlinear generation behavior in the Agus cascaded hydropower system.

2.4.4. Comparison of State-of-the-Art Forecasting Methods

Recent studies on hydropower and energy forecasting show that forecasting methods have evolved from conventional statistical models to advanced artificial intelligence and machine learning approaches. Traditional methods such as ARIMA, SARIMA, and multiple linear regression are useful for simple and interpretable forecasting problems, but they are limited when the system contains nonlinear, nonstationary, and multivariable behavior. In contrast, machine learning and neural network models are more suitable for complex energy forecasting problems because they can learn nonlinear relationships directly from historical operational and environmental data. Table 2.1 summarizes the major forecasting approaches reviewed in this study and explains their relevance to the selected models.

Table 2.1. Comparison of State-of-the-Art Methods for Short-Term Energy and Hydropower Forecasting

Method	Main Strength	Limitation	Role in This Study
ARIMA/ SARIMA	Simple, interpretable, and useful for linear time-series forecasting.	Limited in modeling nonlinear and multivariable hydropower behavior.	Discussed as a conventional forecasting method.
Multiple Linear Regression	Easy to interpret and useful for simple relationships between variables.	Assumes mostly linear relationships and may fail under complex cascade dynamics.	Discussed as a conventional forecasting method.

Continued on next page

Method	Main Strength	Limitation	Role in This Study
ANN/MLP	Can model nonlinear input-output relationships using hidden layers.	Requires tuning and may overfit without proper validation.	Used as part of the literature foundation for AI-based forecasting.
RBFNN	Learns localized nonlinear patterns and is suitable for short-term nonlinear forecasting.	Requires proper center selection, scaling, and hyperparameter tuning.	Main proposed model of this study.
Random Forest	Handles nonlinear regression, reduces variance through ensemble averaging, and works with mixed predictors.	May require careful feature engineering and may not naturally capture smooth sequential dynamics.	Benchmark model used for comparison with the proposed RBFNN.
XGBoost	Provides strong predictive accuracy through gradient boosting, regularization, and sequential error correction.	Sensitive to hyperparameter tuning and may overfit if not properly validated.	Benchmark model used for comparison with the proposed RBFNN.
LSTM/GRU	Designed to capture sequential and long-term temporal dependencies.	Requires larger datasets, longer training time, and more complex tuning.	Recommended for future deep-learning-based comparison.

Continued on next page

Method	Main Strength	Limitation	Role in This Study
Hybrid or Physics-Informed Models	Can combine data-driven learning with physical hydropower constraints.	More complex to develop and requires detailed physical system information.	Recommended for future improvement of cascade-aware forecasting.

Based on the reviewed methods, the RBFNN was selected as the primary model because it provides a balance between nonlinear modeling capability, computational efficiency, and suitability for short-term localized forecasting. Random Forest and XGBoost were selected as benchmark models because they represent strong and commonly used ensemble machine learning methods. Their inclusion strengthens the evaluation of the proposed RBFNN framework by comparing it against established state-of-the-art nonlinear regression models.

2.4.5. Comparative Performance of AI-Based Forecasting Models

AI models consistently demonstrate performance advantages over conventional statistical techniques in hydropower generation (HPG) time-series forecasting (Kontopoulou et al., 2023). This superiority stems from several key capabilities.

Modeling Non-linearity and Complex Dynamics. AI models, particularly Artificial Neural Networks (ANNs) and Support Vector Regression (SVR), exhibit superior capability in modeling the inherently non-linear processes of HPG data. This stands in contrast to traditional statistical techniques such as ARIMA, which often struggle when confronted with complex and highly dynamic structures (Hanoon et al., 2023; Kontopoulou et al., 2023).

Generalization and Accuracy. Machine learning approaches are consistently recognized as powerful tools that provide improved accuracy and generalization compared to conventional methods (Hanoon et al., 2023). Comparative studies frequently demonstrate that ANN and SVM models outperform ARIMA/ARIMAX techniques in predicting hydro-

electric production across various forecasting horizons (Barzola-Monteses et al., 2022).

Adaptability to Complex Systems. The AI approach is generally preferred due to its stronger capability to model time-varying and complex systems. This is particularly important in cascaded hydroelectric power plants, where the operation of one reservoir non-linearly influences downstream units, creating highly coupled system dynamics (Bernardes Jr et al., 2022).

While deep learning models may sometimes require careful tuning of numerous hyperparameters, when properly optimized, AI algorithms generally achieve better prediction performance in most forecasting applications (Kontopoulou et al., 2023).

2.5. Artificial Neural Networks for Generation Forecasting

The increasing complexity and dynamic operation of modern power systems, particularly those incorporating renewable energy sources, have necessitated the development of highly accurate and adaptive forecasting methods (Baliyan et al., 2015). ANNs have emerged as one of the most effective families of models for handling the inherent non-linearity and complexity found in load and generation time series data (Baliyan et al., 2015; Bucci, 2020).

2.5.1. Conceptual Foundations and Suitability of ANNs

An ANN is defined as a highly connected array of elementary processors, or neurons, loosely inspired by the structure and distributed processing of the biological human brain (Baliyan et al., 2015). Fundamentally, an ANN functions as a non-parametric model designed to approximate the unknown function that generates observed data. The typical architecture includes an input layer, one or more hidden layers, and an output layer (Baliyan et al., 2015). The network is trained by iteratively updating the synaptic weights which store the acquired knowledge through algorithms like backpropagation, minimizing a cost function based on the output error (Baliyan et al., 2015). ANNs are exceptionally well-suited for short-term forecasting because they possess the capacity to approximate arbitrarily well a wide class of linear and nonlinear functions without requiring restrictive prior knowledge

of the data-generating process (Bucci, 2020; Lu et al., 2022). This capability is critical when facing non-stationary time series and abrupt environmental or operational changes that render traditional statistical methods ineffective (Baliyan et al., 2015).

2.5.2. ANN Architectures and Applications in Energy Forecasting

The ANN family includes various architectures broadly divided into Feedforward Neural Networks (FFNNs), where signals propagate in a single direction (e.g., Multilayer Perceptron (MLP) and Radial Basis Function (RBF) networks), and Recurrent Neural Networks (RNNs), which incorporate internal feedback loops to retain state information and model temporal dependencies (e.g., Long Short-Term Memory (LSTM) networks) (Lu et al., 2022). ANNs have been extensively employed across power system operations due to their ability to capture complex and non-linear relationships within time-series data.

Load and Renewable Forecasting. ANNs are among the most prominent non-parametric tools for short-term load forecasting (STLF) (Baliyan et al., 2015). Multilayer Perceptron Neural Networks (MLPNNs) have also been successfully applied to solar irradiance and rainfall forecasting by effectively modeling complex non-linear meteorological variables (Madhiarasan & Louzazni, 2022; SV et al., 2020).

Hydropower Prediction. ANNs have been successfully applied in numerous studies to predict hydroelectric power generation and reservoir inflow (Barzola-Monteses et al., 2022; Kaewarsa & Kongpaseuth, 2024). Furthermore, hybrid models that integrate ANNs with optimization algorithms have demonstrated superior accuracy in forecasting available energy, highlighting the adaptability and effectiveness of ANN-based approaches in hydropower systems (Kaewarsa & Kongpaseuth, 2024).

2.5.3. Advantages and Typical Limitations of ANNs

ANNs offer several key advantages over conventional statistical techniques in time-series forecasting applications.

Non-linear Modeling. The primary strength of ANNs lies in their ability to accurately approximate and model complex non-linear behaviors inherent in fluctuating and

non-stationary time-series data, consistently producing more accurate results compared to traditional approaches (Barzola-Monteses et al., 2019; Bucci, 2020).

Data-Driven Generalization. ANNs are inherently data-driven and do not impose restrictive assumptions on the underlying data distribution, enabling them to implicitly detect and learn complex non-linear relationships between variables (SV et al., 2020).

Temporal Memory. Recurrent neural network architectures, such as RNNs and LSTM networks, utilize internal feedback mechanisms to capture and integrate historical dependencies into current predictions, a feature that is particularly crucial for time-series forecasting tasks (Lu et al., 2022).

Despite their advantages, ANNs also exhibit several limitations that must be considered in practical applications.

Data Requirements. ANNs are often described as “greedy” models because they typically require a substantial volume of historical data to train effectively and achieve reliable generalization (Bucci, 2020).

Overfitting and Complexity. Improper selection of network parameters can lead to overfitting, which significantly compromises the model’s ability to generalize to unseen data (Madhiarasan & Louzazni, 2022). Additionally, extensive trial-and-error is often required to determine the optimal network design, such as the appropriate number of hidden neurons (Madhiarasan & Louzazni, 2022).

Training and Computational Cost. Models with multiple hidden layers or complex recurrent structures, such as LSTMs, are inherently more complicated, require longer training times, and demand greater computational resources (Barzola-Monteses et al., 2019).

Hybrid modeling techniques and optimization algorithms often mitigate these limitations (Kaewarsa & Kongpaseuth, 2024). The suitability of the specific architecture, such as the Radial Basis Function Network, must be evaluated based on the complexity and volume of the available cascaded hydropower data.

2.6. The Radial Basis Function Neural Network (RBFNN)

The accurate short-term generation forecasting of cascaded hydroelectric power plants requires predictive models capable of handling complex, nonlinear system dynamics and rapidly processing large volumes of time-series data (Slater et al., 2023). Within the general ANN framework, the RBFNN architecture provides distinct, specialized advantages for high-precision function approximation tasks such as forecasting.

2.6.1. RBFNN Architecture and Differentiation from MLP

The RBFNN is a particular type of feed-forward network recognized as a universal function approximator, capable of approximating complex mathematical functions to any arbitrary degree of precision (Soper, 2023). This capability is shared with the widely used Multi-Layer Perceptron (MLP) network. However, the RBFNN architecture fundamentally differs from conventional MLP networks in its structure and core function (Soper, 2023).

The RBFNN relies on a simple three-layer structure: an input layer, a single hidden layer using non-linear radial basis functions (RBFs) as activation functions, and an output layer (Soper, 2023). In contrast, the MLP typically requires multiple hidden layers to achieve comparable high-precision function approximation (Soper, 2023). The core functional difference lies in the localized response of the hidden layer: each hidden neuron calculates a Gaussian function based on the Euclidean distance between the input vector and the neuron's stored "center" (Slater et al., 2023). This means RBFs function as local receptors, providing a strong output only when the input value is close to the node's center, differentiating it from the global response characteristic of MLP neurons (Soper, 2023).

2.6.2. Fundamentals of RBFNN Parameter Determination

The overall training task of an RBFNN is typically implemented through two-phase learning (Schwenker et al., 2001).

RBF Layer Parameters (Centers and Widths). The first phase involves determining the centers and widths or spreads of the RBF layer, often implemented through unsuper-

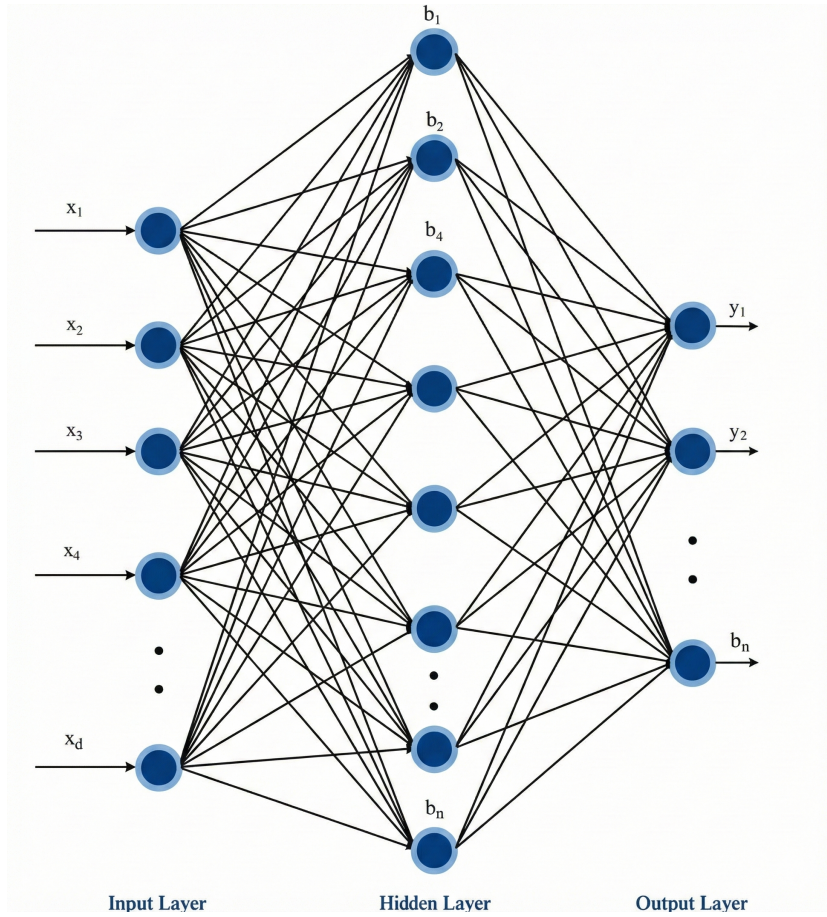


Figure 2.1. Architecture of the RBFNN (Chang et al., 2015).

vised learning (Montazer et al., 2018).

Centers: are vectors representing prototypes of the input data, and their choice is critical (Soper, 2023). Common methods for setting centers include random sampling or k-means clustering, which places centers in areas that best represent the overall data distribution (Soper, 2023). The *Widths (Spreads)* parameter controls the shape of the Gaussian basis function, determining the extent of the node's localized influence (Soper, 2023).

Output Weights. The second phase involves determining the weights between the RBF layer and the output layer (Soper, 2023). This is the only part of the RBFNN that requires supervised learning. Optimal weights can be determined either through iterative methods like gradient descent or, critically, by direct calculation using a pseudoinverse approach, which efficiently finds a minimum-norm weight solution for the given data (Soper, 2023).

2.6.3. RBFNN Success in Forecasting and Function Approximation

RBFNNs have proven versatile and powerful across a broad array of practical applications, confirming their utility in high-precision tasks (Soper, 2023). Successes span numerous domains, including the solution of partial differential equations, computer vision, and medical classification (Montazer et al., 2018).

For predicting dynamic systems, RBFNNs demonstrate strong performance in various time-series analysis and forecasting contexts:

Load Forecasting. RBFNNs have been successfully implemented for short-term load forecasting in electric power systems, demonstrating high accuracy in predicting loads 10 minutes ahead, an application that inherently deals with non-linearity and periodicity (Slater et al., 2023).

System Prediction. RBFNNs have been utilized for forecasting power in wind energy conversion systems and for categorizing high-resolution electrocardiograms (ECGs), demonstrating their effectiveness in processing 1D time-series data (Schwenker et al., 2001; Slater et al., 2023).

Complex Function Approximation. RBFNNs have undergone extensive testing on intricate non-linear mathematical optimization functions and real-world datasets, consistently producing precise predictions (Soper, 2023).

2.6.4. Suitability for Short-Term Hydro Generation Forecasting

The intrinsic characteristics of RBFNNs make them particularly well-suited for the specific demands of short-term generation forecasting in cascaded hydroelectric systems:

Fast and Efficient Training. The rapid training process of RBF networks, often orders of magnitude faster than MLPs is a critical asset for short-term forecasting models that require frequent retraining or rapid iterative adjustment to reflect current operational conditions (Slater et al., 2023; Soper, 2023). Furthermore, RBF networks typically avoid the issue of being trapped in local minima often encountered in backpropagation-trained ANNs (Slater et al., 2023).

High Capacity for Non-Linear Dynamics. As universal function approximators,

RBF networks excel at accurately approximating the highly complex, non-linear dynamics characteristic of cascaded hydro systems, which are governed by reservoir interactions and hydraulic constraints (Soper, 2023). This allows for crucial non-linear mapping between input stimuli and power output.

Local Approximation and Interpretability. The localized nature of the RBF functions allows individual nodes to specialize in approximating the function within a specific region of the input space (Soper, 2023). This local tuning helps the network effectively handle noisy or sparse data points common in real-world operational environments. Additionally, the RBF centers can often be interpreted as representative patterns of the data set, providing a degree of interpretability that is valuable in technical systems (Schwenker et al., 2001).

The combination of computational speed, powerful non-linear modeling capability, and localized learning makes the RBFNN architecture a robust and advantageous choice for the specific demands of short-term hydroelectric generation forecasting, prioritizing efficiency and accuracy in handling complex operational data.

2.6.5. RBFNN Training and Hyperparameter Tuning Methodology

The successful application of the RBFNN for short-term generation forecasting requires a rigorous training protocol focusing on effective parameter optimization and systematic hyperparameter tuning. The RBFNN's efficiency stems from its ability to employ hybrid learning where the hidden node parameters are set prior to optimizing the output weights (Aik et al., 2019).

A. Training RBFNN Parameters. The RBFNN training aims to find the optimal configuration of the centers, spreads, and output weights that minimize the Mean Square Error (MSE) (Horng et al., 2012).

Centers Selection and Clustering. The K-means (KM) algorithm is widely adopted for unsupervised center selection due to its speed and simplicity, grouping input cases into clusters where the centroids become the RBF centers (Aik et al., 2019). However, standard K-means is susceptible to the initial selection of centers and may not always extract the

most accurate prototypes from complex time-series datasets, prompting the investigation of improved methods, such as the Distance-Weighted K-means (DWKM) algorithm (Aik et al., 2019).

Traditional and Gradient-Based Optimization. The Gradient Descent (GD) algorithm is a conventional derivative-based method often used for RBFNN training (Horng et al., 2012). However, like many derivative-based methods, GD frequently gets trapped in local minima and is generally outperformed by more advanced search techniques (Hamadneh, 2020).

Metaheuristic and Hybrid Optimization. To achieve global optimization and efficiently determine all complex parameters simultaneously, metaheuristic and swarm intelligence algorithms have become powerful alternatives (Hamadneh, 2020; Horng et al., 2012). Algorithms like the Genetic Algorithm (GA), Particle Swarm Optimization (PSO), and the Firefly Algorithm (FFA) model the entire set of RBFNN parameters (weights, spreads, centers) as a single solution vector. These techniques utilize randomized search to efficiently explore the solution space, demonstrating competitive or superior performance to traditional GD and GA algorithms when training RBFNNs for time-series prediction (Hamadneh, 2020).

B. Hyperparameter Tuning Strategies. Hyperparameter tuning focuses on selecting values for parameters that control the learning process, such as the number of hidden neurons (which equals the number of centers) and the spread parameter (σ) (Kahlout & Ekler, 2021).

Tuning via Experimentation and Validation. The most practical and common strategy involves extensive trial-and-error experimentation combined with the use of a validation dataset (Horng et al., 2012). Researchers typically test RBFNNs with varying numbers of hidden neurons to identify the configuration that yields the optimal performance metric on the validation set (Horng et al., 2012). This approach ensures the hyperparameter selection is robust and prevents overfitting to the training data.

Tuning via Optimization Algorithms and Criteria. In implementations utilizing metaheuristic training (PSO, FFA), the hyperparameter selection is often intrinsically part

of the optimization process, as the algorithm directly searches for the best combination of centers and spreads alongside the weights (Horng et al., 2012). To formally evaluate the best resulting architecture, model selection criteria like the Schwarz Bayesian criterion (SBC) are utilized to assess the trade-off between minimal error and network complexity (Hamadneh, 2020). Cross-validation methods may also be used to ensure the chosen architecture is not biased towards a particular subset of the data (Horng et al., 2012).

2.6.6. Dataset Splitting in Time-Series Forecasting

In studies that use time-series forecasting, the dataset must be split up in strict chronological order so that the model only learns from the past to make predictions about the future. Unlike general machine learning problems, random splitting is inappropriate, as it introduces future information into the training phase, leading to optimistically biased performance estimates.

The dataset is segmented into three distinct, non-overlapping subsets based on time: the training set (70%), which comprises the earliest data and is used for fitting the model parameters (centers, spreads, and weights); the validation set (15%), which follows the training set chronologically and is used to tune the model hyperparameters, such as selecting the optimal number of hidden neurons or spread value; and the testing set (15%), which consists of the most recent data and is reserved exclusively to provide a final, unbiased evaluation of the model's generalization capability on entirely unseen conditions (Kahlout & Ekler, 2021).

In quantitative studies, common ratios like 70% for training and 30% for testing (or 70–15–15% when a separate validation set is used) are frequently employed, provided this splitting is done sequentially (Hamadneh, 2020). This rigorous partitioning methodology is essential for developing a reliable short-term generation forecasting model for cascaded hydroelectric power plants.

2.7. Justification for the Selection of RBFNN, Random Forest, and XGBoost

The selection of Radial Basis Function Neural Network, Random Forest, and XGBoost was based on the forecasting requirements of the Agus cascaded hydropower system. The generation output of the Agus plants is affected by nonlinear interactions among historical generation, unit availability, forebay elevation, gate operation, rainfall, Lake Lanao outflow, and upstream-downstream cascade behavior. Because of these interacting variables, a purely linear forecasting method may not be sufficient to represent the actual behavior of the system.

The RBFNN was selected as the primary forecasting model because it is suitable for nonlinear function approximation and localized learning. Its radial basis neurons respond strongly to input patterns that are close to their learned centers, allowing the model to represent different operating conditions of the hydropower cascade. This is relevant to the Agus system because plant generation may behave differently under high-flow, low-flow, normal-operation, and outage-influenced conditions. The RBFNN therefore provides a practical structure for learning nonlinear generation behavior from historical data while remaining simpler than deeper recurrent architectures.

Random Forest was selected as the first benchmark model because it is a strong ensemble regression method that can handle nonlinear relationships and mixed operational predictors. It is also relatively easy to train and less sensitive to scaling compared with many neural-network-based methods. Since hydropower forecasting uses a combination of numerical and binary features, including generation lags, rolling statistics, time indicators, and outage states, Random Forest provides a useful comparison model for evaluating whether the proposed RBFNN can outperform a robust tree-based ensemble baseline.

XGBoost was selected as the second benchmark model because it is a state-of-the-art gradient boosting method known for strong predictive performance in structured data forecasting problems. Unlike Random Forest, which averages many independent trees, XGBoost builds trees sequentially so that each new tree improves the previous prediction errors. This makes XGBoost a strong benchmark for testing whether the RBFNN provides

better accuracy and stability than a highly competitive boosting-based model. Prior work has shown that XGBoost can achieve high short-term forecasting accuracy and shorter training time compared with SVM and LSTM in power load forecasting applications (Zheng et al., 2020).

Therefore, the three methods served different roles in this study. The RBFNN was used as the main proposed model because it directly supports localized nonlinear learning for short-term hydropower generation forecasting. Random Forest and XGBoost were used as benchmark models because they are established state-of-the-art tree-based machine learning methods. This model selection allowed the study to evaluate the proposed RBFNN against strong nonlinear alternatives and determine whether it provides improved forecasting accuracy for the Agus cascaded hydropower plants.

2.8. Data Preprocessing and Quality Control

The performance, stability, and reliability of AI models, especially the RBFNN, are fundamentally contingent upon the volume, quality, and diversity of the input data (Maharana et al., 2022). Raw data collected from real-world applications are susceptible to numerous issues, necessitating robust preprocessing before being utilized for short-term generation forecasting (Maharana et al., 2022).

2.8.1. Data Problems in Hydropower Time-Series

Hydropower generation and associated meteorological and operational observations manifest as complex multivariate time-series data (Zhou et al., 2020). These datasets frequently exhibit intrinsic issues that complicate the forecasting process:

Missing Values and Irregular Sampling. Missing data are a common problem resulting from technical difficulties, sensor failures, or network disconnections (Petrusevich, 2021). These failures can create large gaps, sometimes stretching for weeks (Cho et al., 2020). Additionally, data acquisition may suffer from irregular sampling or asymmetric measurements due to transmission delays, complicating the time-alignment of observations (Zhou et al., 2020).

Sensor Noise and Inconsistency. Raw data can be affected by noise and corruption caused by how sensors work or how they communicate (Cho et al., 2020). Data inconsistency occurs when information gathered from various heterogeneous sources differs in standards or detail (Maharana et al., 2022). For sensitive machine learning models like ANNs, the presence of noisy and unreliable data significantly impedes the knowledge discovery process during training (Kotsiantis et al., 2006).

2.8.2. Strategies for Handling Missing Values and Imputation

Addressing missing data is critical because models that rely on continuous time series, such as RBFNNs, require input vectors that are complete and representative (Cho et al., 2020).

Simple Univariate Interpolation. For filling small data gaps, Linear Interpolation (LIN) is highly effective, assuming a linear function between the nearest good measurements (Cho et al., 2020). Spline Interpolation (SPL), such as using cubic splines, expands on this by creating a smooth, piecewise polynomial function that is less susceptible to high-variance periods (Petrusevich, 2021). Other basic methods include replacement using the mean, median, or mode (Maharana et al., 2022).

Advanced Model-Based and Multivariate Imputation. For handling larger gaps or complex patterns, more sophisticated methods are required. Multivariate approaches often yield superior results for large gaps by leveraging inherent periodicity and relationships within the dataset (Cho et al., 2020). These methods include K-Nearest Neighbors (KNN), which imputes values using a weighted mean of the k most similar data rows and has been shown to be accurate for medium gaps (Cho et al., 2020); Matrix Factorization (MF), which decomposes the time-series matrix to infer missing values and provides the most accurate imputations for large gaps across different sensor categories and missing rates (Cho et al., 2020); and Autoregressive (AR) models, which estimate missing values as forecasts but require uninterrupted data segments for reliable coefficient estimation (Petrusevich, 2021).

2.8.3. Noise Reduction and Data Smoothing Techniques

Minimizing the presence of noise is essential prior to training sensitive AI models such as RBFNNs, as noise significantly impedes the knowledge discovery process (Maharana et al., 2022). Filtering and smoothing techniques improve trend detection and generalization performance by highlighting essential features (Hota et al., 2017).

Moving Average and Sliding Window. The Weighted Moving Average (WMA) is widely adopted for smoothing highly non-linear time series data, where the highest weight is assigned to the most recent data to enhance trend detection (Hota et al., 2017). This is often combined with a sliding window concept, where a fixed-size window aggregates historical data points to generate new, smoother input observations for the RBFNN (Hota et al., 2017).

General Filtering. Other methods include binning, which reduces the impact of minor observation errors by replacing binned values with derived general values (Maharana et al., 2022). More generally, sophisticated algorithms can be used that explicitly filter, modify, or relabel noisy data while preserving essential data features (Maharana et al., 2022).

2.8.4. The Crucial Role of Robust Data Preprocessing

Robust preprocessing is not merely a preliminary step but a critical determinant of the eventual success, stability, and reliability of AI-based forecasting models (Kotsiantis et al., 2006). Poor data quality significantly affects the accuracy and reliability of predictions, potentially leading to false conclusions (Maharana et al., 2022).

The substantial time dedicated to initial data preparation (estimated at 50% to 80% of the entire process) underscores its importance (Maharana et al., 2022). Data cleaning ensures the resultant dataset meets critical quality criteria, including validity, accuracy, completeness, consistency, and uniformity (Maharana et al., 2022). Ultimately, ensuring that the training data is clean and representative maximizes the RBFNN's ability to extract meaningful non-linear information, thereby guaranteeing stable and reliable AI-based forecasts necessary for cascaded hydropower operation.

2.9. Evaluation Metrics

The validation of forecasting models requires robust statistical metrics to assess prediction performance. This study employs standard metrics commonly used across hydrological and energy forecasting literature: Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), Mean Absolute Percentage Error (MAPE), and the Coefficient of Determination (R^2) (Chicco et al., 2021; De Myttenaere et al., 2016; Hodson, 2022).

Root Mean Squared Error is derived from the Mean Squared Error (MSE) and uses the L_2 -norm, involving the squaring of prediction errors (Chicco et al., 2021; Hodson, 2022). Because errors are squared, RMSE is inherently more sensitive to outliers, giving larger weights to points with high deviations (Chicco et al., 2021). Minimizing RMSE is statistically optimal if errors follow a normal (Gaussian) distribution (Hodson, 2022). RMSE yields a metric with the same units as the target variable (Hodson, 2022).

Mean Absolute Error minimizes the L_1 -norm, which uses a linear treatment of errors (Hodson, 2022). MAE is recognized as a robust measure because the L_1 -norm tends to smooth out the penalty applied to extreme outliers, making it less sensitive to severe forecasting mistakes compared to RMSE (Chicco et al., 2021; Hodson, 2022). MAE is mathematically optimal when errors follow a Laplacian distribution (Hodson, 2022).

Mean Absolute Percentage Error is widely utilized and provides a highly intuitive interpretation in terms of relative error (Chicco et al., 2021; De Myttenaere et al., 2016). This metric is particularly useful when the application emphasizes sensitivity to proportional variations (De Myttenaere et al., 2016). A critical consideration is that MAPE is restricted to cases where the target variable is strictly positive and is known to be heavily biased towards low forecasts (Chicco et al., 2021).

Coefficient of Determination quantifies the proportion of variance in the dependent variable that is predictable from the independent variables (Chicco et al., 2021). R^2 offers a bounded, scale-independent assessment of fit (Chicco et al., 2021), ranging from negative infinity up to 1 (perfect prediction). A value like $R^2 = 0.8$ clearly suggests a very good regression model performance, irrespective of the predicted generation output's scale (Chicco

et al., 2021) .

These metrics are essential for a comprehensive evaluation, where MAPE is critical for assessing relative accuracy (desirable in generation forecasting) and R^2 provides an absolute measure of goodness-of-fit for comparing the RBFNN against alternative methods (Chicco et al., 2021; De Myttenaere et al., 2016).

2.10. Research Gaps

Despite the proven success of AI in general forecasting, several specific methodological and domain-focused gaps remain when approaching AI-based short-term generation forecasting for cascaded hydroelectric power plants. These gaps define the need for the present thesis:

Limited Focus on Cascaded Systems as a Forecasting Problem. The existing literature often treats cascaded hydropower as an optimization problem where forecasts are merely external inputs (Balan et al., 2024; Ji et al., 2018). Conversely, most hydropower forecasting studies focus on single-reservoir or individual station output, neglecting the core challenge of explicitly modeling the hydraulic coupling, routing delays, and head dependencies simultaneously across an entire cascade (Barzola-Monteses et al., 2019; Bernardes Jr et al., 2022). This leaves a gap in dedicated data-driven models for concurrent, coupled generation forecasting in cascaded systems.

Lack of Focused One-Day-Ahead Operational Resolution. While short-term horizons are recognized as critical, most hydropower forecasting is geared toward daily or longer resolutions for medium-term planning (Essenfelder et al., 2020; Sapitang et al., 2020). Relatively limited attention has been given to one-day-ahead hydropower generation forecasting at an hourly resolution (Essenfelder et al., 2020; Sapitang et al., 2020). This specific resolution is essential for crucial tasks like market bidding, spill management, and real-time coordination of reservoir releases, a gap highlighted by comparisons with advanced load and wind/solar forecasting (Akhtar et al., 2023; Tsai et al., 2023).

Underutilization of RBFNNs. Despite their documented strengths as universal approximators with fast training speed, localized basis functions, and robustness in non-

linear regression (Schwenker et al., 2001; Soper, 2023), RBFNNs are underutilized as the primary architecture for hydropower generation forecasting. Most studies favor MLPs, SVMs, or tree-based models (Hanoon et al., 2023; Kaewarsa & Kongpaseuth, 2024). This presents a gap in systematically applying and comparing the RBFNN architecture for one-day-ahead generation forecasting in complex cascaded systems.

Simple Feature Integration Lacking Cascade Physics. Many existing AI models for hydropower rely on relatively simple input sets. This often fails to fully exploit multivariate operational features that encode the physical coupling central to cascaded operation. Specifically, there is a gap in models that comprehensively integrate upstream releases, inter-reservoir travel time, and dynamic reservoir head across the cascade, which are critical physical drivers of short-term generation (Amitaba et al., 2024; Balan et al., 2024).

Preprocessing and Evaluation. A methodological gap exists concerning the documentation and consistent application of complete data quality pipelines. While the need for handling missing values, sensor noise, and chronological data splitting is stressed (Cho et al., 2020; Kahloot & Ekler, 2021), few studies provide a unified, reproducible preprocessing protocol tailored for multivariate cascaded hydropower data, combining advanced imputation, smoothing, sliding-window feature construction, and strictly chronological splitting for reliable time-series evaluation.

In response to these gaps, the present study develops an AI-based short-term (one-day-ahead) generation forecasting framework for a cascaded hydropower system using a Radial Basis Function Neural Network (RBFNN). The proposed model (i) treats the cascade as a coupled forecasting problem, (ii) explicitly incorporates cascade-relevant physical features such as upstream releases, travel time, and reservoir head, and (iii) employs a rigorous, time-consistent data preprocessing and evaluation protocol using MAPE. This framework is intended to serve as a practical and methodologically transparent benchmark for operational forecasting in cascaded hydroelectric power plants.

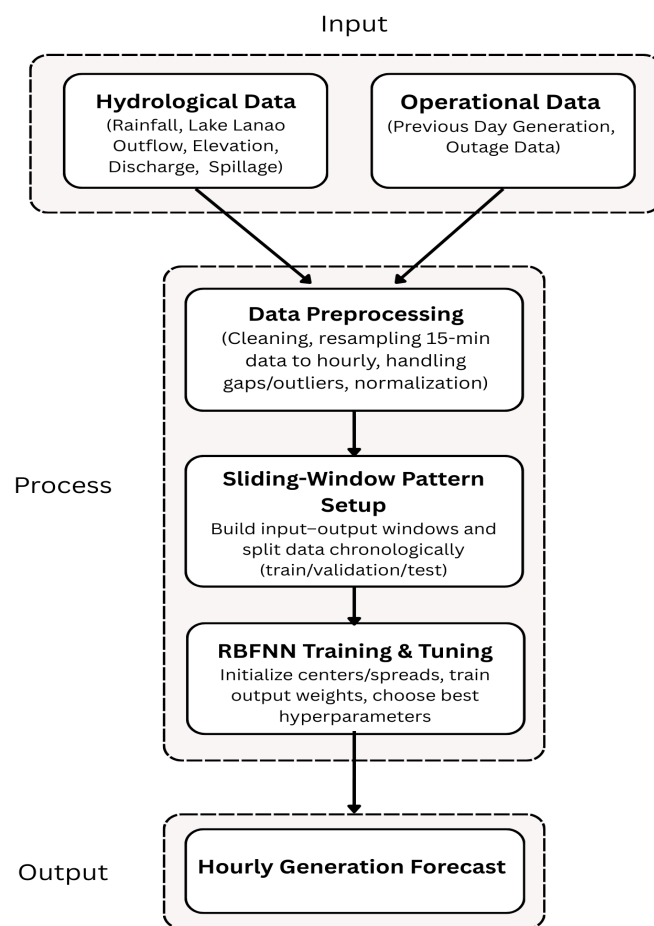


Figure 2.2. Conceptual framework for hourly generation forecasting.

CHAPTER 3

METHODOLOGY

This chapter presents the methodological framework of the study, where historical operational and hydrological data from the Agus 1–7 cascade were processed through a structured data-driven pipeline to develop a short-term forecasting system. The approach began with data collection, cleaning, and resampling into a consistent hourly dataset, followed by feature engineering that captured temporal patterns, cascade relationships, and unit availability through planned outages. These inputs were used to train a Radial Basis Function Neural Network (RBFNN) to learn the nonlinear behavior of the system and generate 24-hour ahead predictions for each plant and the total cascade. The model performance was then evaluated using statistical metrics and compared against benchmark models to assess its accuracy and operational suitability. The overall process is illustrated in Figure 3.1.

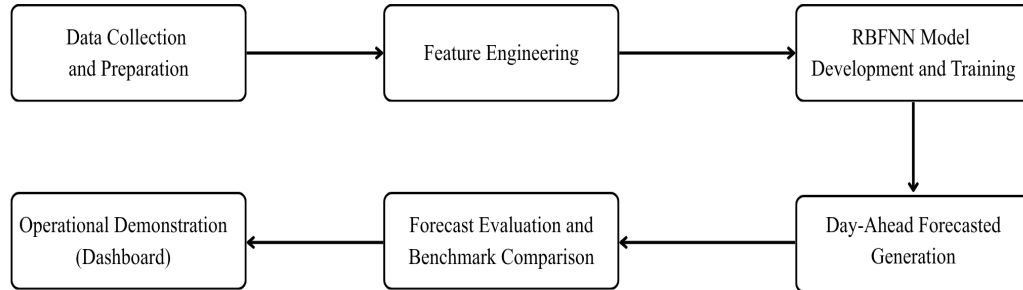


Figure 3.1. Workflow of the Study.

3.1. Research Design

This study adopted a data-driven, supervised learning-based forecasting design, where historical operational data were used to train and evaluate a Radial Basis Function Neural Network (RBFNN) to model the nonlinear relationship between input variables and next-day hourly generation in the Agus 1–7 cascade.

RBFNNs were used because they have demonstrated strong performance in nonlinear time-series forecasting across power and water resources applications, including short-term PV power forecasting (Huang et al., 2015; Xu et al., 2019), short-term load forecasting (Chang et al., 2015), and rainfall prediction (El Shafie et al., 2012).

3.2. Data Collection and Processing

Data collection for this study was conducted using historical operational records obtained from NPC–MinGen, covering the Agus 1–7 hydropower cascade. The dataset included multiple operational variables describing the behavior of the cascade system, such as power generation, reservoir conditions, hydraulic flow, gate operations, environmental inputs, and unit availability. These data captured the nonlinear and time-dependent characteristics of hydropower generation. Since the dataset originated from a unified operational source, consistency and reliability across all variables were ensured without discrepancies arising from multiple data providers.

However, the collected data were recorded at varying temporal resolutions, including 5-minute, hourly, and daily intervals. To support the study objective of preparing a reliable dataset for short-term forecasting, all variables were standardized into a common hourly resolution. This unified structure served as the foundation for preprocessing, feature engineering, model development, and performance evaluation. A summary of the data sources used in this study is presented in Table 3.1.

Table 3.1. Summary of Data Sources

Data Type	Source	Resolution	Variables
Plant Generation	NPC–MinGen Operational Records	5-min (hourly aggregated)	Unit generation (MW), total generation per plant
Outage Data	NPC–MinGen Operational Records	Hourly (derived)	Unit availability (on/off status)
Reservoir Data	NPC–MinGen Operational Records	Hourly	Forebay elevation
Hydraulic Data	NPC–MinGen Operational Records	Daily (hourly upsampled)	Lake Lanao outflow
Gate Operations	NPC–MinGen Operational Records	Hourly	Spillway gate opening
Environmental Data	NPC–MinGen Operational Records	Daily (hourly upsampled)	Rainfall

3.2.1. Data Cleaning and Processing

The collected datasets underwent a structured preprocessing pipeline to ensure data quality, consistency, and suitability for machine learning modeling. Initial steps included cleaning and standardization of column names, conversion of variables into numerical formats, and reconstruction of datetime indices from separate date and time fields. Duplicate entries and invalid records were removed to maintain data integrity.

Consistency checks were performed using basic physical rules, such as non-negative generation values, acceptable reservoir levels, and continuity of flow-related variables. Records that violated these conditions were flagged and either corrected, if caused by obvious logging errors, or treated as missing values.

Missing values were handled using interpolation for short temporal gaps, while re-

maintaining missing entries were addressed using imputation methods. Outliers and abnormal spikes were identified and treated using smoothing and data-cleaning rules to preserve physically meaningful behavior. Near-zero generation values were also carefully treated to distinguish between actual low production and outage conditions.

Unit outage status was not directly available in the dataset but was derived from generation data. Units with zero or near-zero generation were encoded as unavailable (0), while operating units were encoded as available (1). This binary representation allowed the model to incorporate operational constraints into the forecasting process.

3.2.2. Data Consistency and Temporal Alignment

To ensure consistency across all variables, the datasets were aligned to a common hourly timestamp index. High-frequency plant-generation records were aggregated from 5-minute intervals into hourly values, while rainfall and Lake Lanao outflow were retained as daily-derived hourly-equivalent hydrologic context variables. This procedure allowed the operational, hydrologic, and cascade-related variables to be organized into a unified time-series structure suitable for short-term forecasting.

Temporal alignment was necessary because the forecasting model used multiple variables describing the condition of the Agus cascade at each time step. In a cascaded hydropower system, upstream and downstream plant behavior is interconnected, and changes in generation, water release, gate operation, and hydrologic conditions may affect succeeding plants over time. Therefore, synchronizing the data into a common hourly format helped ensure that the model inputs represented temporally comparable operating conditions for each forecasting instance.

Rainfall and Lake Lanao outflow were not treated as direct hourly sensor measurements. The available records represented daily hydrologic conditions that had already been converted into hourly-equivalent values before the modeling stage. Therefore, the preprocessing procedure did not divide these variables by 24 again.

Repeated hourly entries within the same day did not represent 24 independent measurements. Instead, they represented the same daily-derived hydrologic condition asso-

ciated with that date. These variables were therefore retained as coarse daily hydrologic context rather than as representations of within-day rainfall or outflow variability.

During forecasting, rainfall and Lake Lanao outflow were used only through lagged values or the most recently known values available up to 23:00 of the preceding day. Actual values from the forecast day were excluded from the model inputs during validation and testing. This procedure prevented future-information leakage and avoided artificially improving the reported day-ahead forecasting performance.

Although this treatment allowed the available hydrologic records to provide contextual information regarding prevailing water conditions, it introduced a temporal-resolution limitation. The daily-derived values could not represent short-duration rainfall events, within-day outflow changes, or abrupt hydrologic variations. This limitation was considered when interpreting the forecasting results.

3.2.3. Normalization

To improve numerical stability and model convergence, all continuous input features were standardized before model training, consistent with the role of preprocessing in improving the reliability of machine-learning models (Kotsiantis et al., 2006; Maharana et al., 2022). Standardization was performed using the mean and standard deviation calculated exclusively from the training set. For feature j and observation i , the standardized value was computed as

$$x_{i,j}^{\text{std}} = \frac{x_{i,j} - \mu_j^{\text{train}}}{\sigma_j^{\text{train}}}, \quad (3.1)$$

where $x_{i,j}$ is the original value of feature j for observation i , μ_j^{train} is the mean of feature j calculated from the training set, and σ_j^{train} is its corresponding training-set standard deviation. The resulting training features had a mean approximately equal to zero and a standard deviation approximately equal to one.

The same training-derived mean and standard deviation were subsequently applied to the validation and testing sets. No normalization parameters were recalculated using validation or testing data, thereby preserving the chronological separation of the datasets

and preventing information leakage.

3.2.4. Input–Output Pattern Construction

The forecasting problem was formulated using a sliding-window approach, which is commonly used to construct time-series forecasting inputs from historical observations (Hota et al., 2017; Zhou et al., 2020). In this approach, the input vectors were constructed from historical observations available before the forecast time. A separate feature vector was generated for each hydropower plant. For plant p at time t , the model input vector was defined as

$$\mathbf{x}_{p,t} = \left[\mathbf{x}_t^{\text{time}}, \mathbf{x}_{p,t}^{\text{lag}}, \mathbf{x}_{p,t}^{\text{roll}}, \mathbf{x}_{p,t}^{\text{unit}}, \mathbf{x}_{p,t}^{\text{outage}}, \mathbf{x}_{p,t}^{\text{hydro}}, \mathbf{x}_{p,t}^{\text{cascade}} \right]^T, \quad (3.2)$$

where $\mathbf{x}_t^{\text{time}}$ contains calendar and temporal indicators; $\mathbf{x}_{p,t}^{\text{lag}}$ contains lagged plant-generation values and generation-difference features; $\mathbf{x}_{p,t}^{\text{roll}}$ contains rolling means, standard deviations, minimum values, and maximum values; $\mathbf{x}_{p,t}^{\text{unit}}$ contains unit-level generation values and generation-share variables; $\mathbf{x}_{p,t}^{\text{outage}}$ contains unit-availability and outage-related indicators; $\mathbf{x}_{p,t}^{\text{hydro}}$ contains lagged rainfall, Lake Lanao outflow, forebay elevation, gate-opening, and spillway-related variables; and $\mathbf{x}_{p,t}^{\text{cascade}}$ contains lagged upstream-generation variables representing the hydraulic relationship between consecutive plants.

The corresponding model target was the next-hour change in plant generation, defined as

$$\Delta y_{p,t+1} = y_{p,t+1} - y_{p,t}, \quad (3.3)$$

where $y_{p,t}$ is the total generation of plant p at time t , and $\Delta y_{p,t+1}$ is the change in generation between the current hour and the succeeding hour. The RBFNN was therefore trained to estimate $\Delta y_{p,t+1}$ rather than the absolute generation level directly.

Only variables available at or before time t were included in $\mathbf{x}_{p,t}$. During validation and testing, the actual generation, rainfall, Lake Lanao outflow, and outage status from the forecast day were not used as model inputs. This ensured that the input–output construction

remained consistent with the leakage-free day-ahead forecasting procedure.

3.2.5. Dataset Splitting

To ensure realistic model evaluation and to prevent data leakage, the dataset was divided chronologically into three subsets: training, validation, and testing, following the recommended sequential partitioning of time-series forecasting data (Hamadneh, 2020; Kahlout & Ekler, 2021). The training set comprised 70% of the data and was used to fit the model parameters. The validation set, consisting of 15% of the data, was used for hyperparameter tuning and model selection. The remaining 15% was reserved as the testing set for final performance evaluation.

This chronological splitting approach ensured that the model was trained on past data and evaluated on future data, reflecting real-world forecasting conditions.

3.3. Research Instruments

The study used software tools and computational environments for data organization, preprocessing, model development, evaluation, visualization, and prototype system implementation.

Microsoft Excel was used to organize, inspect, and verify the historical operational and hydrologic records obtained from the plant operator. It was also used to review the cleaned datasets, planned-outage inputs, forecast outputs, and model-performance summaries.

Python served as the primary programming environment for implementing the forecasting workflow. The Pandas and NumPy libraries were used for data manipulation, numerical processing, temporal alignment, and feature construction. Scikit-learn was used for preprocessing, standardization, clustering, metric calculation, and the implementation of the Random Forest benchmark model. TensorFlow and Keras were used to construct, train, and evaluate the Radial Basis Function Neural Network. XGBoost was used to implement the Extreme Gradient Boosting benchmark model.

Matplotlib was used to generate the statistical and forecasting visualizations pre-

sented in the results and discussion. The graphical user interface was developed as a prototype environment for reviewing input data, managing planned-outage assumptions, generating day-ahead forecasts, and displaying plant-level and cascade-level forecast results.

Visual Studio Code was used as the integrated development environment for developing, debugging, and executing the computational procedures. All model training, validation, testing, calibration, and forecast generation were performed in a local computational environment using fixed random seeds and saved model artifacts to support reproducibility.

The specific preprocessing, feature-engineering, model-training, calibration, benchmark-evaluation, and rolling day-ahead backtesting procedures are described in the succeeding methodological sections. Explicit backend filenames are provided only in the reproducibility appendix and are not used as substitutes for methodological descriptions in the main body.

3.4. Feature Engineering

To improve the predictive capability of the model, additional features were derived from the processed dataset. Temporal features such as hour-of-day, day-of-week, and seasonal indicators were included to represent periodic patterns in power generation. Lagged variables were generated to incorporate historical dependencies, enabling the model to learn from past system behavior.

Rolling statistical features, including moving averages and standard deviations, were computed to represent short-term trends and variability in the data. Operational features such as unit availability, derived from outage data, were also incorporated to reflect real-world plant conditions.

Cascade relationships were explicitly modeled through upstream generation features, allowing downstream plants to utilize information from preceding plants in the cascade. This represents the physical dependency of water flow between interconnected plants and improves forecasting performance in cascaded hydropower systems.

These engineered features were selected based on their relevance to hydropower system dynamics and their ability to represent both temporal dependencies and physical

interactions within the cascade, guided by domain knowledge and time-series analysis principles.

3.5. RBFNN Model Development

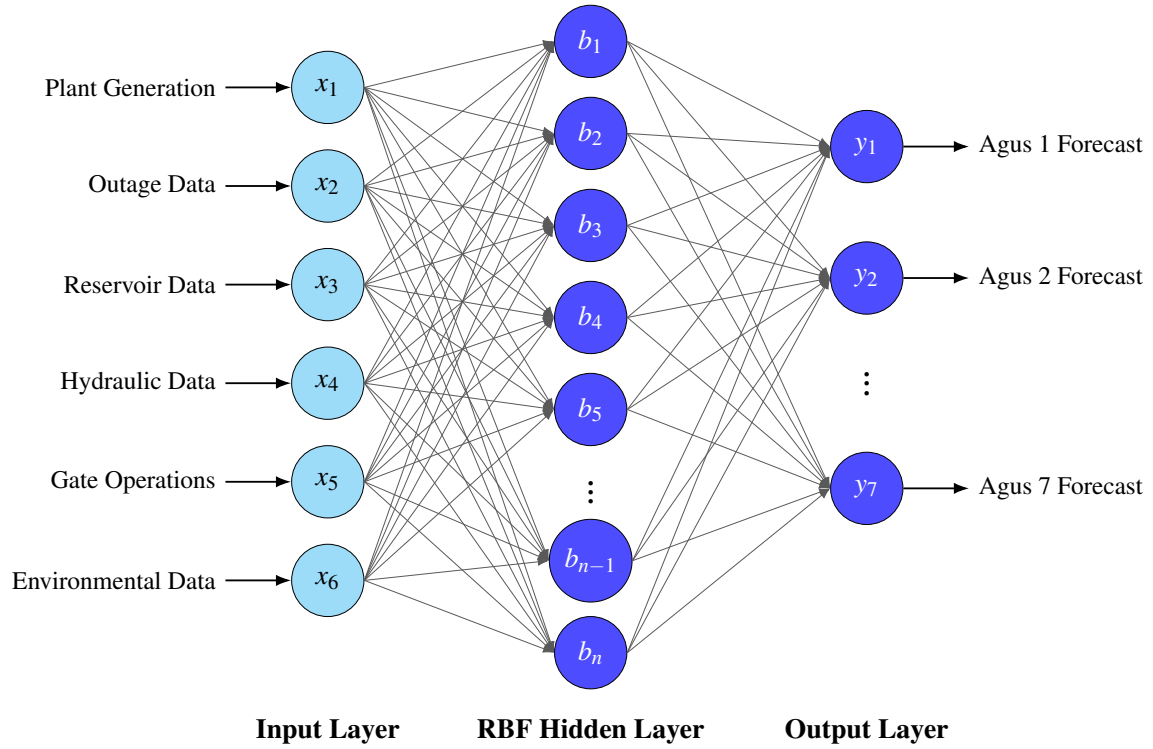


Figure 3.2. RBFNN architecture for day-ahead generation forecasting using hydropower operational and environmental input variables.

The forecasting model was based on a three-layer Radial Basis Function Neural Network (RBFNN), consisting of an input layer, a radial basis hidden layer, and a linear output layer, consistent with the standard RBFNN structure described in prior neural-network forecasting studies (Schwenker et al., 2001; Slater et al., 2023; Soper, 2023). A separate plant-specific model was developed for Agus 1, Agus 2, Agus 4, Agus 5, Agus 6, and Agus 7.

The input layer received the feature vector constructed from the processed dataset. The input variables included temporal indicators, lagged generation values, rolling statistics, unit availability, gate-related variables, forebay elevation, lagged rainfall, Lake Lanao

outflow, and upstream cascade information. The dimension of the input vector depended on the number of selected features for each plant.

For an input-feature vector $\mathbf{x}_t$, the activation of the j -th radial basis neuron was calculated using the Gaussian radial basis function, where the hidden-neuron response depends on the distance between the input vector and the neuron's center (Slater et al., 2023; Soper, 2023):

$$\phi_j(\mathbf{x}_t) = \exp\left(-\gamma_j \|\mathbf{x}_t - \mathbf{c}_j\|_2^2\right), \quad (3.4)$$

where $\mathbf{x}_t$ is the scaled input-feature vector at time t , $\mathbf{c}_j$ is the center vector of the j -th radial basis neuron, $\gamma_j > 0$ is the Gaussian width parameter, and $\phi_j(\mathbf{x}_t)$ is the corresponding hidden-layer activation.

In the implemented RBFNN, the Gaussian width parameter was constrained to remain positive through the relation

$$\gamma_j = \exp(g_j), \quad (3.5)$$

where g_j is the trainable logarithmic width parameter of the j -th radial basis neuron.

The output layer formed a linear combination of the hidden-layer activations, which is the supervised output stage of the RBFNN (Schwenker et al., 2001; Soper, 2023):

$$\widehat{\Delta y}_{t+1} = b_0 + \sum_{j=1}^M w_j \phi_j(\mathbf{x}_t), \quad (3.6)$$

where $\widehat{\Delta y}_{t+1}$ is the predicted change in generation for the succeeding hour, M is the number of radial basis neurons, w_j is the output weight associated with the j -th neuron, and b_0 is the output-layer bias.

3.5.1. Residual Forecasting Formulation

The residual target used in this study was formulated as a study-specific target transformation within the sliding-window time-series forecasting setup. Lagged and rolling historical observations are commonly used to construct time-series forecasting inputs (Hota

et al., 2017; Zhou et al., 2020).

Using the residual target defined in Equation 3.3, each plant-specific RBFNN was trained to predict the next-hour change in generation rather than the absolute generation level directly.

After the RBFNN estimated the residual $\widehat{\Delta y}_{p,t+1}$, the preliminary generation forecast was reconstructed as

$$\widehat{y}_{p,t+1}^{\text{raw}} = y_{p,t} + s_p \widehat{\Delta y}_{p,t+1}, \quad (3.7)$$

where s_p is the validation-selected shrinkage factor for plant p . The bias-corrected forecast was then calculated as

$$\widehat{y}_{p,t+1}^{\text{bc}} = \widehat{y}_{p,t+1}^{\text{raw}} + b_p, \quad (3.8)$$

where b_p is the validation-derived bias correction for plant p .

Finally, the forecast was constrained within the physically plausible plant-capacity range:

$$\widehat{y}_{p,t+1} = \text{clip} \left(\widehat{y}_{p,t+1}^{\text{bc}}, 0, 1.05C_p \right), \quad (3.9)$$

where C_p is the installed capacity of plant p . The upper limit of $1.05C_p$ allowed a small operational tolerance while preventing physically implausible forecasts. The $1.05C_p$ upper bound was used as a study-specific operational guard to allow small tolerance above nominal capacity while preventing unrealistic forecast values.

3.6. Model Training

The RBFNN parameters were estimated using the training subset only. Before model fitting, the input features and residual target were standardized using scaling parameters calculated exclusively from the training data. The same fitted scalers were subsequently applied to the validation and testing subsets.

The network was trained by minimizing the mean squared error between the ob-

served and predicted residual targets, a standard objective for training RBFNN forecasting models (Horng et al., 2012):

$$\mathcal{L}_{\text{MSE}} = \frac{1}{N_{\text{tr}}} \sum_{i=1}^{N_{\text{tr}}} \left(\Delta y_i - \widehat{\Delta y}_i \right)^2, \quad (3.10)$$

where $\mathcal{L}_{\text{MSE}}$ is the training loss, N_{tr} is the number of training observations, Δy_i is the observed generation change for training observation i , and $\widehat{\Delta y}_i$ is the corresponding RBFNN prediction.

The Adam optimizer was used to update the trainable network parameters. Early stopping was applied using validation loss, and the parameter values from the epoch with the lowest validation loss were restored. The RBF centers were initialized through K-means clustering using the scaled training-feature matrix, consistent with common RBFNN center-selection procedures (Aik et al., 2019).

Shrinkage, bias correction, bin calibration, hourly residual correction, and profile blending were not terms in the neural-network loss function. These procedures were applied separately as validation-based forecast calibration and post-processing steps.

3.6.1. Validation-Only Calibration and Model Selection

The validation subset was used to select the model configuration and determine the post-processing parameters. The testing subset was not used during model training, hyperparameter selection, calibration, or post-processing design.

Candidate RBFNN configurations were initially assessed using validation operational MAPE, RMSE, and the coefficient of determination, R^2 . To avoid selecting a configuration with a relatively low percentage error but poor representation of the observed generation pattern, the candidates were ranked using the following validation selection score:

$$S = \text{MAPE}^{\text{op}} + \text{RMSE} + 20(1 - R^2) + 25 \max(0, -R^2), \quad (3.11)$$

where S is the validation selection score. The term $20(1 - R^2)$ penalized configurations with weak explained variance, while $25 \max(0, -R^2)$ imposed an additional penalty

when R^2 was negative. The numerical penalty weights in this score were study-specific validation weights used to discourage models with weak or negative explained variance and were not adopted as general forecasting constants. Candidate configurations with lower values of S were preferred.

The highest-ranked candidate configurations from the preliminary validation screening were subsequently evaluated using the rolling 24-hour day-ahead validation procedure. This ensured that final model selection reflected performance under the same recursive forecasting conditions used during the day-ahead evaluation rather than relying only on one-step validation results.

Only calibration and post-processing procedures that improved the rolling day-ahead validation score were retained. Once selected, the corresponding model and calibration parameters were fixed and applied without modification during testing. Testing observations were not used to accept, reject, or tune any model configuration or calibration procedure.

The post-processing equations in the following subsections were implemented as validation-only calibration rules for the developed forecasting pipeline. Their use was guided by chronological validation and time-series forecasting practice (Hota et al., 2017; Kahloot & Ekler, 2021; Zhou et al., 2020), while the exact grids, thresholds, clipping limits, and correction scales were study-specific and were selected only when they improved rolling day-ahead validation performance.

3.6.2. Shrinkage Factor and Bias Correction

The shrinkage factor controlled the magnitude of the predicted generation change applied to the most recent generation value. For each plant p , the shrinkage factor was selected from the following fixed validation grid:

$$s_p \in \{0.05, 0.10, 0.20, 0.35, 0.50, 0.75, 1.00, 1.15\}. \quad (3.12)$$

A value of $s_p < 1.00$ reduced the magnitude of the predicted residual and kept the reconstructed forecast closer to persistence. A value of $s_p > 1.00$ permitted a slightly stronger

generation change when supported by validation performance. The selected value was the candidate that produced the best validation score under the model-selection procedure.

After selecting s_p , the unbounded bias correction for plant p was computed from the median validation residual:

$$b_p^{\text{raw}} = \text{median}_{i \in \mathcal{V}} (y_{p,i} - \hat{y}_{p,i}^{\text{raw}}), \quad (3.13)$$

where $\mathcal{V}$ denotes the validation set, $y_{p,i}$ is the actual generation of plant p for validation observation i , and $\hat{y}_{p,i}^{\text{raw}}$ is the corresponding preliminary forecast obtained using Equation 3.7.

To prevent an excessively large adjustment, the maximum allowable absolute bias was defined using the plant-specific operational threshold:

$$\tau_p = \max(1, 0.01C_p), \quad (3.14)$$

where τ_p is expressed in megawatts. The final bias correction was then obtained as

$$b_p = \text{clip}(b_p^{\text{raw}}, -\tau_p, \tau_p). \quad (3.15)$$

The selected shrinkage factor s_p and bias correction b_p were derived exclusively from the validation set. Testing observations were not used to select or modify either parameter.

3.6.3. Bin Calibration and Hourly Residual Correction

Additional validation-based calibration was considered to correct systematic forecast errors associated with generation level and forecast hour.

For value-bin calibration, the validation observations were divided into quantile-based bins. For bin B_k , the median residual correction was calculated as

$$c_k = \text{median} \{y_i - \hat{y}_i \mid z_i \in B_k\}, \quad (3.16)$$

where c_k is the median residual correction for bin k , z_i is the selected binning variable, and B_k is the corresponding quantile interval. Depending on the plant-specific configuration, the binning variable represented either the current generation or the preliminary model forecast.

The bin-calibrated forecast was computed as

$$\hat{y}_i^{\text{bin}} = \text{clip}(\hat{y}_i + \alpha c_{k(i)}, 0, 1.05C_p), \quad (3.17)$$

where α is the validation-selected bin-correction scale and $k(i)$ identifies the bin containing observation i .

Hourly residual correction was used to address systematic time-of-day errors. For forecast hour h , the median validation residual was defined as

$$q_h = \text{median}\{y_i - \hat{y}_i \mid h_i = h\}, \quad (3.18)$$

where h_i is the target hour associated with validation observation i .

The hourly corrected forecast was calculated as

$$\hat{y}_i^{\text{hour}} = \text{clip}(\hat{y}_i + \beta q_{h_i}, 0, 1.05C_p), \quad (3.19)$$

where β is the validation-selected hourly correction scale.

The scale β was selected from

$$\beta \in \{0, 0.25, 0.50, 0.75, 1.00, 1.25\}. \quad (3.20)$$

Bin calibration and hourly residual correction were retained only when they improved the rolling day-ahead validation performance. Testing observations were not used to calculate the correction values or select their scale factors.

For value-bin calibration, the number of quantile bins was selected from

$$q \in \{3, 4, \dots, 20\}, \quad (3.21)$$

for most plants. For Agus 1, a wider candidate range from 3 to 40 quantile bins was considered to allow finer separation of generation levels. The bin-correction scale was selected from

$$\alpha_p \in \{0.50, 0.55, \dots, 2.50\}, \quad (3.22)$$

where α_p is the plant-specific scale applied to the median residual correction of the selected bin. Depending on the plant-specific configuration, the binning basis was either the current generation level or the preliminary model forecast. Each combination of bin count, binning basis, and correction scale was evaluated using the validation set only. The selected configuration was retained only when it improved the rolling day-ahead validation score.

Bin calibration and hourly residual correction were not automatically applied to every plant. Each correction was evaluated separately for each plant and was retained only when the validation results improved. The selected plant-specific calibration settings are reported in Appendix C.

Candidate calibration configurations were ranked using a validation score that combined operational MAPE, RMSE, and R^2 , with an additional penalty for negative R^2 . Testing data were not used to accept, reject, or modify any calibration configuration.

The model forecast was also evaluated for possible blending with the generation observed at the same target hour on the preceding day. This procedure was intended to preserve recurring daily generation patterns when the preceding-day profile provided useful forecasting information. The blended forecast was defined as

$$\hat{y}_{t+1}^{\text{blend}} = (1 - w)\hat{y}_{t+1} + wy_{t+1-24}, \quad (3.23)$$

where $\hat{y}_{t+1}$ is the model forecast before profile blending, y_{t+1-24} is the observed generation at the same target hour on the preceding day, and w is the profile-blending weight. The weight was selected exclusively using the rolling day-ahead validation results. A blending configuration was retained only when it improved the validation-based model-

selection score. Testing observations were not used to determine the blending weight.

3.7. Benchmark Models

Random Forest and XGBoost were implemented as benchmark models to evaluate the performance of the proposed RBFNN under comparable forecasting conditions. The benchmark models were not treated as alternative proposed frameworks; rather, they served as established nonlinear regression methods for performance comparison (Dudek, 2022; Zheng et al., 2020).

For each Agus plant, the RBFNN, Random Forest, and XGBoost models used the same cleaned hourly dataset, chronological training–validation–testing split, plant-specific target definition, operational MAPE formulation, and rolling 24-hour day-ahead evaluation procedure. All models predicted the next-hour generation change relative to the latest available plant generation value. The resulting residual predictions were reconstructed into generation-level forecasts before evaluation.

The benchmark models also received the same plant-specific forecasting feature families used by the RBFNN, including temporal indicators, lagged generation, rolling statistics, generation differences, unit-level generation shares, outage-related variables, lagged hydrologic context, gate and elevation variables, and upstream cascade features. Feature columns were aligned to the corresponding plant-specific model input structure before training and prediction. No model was permitted to use actual forecast-day generation, forecast-day rainfall, Lake Lanao outflow, or actual same-day outage status.

Model parameters were estimated using the training set, while hyperparameter selection and any accepted calibration settings were based only on validation performance. Testing data were reserved for final evaluation. The same recursive 24-hour day-ahead procedure was applied to all three models, in which predicted generation values were appended to the temporary forecasting history and used to reconstruct the lagged and rolling features for succeeding hours.

Therefore, the comparison was interpreted only within the implemented Agus dataset, feature construction, model-selection procedure, and rolling day-ahead evaluation.

The results were not intended to establish universal superiority of one forecasting algorithm over another.

Table 3.2. Common conditions used for RBFNN, Random Forest, and XGBoost benchmarking.

Comparison element	Common condition
Dataset and split	Same cleaned hourly dataset and chronological training, validation, and testing periods.
Forecast target	Same plant-specific residual target representing next-hour generation change.
Input information	Same plant-specific feature families and leakage-control restrictions.
Evaluation procedure	Same recursive rolling 24-hour day-ahead validation and testing procedure.
Metrics	Same operational MAPE, MAE, RMSE, and R^2 definitions.
Model selection	Training data used for fitting, validation data used for selection and calibration, and testing data reserved for final evaluation.

3.8. Evaluation Metrics

The forecasting performance of the proposed RBFNN and the benchmark models was evaluated using operational Mean Absolute Percentage Error (MAPE), Mean Absolute Error (MAE), Root Mean Square Error (RMSE), and the coefficient of determination (R^2), which are commonly used error and goodness-of-fit metrics in forecasting evaluation (Chicco et al., 2021; De Myttenaere et al., 2016; Hodson, 2022). In the following equations, $y_{p,i}$ denotes the actual generation of plant p for observation i , $\hat{y}_{p,i}$ denotes the corresponding

forecast generation, n_p is the total number of evaluated observations for plant p , and $\bar{y}_p$ is the mean actual generation of plant p .

3.8.1. Mean Absolute Error

The Mean Absolute Error measures the average absolute difference between the actual and forecast generation values (Chicco et al., 2021; Hodson, 2022):

$$\text{MAE}_p = \frac{1}{n_p} \sum_{i=1}^{n_p} |y_{p,i} - \hat{y}_{p,i}|, \quad (3.24)$$

where a lower MAE indicates that the forecasts were, on average, closer to the observed generation values.

3.8.2. Root Mean Square Error

The Root Mean Square Error assigns greater emphasis to larger forecast errors (Chicco et al., 2021; Hodson, 2022):

$$\text{RMSE}_p = \sqrt{\frac{1}{n_p} \sum_{i=1}^{n_p} (y_{p,i} - \hat{y}_{p,i})^2}, \quad (3.25)$$

Because forecast errors are squared before averaging, RMSE is more sensitive than MAE to relatively large hourly prediction errors.

3.8.3. Coefficient of Determination

The coefficient of determination measures how well the forecast represents the variability of the observed generation (Chicco et al., 2021):

$$R_p^2 = 1 - \frac{\sum_{i=1}^{n_p} (y_{p,i} - \hat{y}_{p,i})^2}{\sum_{i=1}^{n_p} (y_{p,i} - \bar{y}_p)^2}, \quad (3.26)$$

where

$$\bar{y}_p = \frac{1}{n_p} \sum_{i=1}^{n_p} y_{p,i}. \quad (3.27)$$

An R_p^2 value closer to 1 indicates stronger agreement between the forecast and the observed variation. A value near 0 indicates performance similar to predicting the mean observed generation, while a negative value indicates performance below that baseline.

3.8.4. Operational Mean Absolute Percentage Error

Percentage errors become unstable when actual generation is zero or close to zero. Therefore, operational MAPE was calculated only for observations that satisfied the plant-specific operational threshold τ_p , previously defined in Equation 3.14, consistent with known limitations of percentage-error metrics near zero-valued observations (Chicco et al., 2021; De Myttenaere et al., 2016).

For plant p , the set of included observations was defined as

$$\mathcal{J}_p = \{i : |y_{p,i}| \geq \tau_p\}, \quad (3.28)$$

where $\mathcal{J}_p$ contains only the observations for which the absolute actual generation was greater than or equal to the plant-specific threshold. The number of included observations was

$$n_p^{\text{op}} = |\mathcal{J}_p|. \quad (3.29)$$

Operational MAPE was then calculated as

$$\text{MAPE}_p^{\text{op}} = \frac{100}{n_p^{\text{op}}} \sum_{i \in \mathcal{J}_p} \left| \frac{y_{p,i} - \hat{y}_{p,i}}{y_{p,i}} \right|. \quad (3.30)$$

MAE, RMSE, and R_p^2 were calculated using all available evaluation observations. Only operational MAPE excluded observations below the plant-specific threshold. This prevented near-zero generation values from producing unstable or disproportionately large percentage errors.

3.9. Forecasting Procedure

After the RBFNN models were trained and validated, they were used to generate day-ahead forecasts for the Agus 1–7 cascade. For a given forecast day ($D+1$), all required input variables, including recent hourly generation values, reservoir conditions, hydraulic variables, and operational indicators up to the end of day D , were prepared using the same feature engineering pipeline applied during training.

These input features were standardized using the scaling parameters derived from the training stage to ensure consistency in model input representation. The trained plant-specific RBFNN models generated a complete 24-hour day-ahead forecast for each validation and testing day. The forecast origin was set at 23:00 of the day preceding the target day; therefore, only information available before 00:00 of the forecast day was permitted as model input.

The 24 hourly predictions were generated recursively rather than as independent one-step forecasts. After the forecast for hour $t + 1$ was produced, the predicted plant-generation value was appended to a temporary forecasting history. This updated history was then used to reconstruct the lagged generation values, generation differences, rolling statistics, unit-generation shares, and upstream cascade features required for forecasting hour $t + 2$. The same procedure was repeated until all 24 target hours had been predicted.

Observed generation values from the forecast day were not inserted into the temporary history. Actual forecast-day generation was accessed only after the complete forecast had been generated and was used solely to calculate the evaluation metrics. Similarly, actual forecast-day rainfall, Lake Lanao outflow, hydrologic measurements, and actual unit-outage status were excluded from the forecasting inputs.

For historical validation and testing, archived planned-outage schedules were not available. Consequently, the latest unit-status information known before each forecast day was persisted throughout the corresponding 24-hour backtest. For the operational day-ahead demonstration, the model used the planned-outage schedule prepared before the start of the forecast horizon.

To ensure that the day-ahead evaluation remained free from data leakage, the study applied specific control rules for training, validation, calibration, testing, forecast-day inputs, and recursive feature construction, as summarized in Table 3.3.

Table 3.3. Summary of leakage-control rules used in the day-ahead evaluation.

Component	Leakage-control rule
Training and scaling	Model parameters and feature scalers were fitted using the chronological training set only.
Validation and calibration	Hyperparameters, shrinkage, bias correction, and other calibration settings were selected using validation data only.
Testing data	Testing observations were reserved for final performance evaluation and were not used for training, model selection, or calibration.
Forecast-day inputs	Actual forecast-day generation, rainfall, Lake Lanao outflow, and actual outage status were not used as forecasting inputs. Planned or latest-known pre-forecast outage information was used instead.
Recursive forecasting	Lagged and rolling features within the 24-hour horizon were reconstructed using historical observations available before the forecast day and recursively generated predictions.

3.9.1. Capacity and Outage-Aware Post-Processing

After plant-level forecasting, deterministic post-processing was applied to ensure that the outputs remained consistent with physical plant capacities and planned unit availability, reflecting the operational constraints of cascaded hydropower generation (Bernardes

Jr et al., 2022; Polestico, 2019). The plant-level forecast was first constrained to a feasible generation range:

$$\hat{y}_{p,t}^{\text{clip}} = \text{clip}(\hat{y}_{p,t}, 0, 1.05C_p), \quad (3.31)$$

where $\hat{y}_{p,t}$ is the forecast generation of plant p at hour t , $\hat{y}_{p,t}^{\text{clip}}$ is the capacity-clipped forecast, and C_p is the installed capacity of plant p .

For outage-aware adjustment, the baseline available capacity was determined from the latest unit-status information available before the forecast day:

$$C_{p,t}^{\text{base}} = \sum_{u=1}^{U_p} a_{p,u}^{\text{base}} C_{p,u}, \quad (3.32)$$

where U_p is the number of generating units in plant p , $a_{p,u}^{\text{base}}$ is the latest known pre-forecast availability indicator of unit u , and $C_{p,u}$ is the installed capacity of that unit.

The available capacity under the planned outage schedule was calculated as

$$C_{p,t}^{\text{plan}} = \sum_{u=1}^{U_p} a_{p,u,t}^{\text{plan}} C_{p,u}, \quad (3.33)$$

where $a_{p,u,t}^{\text{plan}}$ is equal to 1 when unit u is planned to be available at hour t , and 0 when it is scheduled to be unavailable.

The available-capacity ratio was computed from the planned and baseline available capacities. To avoid division by zero when no unit was available under the baseline condition, the ratio was defined piecewise as

$$r_{p,t}^{\text{cap}} = \begin{cases} \frac{C_{p,t}^{\text{plan}}}{C_{p,t}^{\text{base}}}, & C_{p,t}^{\text{base}} > 0, \\ 0, & C_{p,t}^{\text{base}} = 0, \end{cases} \quad (3.34)$$

where $r_{p,t}^{\text{cap}}$ is the available-capacity ratio for plant p at hour t .

The outage-adjusted plant forecast was then calculated as

$$\hat{y}_{p,t}^{\text{out}} = \text{clip}\left(r_{p,t}^{\text{cap}} \hat{y}_{p,t}^{\text{clip}}, 0, C_{p,t}^{\text{plan}}\right), \quad (3.35)$$

where $\hat{y}_{p,t}^{\text{clip}}$ is the capacity-clipped plant forecast. When $C_{p,t}^{\text{base}} = 0$, the ratio $r_{p,t}^{\text{cap}}$ and the resulting outage-adjusted forecast were both set to zero. Similarly, when no generating unit was planned to be available, $C_{p,t}^{\text{plan}} = 0$, the plant-level forecast was constrained to 0 MW.

After the outage adjustment, the plant-level forecast was distributed only among the available units. Units scheduled for outage were assigned zero generation:

$$\hat{y}_{p,u,t} = 0, \quad \text{if } a_{p,u,t}^{\text{plan}} = 0. \quad (3.36)$$

For available units, the allocated forecast was constrained by the unit capacity:

$$0 \leq \hat{y}_{p,u,t} \leq C_{p,u}. \quad (3.37)$$

The plant-level forecast was allocated among the available units using a weighted combination of the current pre-forecast unit share, the recent median unit share, and the historical median share for the same forecast hour. When these shares were unavailable, allocation was based on the relative installed capacities of the available units. Units scheduled for outage were assigned zero generation, and the forecast allocated to each available unit was constrained by its installed capacity.

The final plant-level forecast was recomputed as

$$\hat{y}_{p,t}^{\text{plant}} = \sum_{u=1}^{U_p} \hat{y}_{p,u,t}, \quad (3.38)$$

where U_p is the number of generating units in plant p .

The total Agus cascade forecast was then obtained by summing the final plant-level forecasts:

$$\hat{y}_t^{\text{cascade}} = \sum_{p \in \mathcal{P}} \hat{y}_{p,t}^{\text{plant}}, \quad (3.39)$$

where $\mathcal{P} = \{1, 2, 4, 5, 6, 7\}$ represents the six Agus plants included in the study.

During validation and testing, actual outage information from the forecast day was not used. Because archived planned-outage schedules were unavailable for the historical evaluation period, the latest unit-status information known before each forecast day was

persisted throughout the corresponding 24-hour backtest. For the operational day-ahead forecast demonstration, unit availability indicators were obtained from the planned-outage schedule prepared before the forecast horizon. The capacity clipping, outage adjustment, unit allocation, and cascade aggregation rules were deterministic and were not tuned using testing observations.

CHAPTER 4

RESULTS AND DISCUSSION

4.1. Overview of Results

This chapter presents the results of the proposed Radial Basis Function Neural Network (RBFNN) forecasting model applied to the Agus 1–7 hydropower cascade. The performance of the model is evaluated using multiple statistical metrics and compared with benchmark models, including Random Forest and XGBoost.

The results are organized to reflect the overall modeling pipeline, beginning with model training behavior, followed by forecasting performance, benchmark comparison, and detailed error analysis. In addition, the operational behavior of the model under cascade interactions and planned outages is examined to assess its practical applicability.

The findings of this study are discussed in relation to the objectives outlined in Chapter 1, highlighting the effectiveness of the proposed approach in improving short-term hydropower generation forecasting.

4.2. Data Preparation Results

4.2.1. Dataset Overview

The cleaned and processed dataset was derived from historical operational records of the Agus hydropower cascade. All variables were standardized into a uniform hourly time resolution to ensure consistency across inputs.

Table 4.1. Summary of Processed Dataset

Description	Value
Time Range	01/01/2024 – 06/30/2025
Total Rows	13131
Number of Features	53
Resolution	Hourly

Table 4.2. Input Variables Used in the Forecasting Model

Input Variable Group	Description
Plant Generation	Hourly unit and total generation per plant.
Reservoir Variables	Forebay elevation.
Hydraulic Variables	Lake Lanao outflow.
Gate Operation Variables	Total spillway gate openings.
Outage Indicators	Unit availability (1 = on, 0 = off).

4.2.2. Cleaned Data Visualization

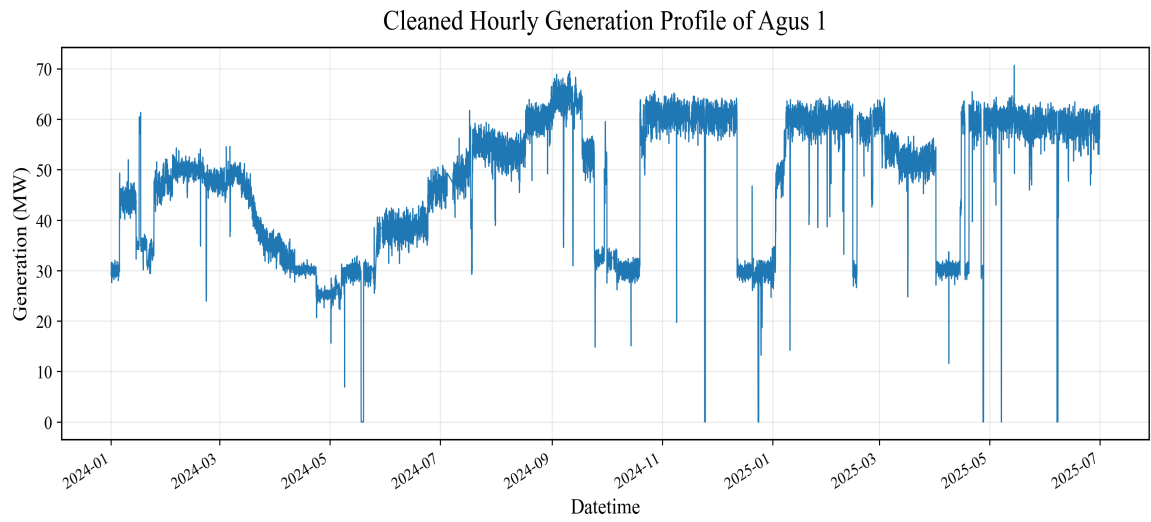


Figure 4.1. Cleaned hourly generation profile of Agus 1.

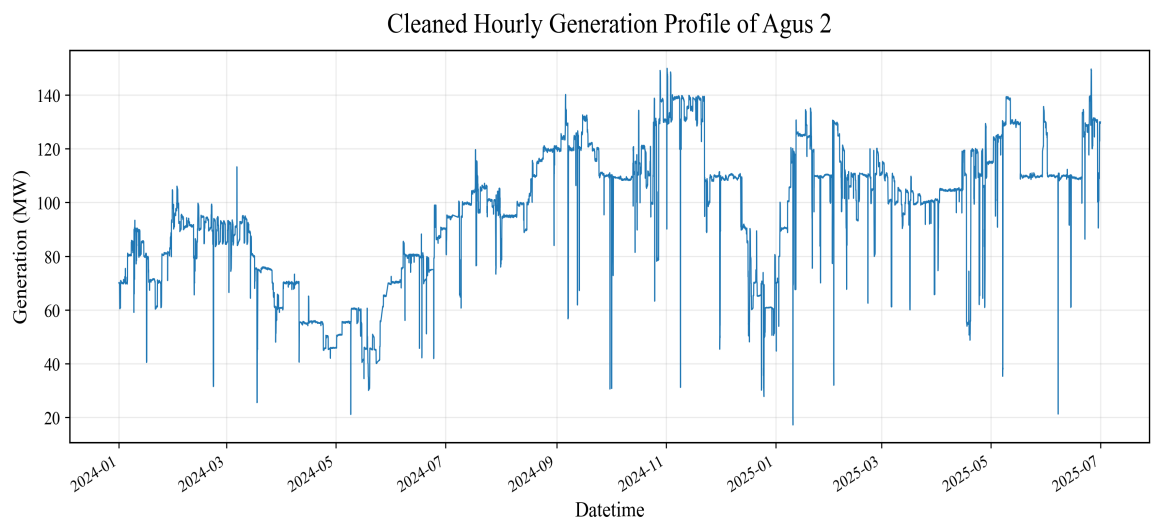


Figure 4.2. Cleaned hourly generation profile of Agus 2.

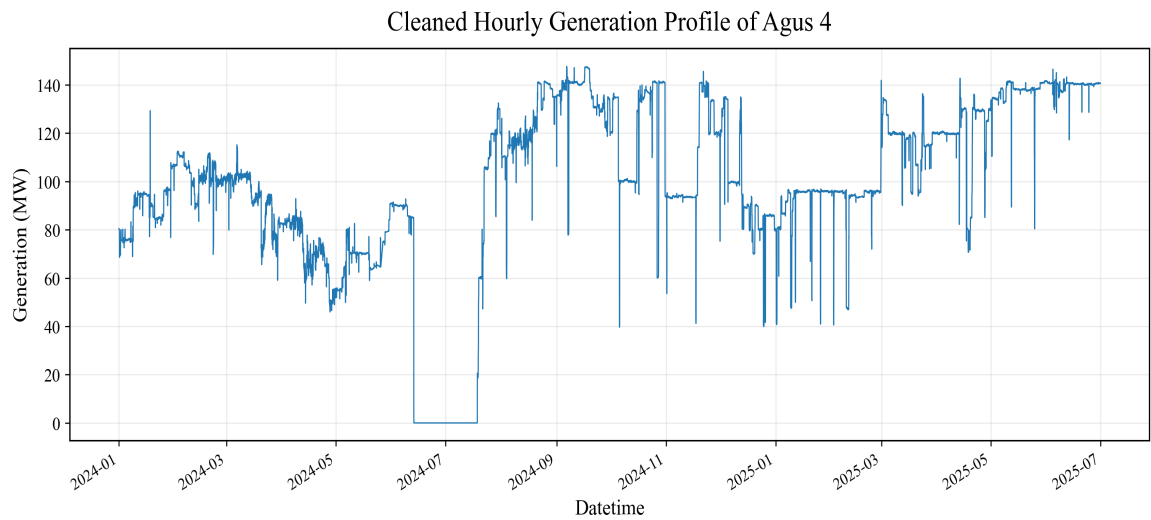


Figure 4.3. Cleaned hourly generation profile of Agus 4.

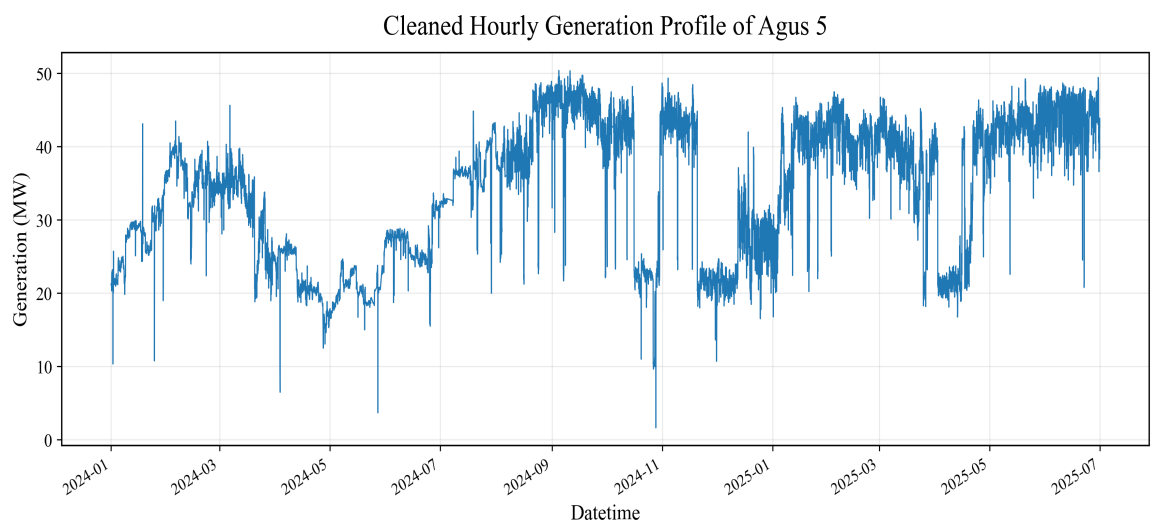


Figure 4.4. Cleaned hourly generation profile of Agus 5.

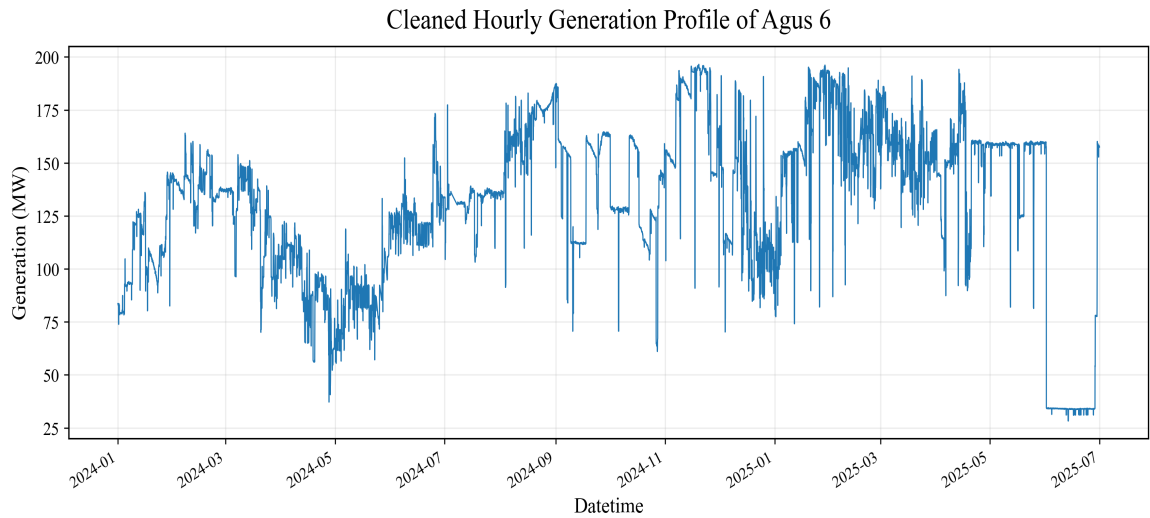


Figure 4.5. Cleaned hourly generation profile of Agus 6.

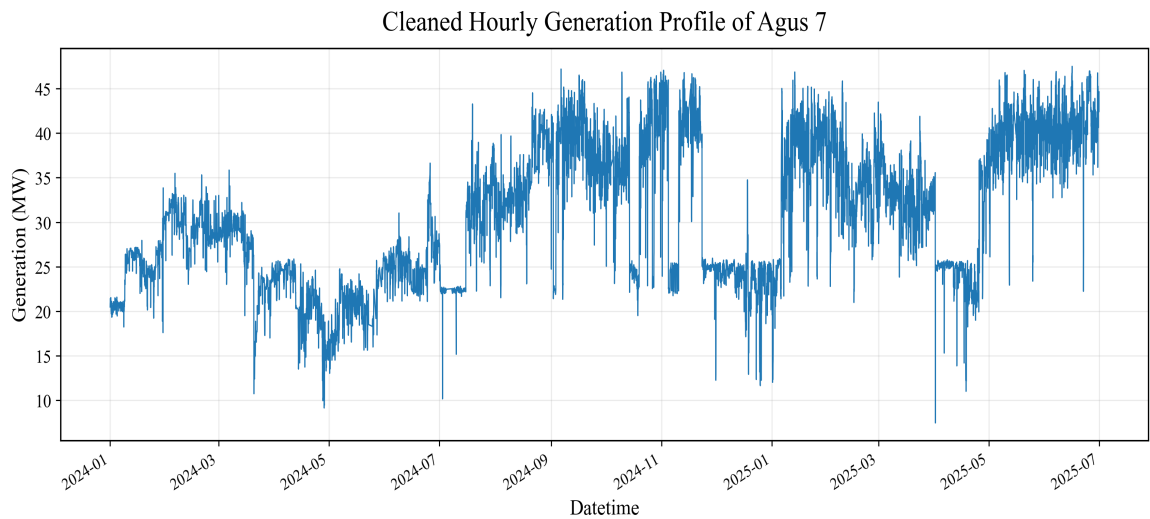


Figure 4.6. Cleaned hourly generation profile of Agus 7.

4.2.3. Discussion of Cleaned Data

The cleaned hourly generation profiles presented above illustrates the operational behavior of each plant in the Agus hydropower cascade after preprocessing. The results confirm that the applied data cleaning procedures successfully produced continuous and consistent time series suitable for modeling.

Across all plants, the generation profiles exhibit expected temporal variability driven by operational scheduling, hydraulic conditions, and demand patterns. Upstream plants such as Agus 1 and Agus 2 show relatively smoother generation trends, while downstream

plants, particularly Agus 5, Agus 6, and Agus 7, display higher variability. This behavior is consistent with the cascading nature of the system, where downstream generation is influenced by upstream discharge and operational adjustments.

Notable fluctuations and sudden drops in generation are observed in several plants. These are attributed to operational events such as unit outages, maintenance schedules, or abrupt changes in water release. The presence of such variations indicates that the dataset captures realistic plant behavior rather than being overly smoothed, which is essential for training robust forecasting models.

Additionally, the absence of prolonged missing segments and the continuity of the time series confirm that the interpolation and data repair techniques were effective. Minor spikes and short-term irregularities remain in the data, reflecting real-world measurement noise and operational dynamics, which are important for improving model generalization.

Overall, the cleaned dataset preserves the inherent temporal patterns and interdependencies of the Agus cascade while ensuring data quality and consistency. This provides a reliable foundation for subsequent feature engineering and model training stages.

4.3. RBFNN Training and Validation Results

4.3.1. Model Training and Validation Results

This section presents the training and validation performance of the proposed Radial Basis Function Neural Network (RBFNN) models applied to the Agus hydropower cascade. Separate models were developed for each plant, namely Agus 1, Agus 2, Agus 4, Agus 5, Agus 6, and Agus 7, to capture plant-specific operational characteristics.

The evaluation focuses on two key aspects: (1) the selected model configuration and calibration parameters, and (2) validation performance based on standard error metrics. These results provide insight into the capability of the RBFNN models to learn the nonlinear relationships present in the hydropower system and generalize to unseen data.

The optimized hyperparameters and calibration settings used for each plant are summarized in Table 4.3. These parameters were selected to balance model accuracy and stability across the different plants in the cascade.

The RBFNN configuration in Table 4.3 shows that each Agus plant required a plant-specific model structure and calibration setting. This is expected because the plants differ in installed capacity, number of generating units, operating range, upstream influence, and generation variability. Agus 6 used the largest number of input features because it has the highest installed capacity among the modeled plants and a more complex unit structure. In contrast, Agus 1, Agus 5, and Agus 7 used fewer features because their unit configurations and operating ranges are smaller.

The selected number of RBF centers, learning rate, shrinkage factor, and calibration settings were determined using the validation set only. The testing set was not used in selecting the model configuration, shrinkage factor, bias correction, bin calibration, hourly residual correction, or profile blending. This separation was used to preserve the independence of the final testing evaluation. The use of residual or delta-based forecasting also helped anchor the model to the most recent known generation level, while the shrinkage and calibration steps reduced excessive forecast drift during recursive day-ahead prediction.

Table 4.3. RBFNN Model Configuration Parameters per Agus Plant

Plant	Features	MAPE Thr.	Centers	LR	Gamma	Batch	Pat.	Shrinkage
Agus 1	203	1.000	180	0.00030	0.50	16	16	1.00
Agus 2	234	1.800	120	0.00070	0.50	16	16	1.15
Agus 4	234	1.581	180	0.00030	0.50	16	16	1.15
Agus 5	213	1.000	80	0.00050	0.50	16	16	0.10
Agus 6	276	2.190	60	0.00050	0.50	16	16	1.15
Agus 7	213	1.000	120	0.00070	0.50	16	16	1.00

Table 4.4 presents the validation performance of the RBFNN models for each Agus plant. The models were evaluated using Mean Absolute Percentage Error (MAPE), Mean Absolute Error (MAE), Root Mean Square Error (RMSE), and the coefficient of determination (R^2).

The validation results in Table 4.4 were obtained under the strict rolling 24-hour day-ahead evaluation. The overall average validation operational MAPE was 6.1329%, with an average MAE of 4.3216 MW, RMSE of 8.4214 MW, and R^2 of 0.4650. Among the plants, Agus 4 obtained the lowest validation MAPE at 3.5411%, followed by Agus 2

at 4.2965% and Agus 1 at 5.9768%. These results show that the model was able to capture meaningful generation patterns in the validation period, especially for plants with more stable operating behavior.

However, the validation performance also shows that forecasting difficulty was not uniform across the cascade. Agus 6 produced a validation R^2 of 0.0759 despite having an operational MAPE of 7.2936%. This means that the percentage error remained moderate, but the model had difficulty explaining the full variation of Agus 6 during the validation period. This behavior is reasonable because Agus 6 has larger capacity, more units, stronger ramping behavior, and stronger dependence on upstream releases and operational dispatch decisions. Therefore, the validation results should be interpreted using all metrics together rather than relying only on R^2 .

Table 4.4. RBFNN Validation Performance per Agus Plant

Plant	Operational MAPE (%)	MAE (MW)	RMSE (MW)	R^2
Agus 1	5.9768	2.7092	5.2144	0.7172
Agus 2	4.2965	3.9337	8.8917	0.2962
Agus 4	3.5411	2.9848	9.6373	0.6073
Agus 5	8.0941	2.6787	4.2826	0.6014
Agus 6	7.2936	11.1517	18.9782	0.0759
Agus 7	7.5951	2.4717	3.5241	0.4919

To provide an overall assessment of model performance, the average validation metrics across all plants are summarized in Table 4.5. These aggregated results represent the general forecasting capability of the RBFNN model for the Agus cascade.

The validation results confirm that the forecasting task becomes more demanding when evaluated under a full day-ahead forecasting procedure. In the rolling 24-hour validation setup, each forecast day was predicted using only information available before the forecast day, and the model recursively used its previous hourly forecasts to predict the succeeding hours. Because actual same-day generation, rainfall, Lake Lanao outflow, hydrologic conditions, and outage status were not used as inputs, the validation performance reflects a more realistic forecasting condition for day-ahead generation planning.

The relatively low validation MAPE values suggest that the RBFNN was able to maintain useful forecasting accuracy under this stricter condition. At the same time, the moderate average R^2 indicates that some validation periods contained operating patterns that were more difficult to explain, such as dispatch changes, outage-related generation shifts, or hydrologic conditions not fully known before the forecast day. Thus, the validation results support the use of the RBFNN as the main forecasting model, while also showing that the model should be interpreted as a decision-support tool rather than a replacement for operator judgment.

Table 4.5. Overall Average Validation Performance of the RBFNN Model

Plant	MAPE (%)	MAE (MW)	RMSE (MW)	R^2
Agus Cascade	6.1329	4.3216	8.4214	0.4650

4.4. Forecasting Performance of the RBFNN Model

The forecasting performance of the RBFNN model was evaluated using the testing dataset. The model generated day-ahead hourly predictions for each plant in the Agus cascade, as well as the total cascade generation.

The results indicate that the RBFNN generally followed the observed generation patterns and represented both short-term fluctuations and broader generation trends during the historical testing period. The residual forecasting formulation may have contributed to forecast stability by anchoring each prediction to the most recently available generation level.

Tables 4.6 and 4.7 summarize the final testing performance of the RBFNN model under the rolling 24-hour day-ahead backtest. The model obtained an average testing operational MAPE of 5.7949%, MAE of 3.4687 MW, RMSE of 8.5306 MW, and R^2 of 0.6705. These results indicate that the RBFNN retained useful predictive performance on unseen testing data while following the same leakage-free day-ahead forecasting procedure used during validation.

At the plant level, Agus 4 achieved the lowest testing MAPE at 2.2580%, followed by Agus 2 at 5.5062%, Agus 1 at 5.9486%, Agus 6 at 6.9420%, Agus 5 at 7.0145%, and Agus 7 at 7.1001%. Although Agus 5 and Agus 7 had the highest testing MAPE values among the six plants, their errors remained within a moderate range for offline day-ahead forecasting. The testing R^2 values also improved for several plants compared with validation, especially for Agus 6, which obtained a testing R^2 of 0.9189. This suggests that the testing period contained generation patterns that were more consistent with the learned relationship between historical generation, hydrologic context, unit availability, and upstream cascade behavior.

The results should not be interpreted as proof of full operational readiness because the evaluation was still conducted offline using historical data. Nevertheless, the testing results show that the RBFNN can provide informative day-ahead generation estimates and may support capacity nomination review and short-term planning when combined with operator assessment and further validation under actual plant operation.

Table 4.6. Overall Average Testing Performance of the RBFNN Model

Plant	MAPE (%)	MAE (MW)	RMSE (MW)	R^2
Agus Cascade	5.7949	3.4687	8.5306	0.6705

Table 4.7. RBFNN Testing Performance per Agus Plant

Plant	Operational MAPE (%)	MAE (MW)	RMSE (MW)	R^2
Agus 1	5.9486	3.2757	8.0485	0.4751
Agus 2	5.5062	4.6206	11.2335	0.5164
Agus 4	2.2580	2.4129	7.9539	0.6352
Agus 5	7.0145	2.5619	4.0173	0.6890
Agus 6	6.9420	5.4706	16.5399	0.9189
Agus 7	7.1001	2.4703	3.3902	0.7882

4.4.1. Comparison of Actual and Forecast Generation

The quantitative evaluation presented in the previous subsection is further supported through visual analysis of actual and forecasted generation profiles.

This subsection presents the comparison between the actual and forecasted generation values of the RBFNN model for each Agus hydropower plant. The analysis is based on the testing dataset, which consists of unseen chronological data not used during model training and validation.

The purpose of this comparison is to visually evaluate the ability of the model to capture the temporal behavior of hydropower generation, including both short-term variations and overall operational trends.

Figures 4.7 to 4.12 illustrate the comparison between actual and forecasted generation of the RBFNN model for all Agus plants during the testing period.

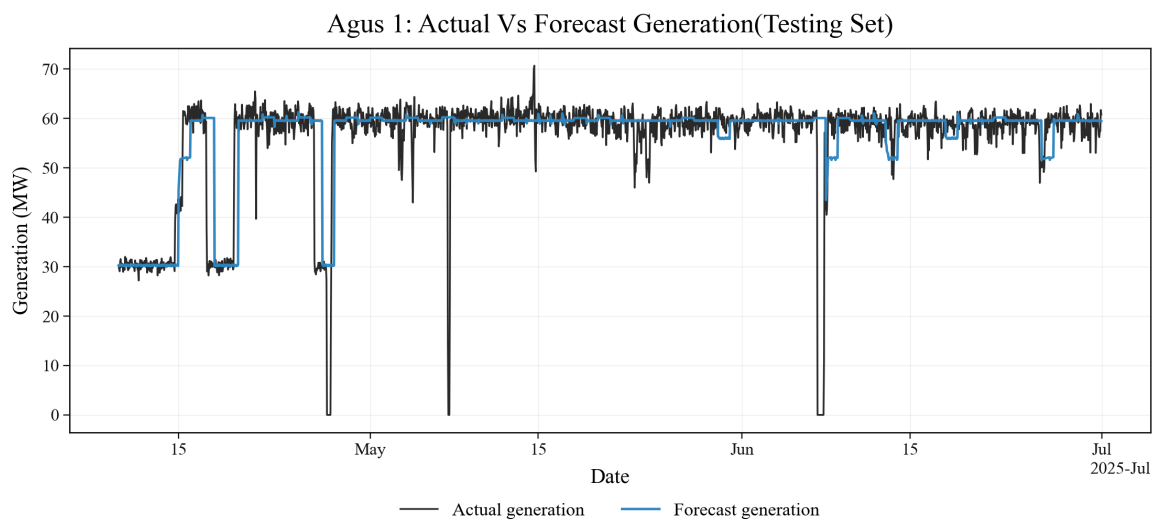


Figure 4.7. Actual and forecast generation of Agus 1 during the testing period.

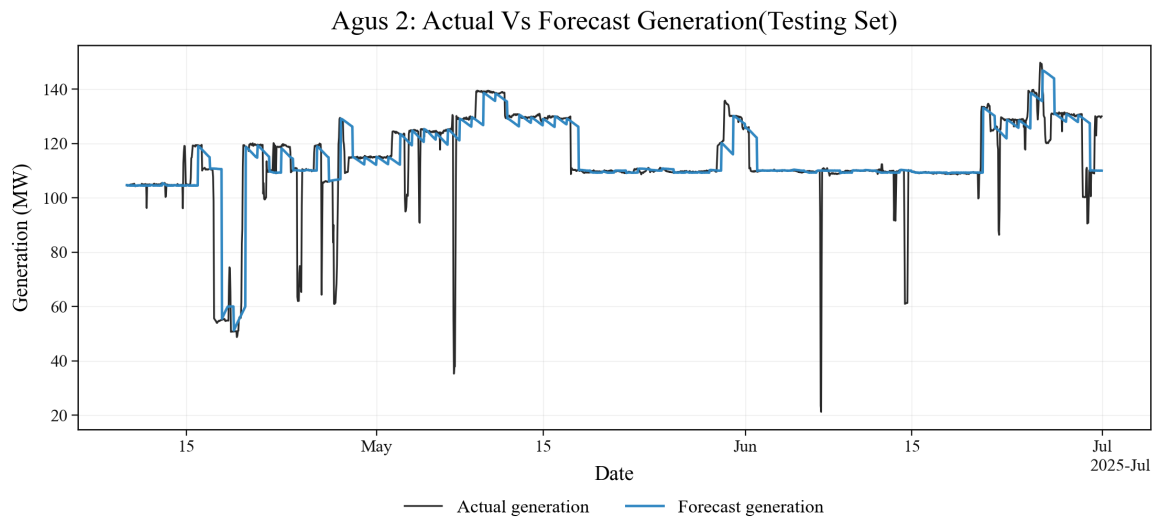


Figure 4.8. Actual and forecast generation of Agus 2 during the testing period.

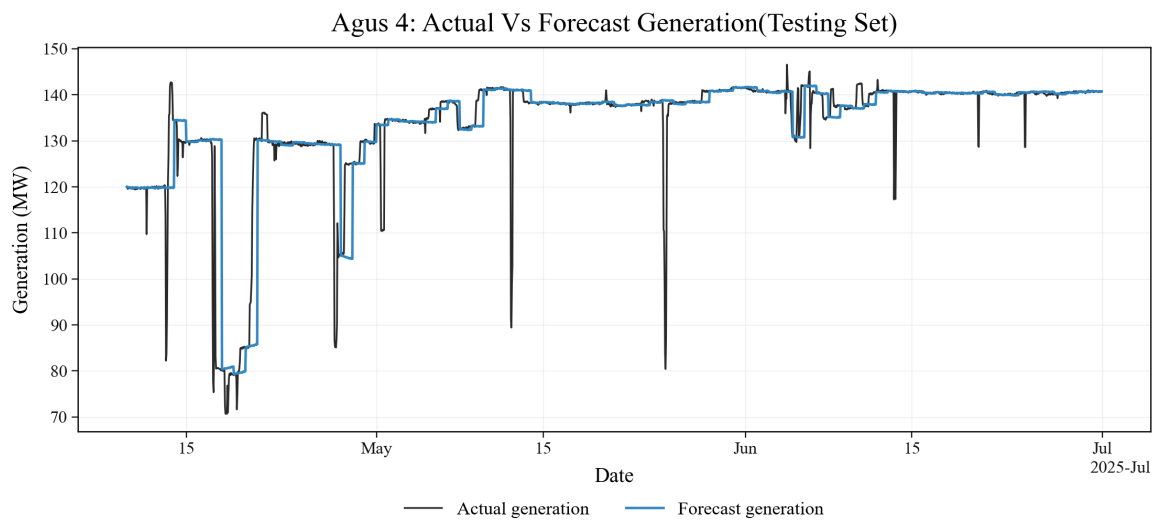


Figure 4.9. Actual and forecast generation of Agus 4 during the testing period.

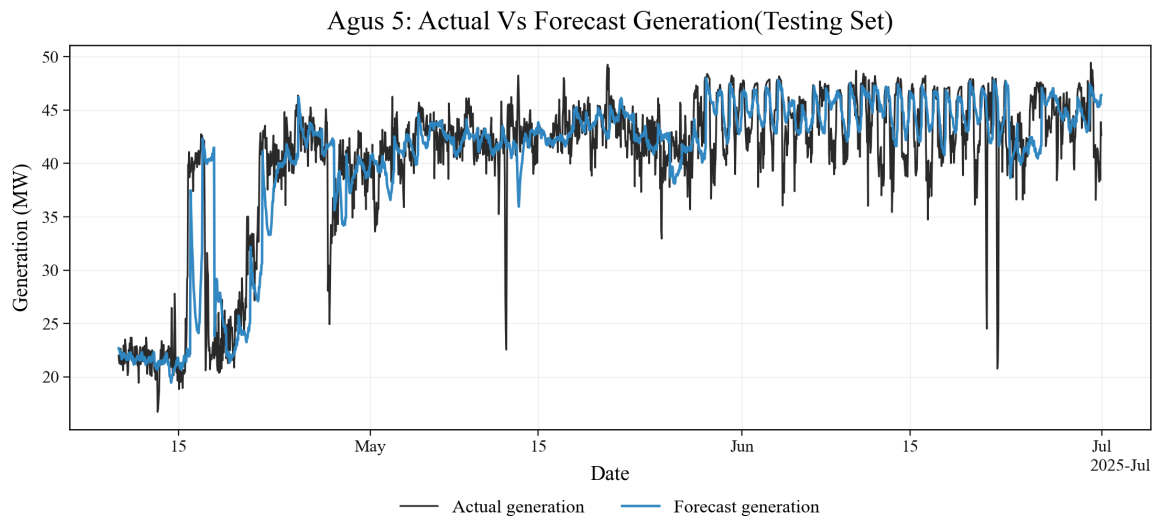


Figure 4.10. Actual and forecast generation of Agus 5 during the testing period.

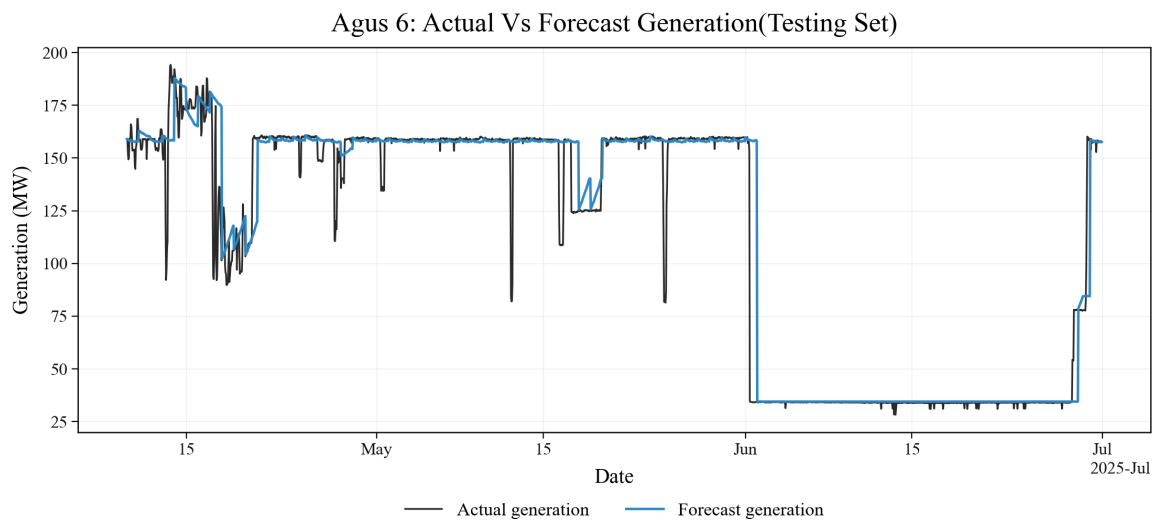


Figure 4.11. Actual and forecast generation of Agus 6 during the testing period.

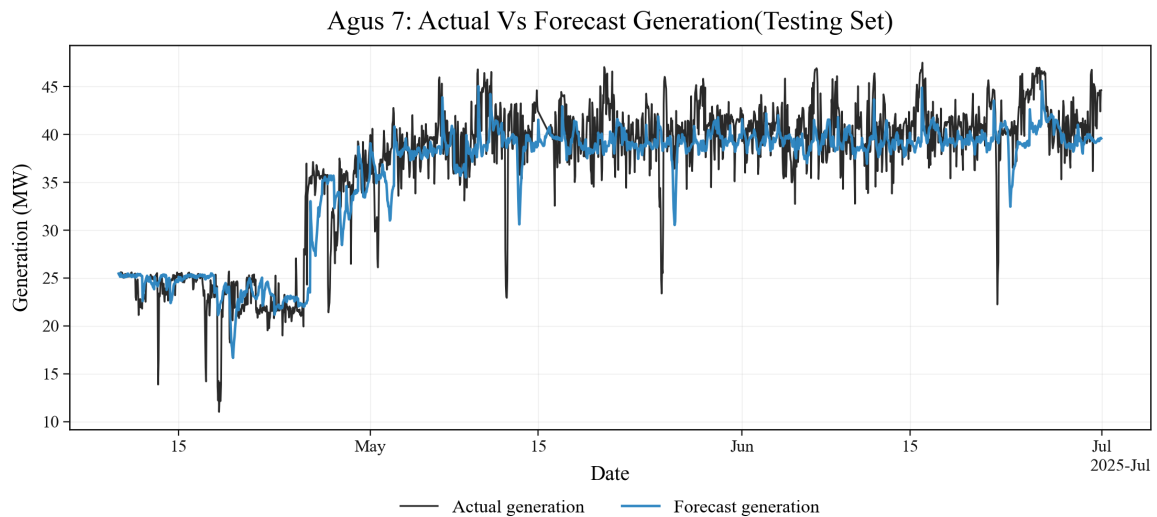


Figure 4.12. Actual and forecast generation of Agus 7 during the testing period.

Figures 4.7 to 4.12 compare the actual and forecasted generation of each Agus plant during the testing period. The plots show that the RBFNN generally followed the main level and direction of the hourly generation profiles. Close overlap between the actual and forecast curves indicates that the model captured the timing and magnitude of common operating patterns. Larger vertical gaps represent hourly MW errors, which are expected during sudden dispatch changes, unit availability changes, ramping events, or hydrologic conditions that were not fully available before the forecast day.

For Agus 1, Agus 2, and Agus 4, the forecast curves generally followed the actual generation levels with moderate deviations. Agus 4 showed the strongest testing performance, which is consistent with its low operational MAPE and relatively strong R^2 . For Agus 5 and Agus 7, the plots show that the model still captured the broad generation trend, although the forecast occasionally deviated from actual values during periods of sharper operating changes. For Agus 6, the model achieved a strong testing R^2 , but the RMSE remained higher in MW terms because Agus 6 has the largest installed capacity and larger possible generation swings.

Overall, the actual-versus-forecast plots support the numerical results by showing that the RBFNN forecasts were not merely close in average error, but also generally consistent with the observed temporal behavior of the plants. The remaining forecast deviations highlight the operational uncertainty of the cascade and the influence of same-day decisions

that cannot be fully known during day-ahead forecasting.

4.4.2. Testing Error Analysis

To further evaluate the performance of the RBFNN model, additional analyses were conducted using scatter plots, residual distributions, and plant-level error comparisons.

Figure 4.13 shows the relationship between actual and predicted generation values for all plants during the testing period. Figure 4.14 presents the distribution of forecast residuals, while Figure 4.15 compares the testing MAPE across the six Agus plants.

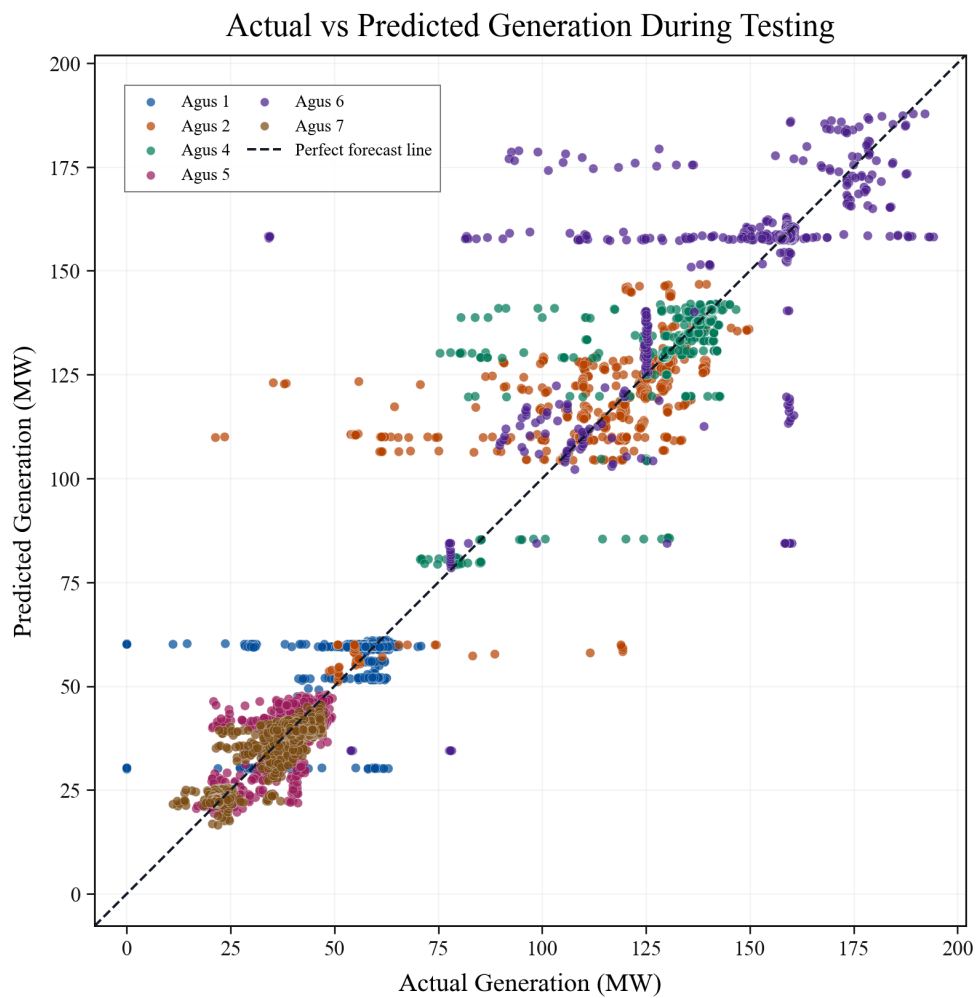


Figure 4.13. Scatter plot of actual and predicted generation values during testing.

The scatter plot of actual and predicted generation values shows how closely the forecasts align with the ideal one-to-one relationship. Points near the diagonal line indicate accurate forecasts, while wider dispersion indicates larger forecast errors. The overall dis-

tribution suggests that the RBFNN captured the main generation range of the Agus plants, although some deviations remained during high-ramp or irregular operating periods.

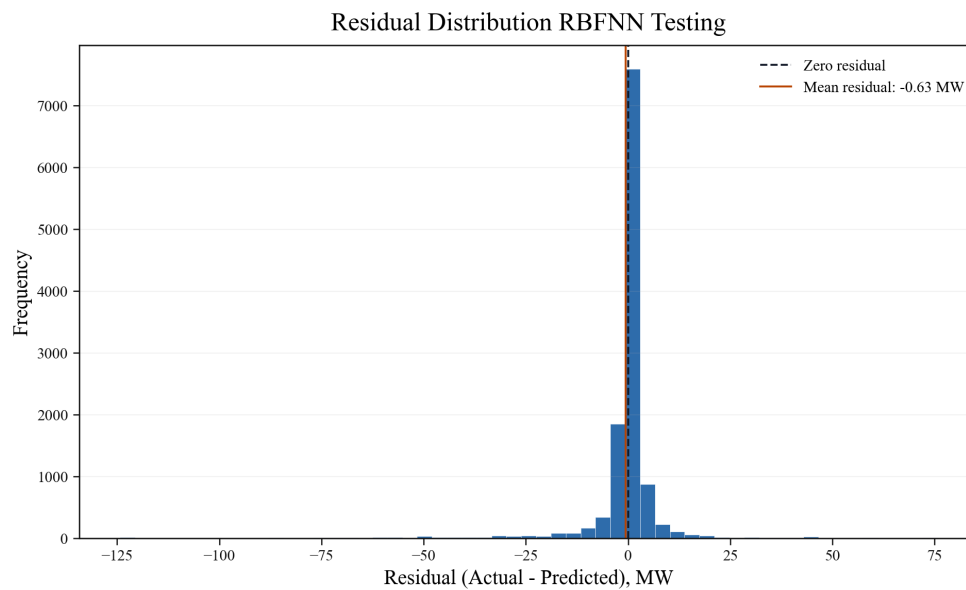


Figure 4.14. Residual distribution of RBFNN forecast errors during testing.

The residual distribution further shows whether the model produced systematic overforecasting or underforecasting. A residual pattern centered near zero indicates that the model did not consistently overestimate or underestimate generation across the testing set. Wider residual spread is expected for plants with larger capacity or more variable operation, especially Agus 6, where a small percentage error can still correspond to a larger MW error.

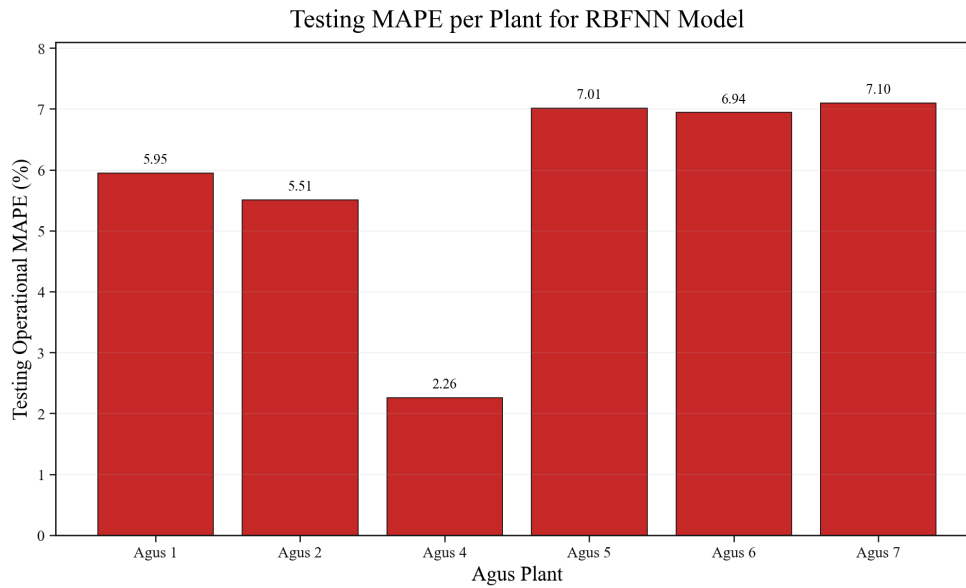


Figure 4.15. Testing MAPE per Agus plant using the RBFNN model.

The plant-level testing MAPE comparison shows that all plants remained within a moderate operational error range under the strict day-ahead backtest. Agus 4 produced the lowest percentage error, while Agus 5 and Agus 7 had relatively higher MAPE values. These differences reflect the plant-specific nature of the cascade, where each plant has different hydraulic conditions, unit configurations, and sensitivity to upstream operation. Therefore, the error analysis supports the conclusion that the RBFNN is useful for day-ahead forecasting, but its forecasts should still be interpreted together with operating conditions and planned outage information.

4.5. Benchmark Model Comparison

This section presents the testing-performance comparison of the RBFNN, Random Forest, and XGBoost models under the common evaluation procedure described in Section 3.7. The comparison focuses on plant-level and overall differences in operational MAPE, MAE, RMSE, and R^2 .

The results were interpreted within the present Agus dataset and implemented day-ahead forecasting setup. Accordingly, the comparison indicates the relative performance of the three models in this study and does not establish universal superiority of one forecasting

method over all hydropower forecasting applications.

4.5.1. Quantitative Performance Comparison

The evaluation focuses on standard forecasting metrics, including Mean Absolute Percentage Error (MAPE), Mean Absolute Error (MAE), Root Mean Square Error (RMSE), and the coefficient of determination (R^2). These metrics provide a comprehensive assessment of both accuracy and variability in model predictions.

Tables 4.8, 4.9, and 4.10 summarize the quantitative performance of the benchmark models and the proposed RBFNN model.

Table 4.8. Random Forest Testing Performance per Agus Plant

Plant	MAPE (%)	MAE (MW)	RMSE (MW)	R^2
Agus 1	5.8155	3.1784	8.0350	0.4769
Agus 2	8.2026	7.7747	13.0351	0.3488
Agus 4	5.5206	6.8281	9.5912	0.4695
Agus 5	6.8928	2.6052	3.8266	0.7178
Agus 6	8.2262	7.6897	20.0458	0.8808
Agus 7	6.7785	2.3266	3.2914	0.8003

Table 4.9. XGBoost Testing Performance per Agus Plant

Plant	MAPE (%)	MAE (MW)	RMSE (MW)	R^2
Agus 1	6.5548	3.5113	8.1891	0.4566
Agus 2	5.3654	4.4356	11.5625	0.4877
Agus 4	5.0376	6.1286	9.3057	0.5007
Agus 5	7.5074	2.8447	4.1027	0.6756
Agus 6	8.2262	7.6897	20.0458	0.8808
Agus 7	7.8623	2.7190	3.8439	0.7277

Table 4.10 shows that the RBFNN obtained the best overall testing performance among the three models. The RBFNN achieved an average operational MAPE of 5.7949%, MAE of 3.4687 MW, RMSE of 8.5306 MW, and R^2 of 0.6705. Random Forest obtained an average operational MAPE of 6.9060%, MAE of 5.0671 MW, RMSE of 9.6375 MW,

and R^2 of 0.6157. XGBoost obtained an average operational MAPE of 6.7590%, MAE of 4.5548 MW, RMSE of 9.5083 MW, and R^2 of 0.6215.

Table 4.10. Overall Testing Performance Comparison of Forecasting Models

Model	Operational MAPE (%)	MAE (MW)	RMSE (MW)	R^2
RBFNN	5.7949	3.4687	8.5306	0.6705
Random Forest	6.9060	5.0671	9.6375	0.6157
XGBoost	6.7590	4.5548	9.5083	0.6215

These results indicate that the RBFNN produced lower average percentage error and lower MW-based errors than both benchmark models. The advantage of the RBFNN is consistent with its residual learning formulation, localized radial basis representation, and validation-only calibration steps. Because all three models used the same residual-generation target and recursive 24-hour evaluation procedure, the observed performance differences are more appropriately attributed to their model structures, learned parameters, and validation-selected calibration procedures.

At the plant level, the comparison shows that no model performed equally well for every plant. Random Forest and XGBoost produced competitive results in some cases, but the RBFNN had the best overall balance of operational MAPE, MAE, RMSE, and R^2 . This supports the selection of RBFNN as the primary model for the proposed forecasting framework, while recognizing that benchmark models remain useful references for evaluating generalization performance.

4.5.2. Error Analysis and Visualization

To further analyze model performance beyond the numerical accuracy metrics, visual comparisons were used to examine the distribution and behavior of testing errors. Figures 4.16 and 4.17 compare the plant-level testing MAPE and RMSE of the RBFNN, Random Forest, and XGBoost models. In addition, Figure 4.18 presents the cumulative distribution function (CDF) of absolute testing forecast errors to evaluate how frequently each model produced small or large MW errors across the testing set.

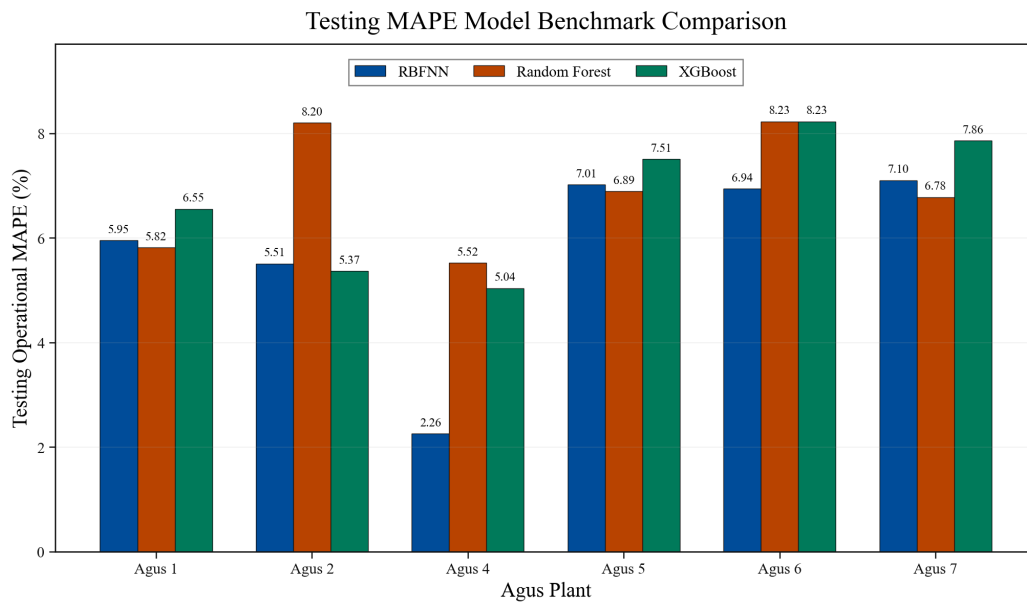


Figure 4.16. Comparison of testing MAPE across Agus plants for RBFNN, Random Forest, and XGBoost models.

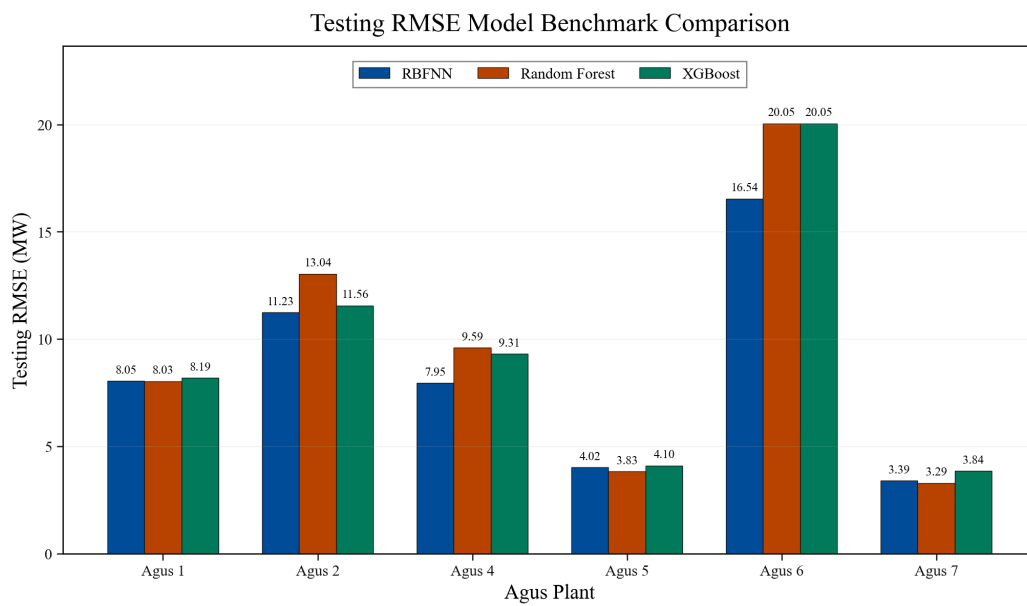


Figure 4.17. Comparison of testing RMSE across Agus plants for RBFNN, Random Forest, and XGBoost models.

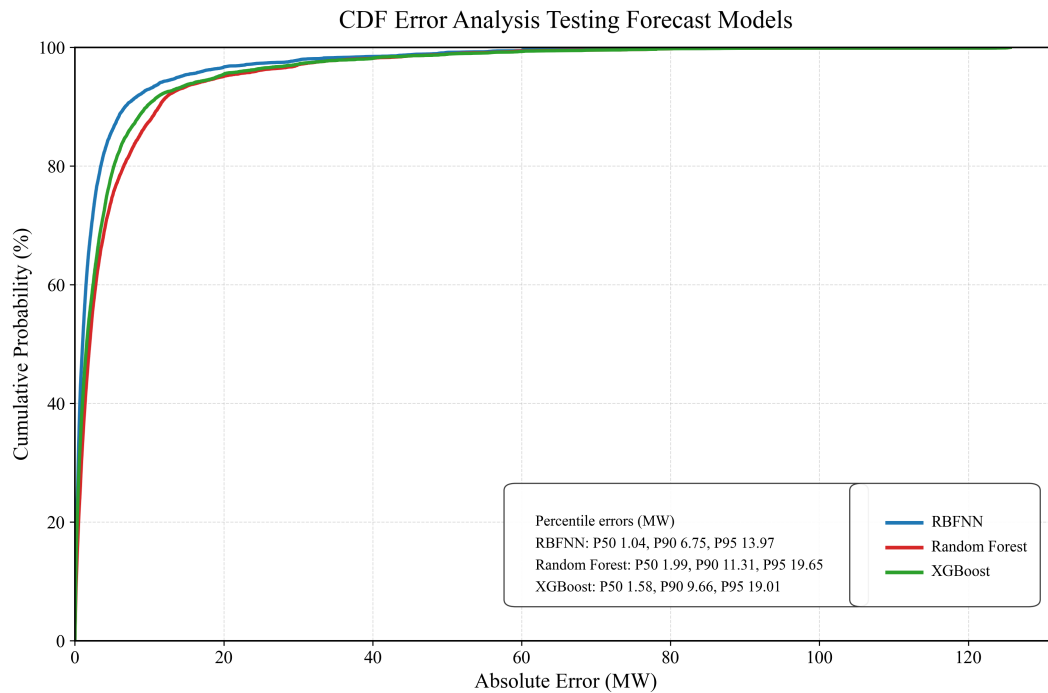


Figure 4.18. Cumulative distribution function of absolute testing forecast errors for RBFNN, Random Forest, and XGBoost models.

The cumulative distribution function (CDF) was used to examine the overall distribution of absolute testing forecast errors. The absolute error was computed as the absolute difference between the actual generation and the predicted generation, expressed in MW. In the CDF plot, the horizontal axis represents the absolute forecast error, while the vertical axis represents the cumulative probability, or the proportion of testing predictions with errors less than or equal to a given MW value. Therefore, a curve that rises more steeply near the origin and remains farther to the left indicates better forecasting performance because a larger portion of its predictions has smaller errors.

As shown in Figure 4.18, the RBFNN curve rises faster and remains generally farther to the left than the Random Forest and XGBoost curves. This indicates that the RBFNN produced smaller absolute errors for a larger proportion of testing observations. The percentile summary supports this result. The RBFNN obtained a median absolute error of 1.04 MW, compared with 1.99 MW for Random Forest and 1.58 MW for XGBoost. At the 90th percentile, 90% of the RBFNN testing errors were within 6.75 MW, while Random Forest and XGBoost required higher error thresholds of 11.31 MW and 9.66 MW, respec-

tively. At the 95th percentile, the RBFNN also showed a lower error threshold of 13.97 MW compared with 19.65 MW for Random Forest and 19.01 MW for XGBoost.

These results suggest that the RBFNN not only achieved favorable average testing performance but also produced a more favorable error distribution under the rolling 24-hour day-ahead testing setup. The faster-rising CDF curve indicates that the proposed model had a higher probability of producing low-MW errors across the testing predictions. This supports the selection of the RBFNN as the main forecasting model in this study, while still recognizing that larger errors may occur during sudden dispatch changes, outage-related variations, ramping conditions, or hydrologic changes that were not fully known before the forecast day.

It should be noted that Figure 4.18 does not represent the total cascade forecast error. Instead, it pools plant-level testing predictions from Agus 1, Agus 2, Agus 4, Agus 5, Agus 6, and Agus 7 into one model-level error distribution. Thus, each curve summarizes the plant-level testing error behavior of one forecasting model across all Agus plants. A separate total-cascade CDF would require summing the actual and predicted generation of all plants by timestamp before computing the cascade-level absolute error.

4.5.3. Sample Forecast Comparison

Furthermore, Figures 4.19–4.24 present sample forecast comparisons for a representative testing day.

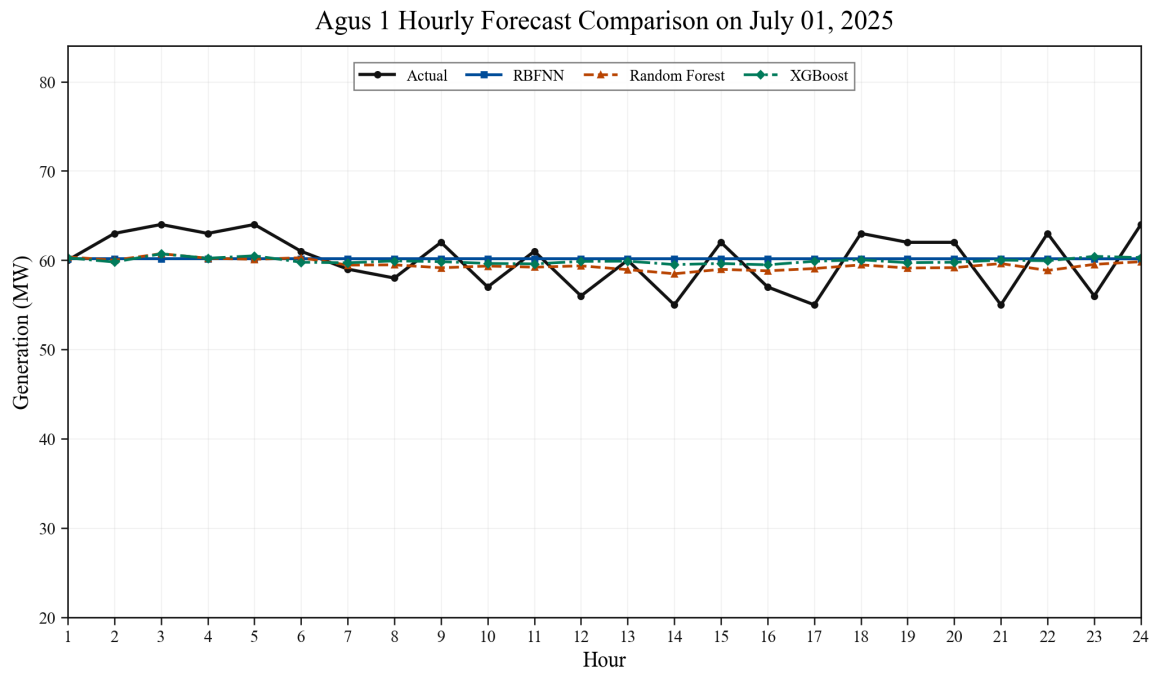


Figure 4.19. Actual and forecast generation for Agus 1 using the RBFNN model.

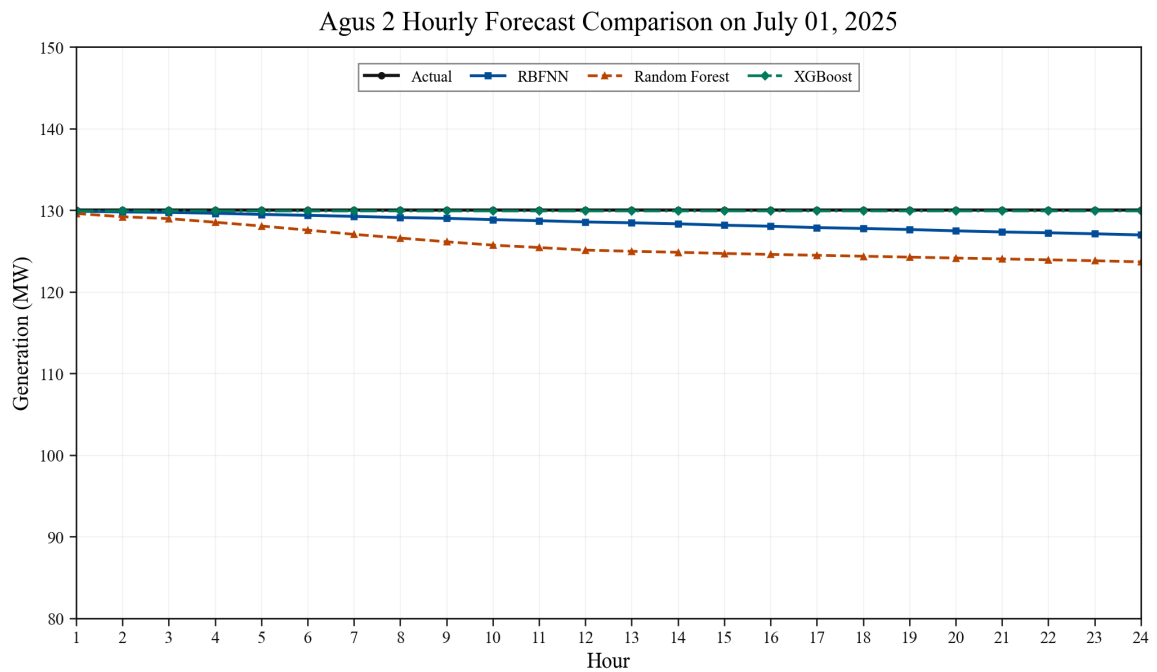


Figure 4.20. Actual and forecast generation for Agus 2 using the RBFNN model.

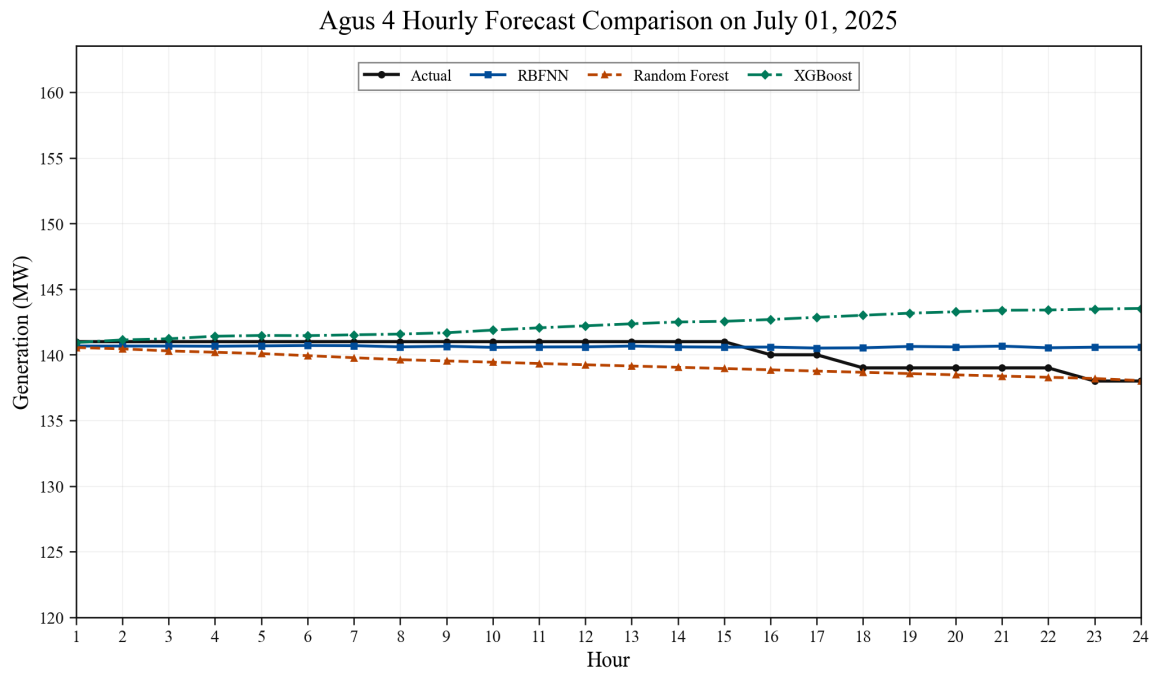


Figure 4.21. Actual and forecast generation for Agus 4 using the RBFNN model.

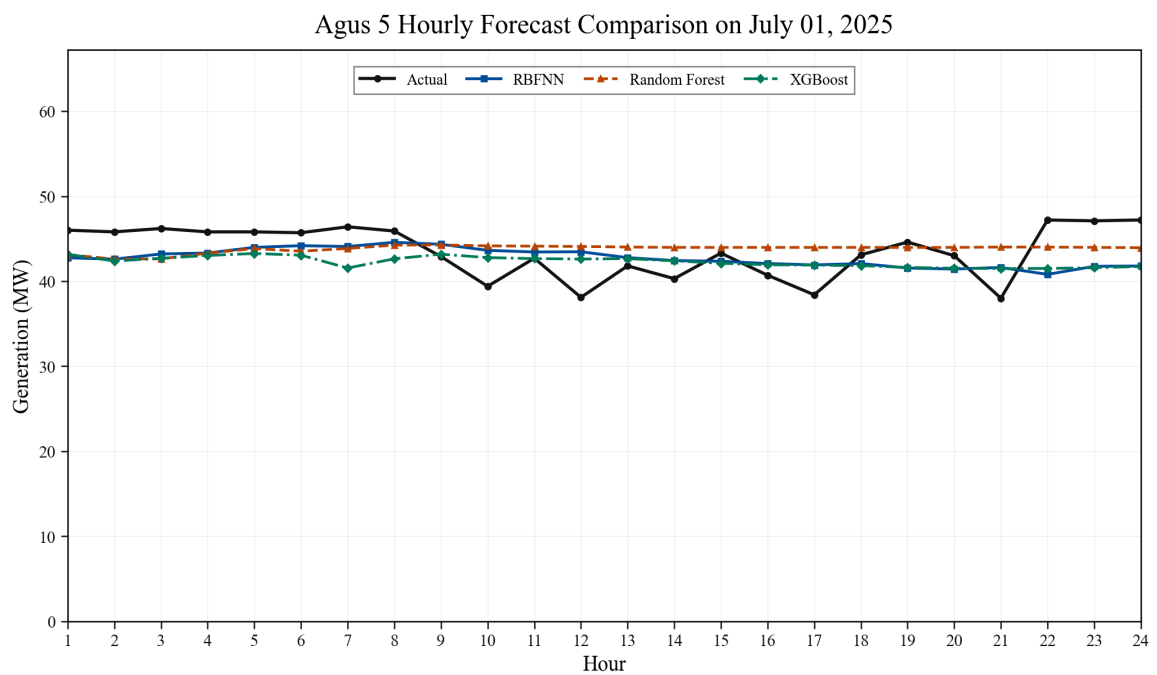


Figure 4.22. Actual and forecast generation for Agus 5 using the RBFNN model.

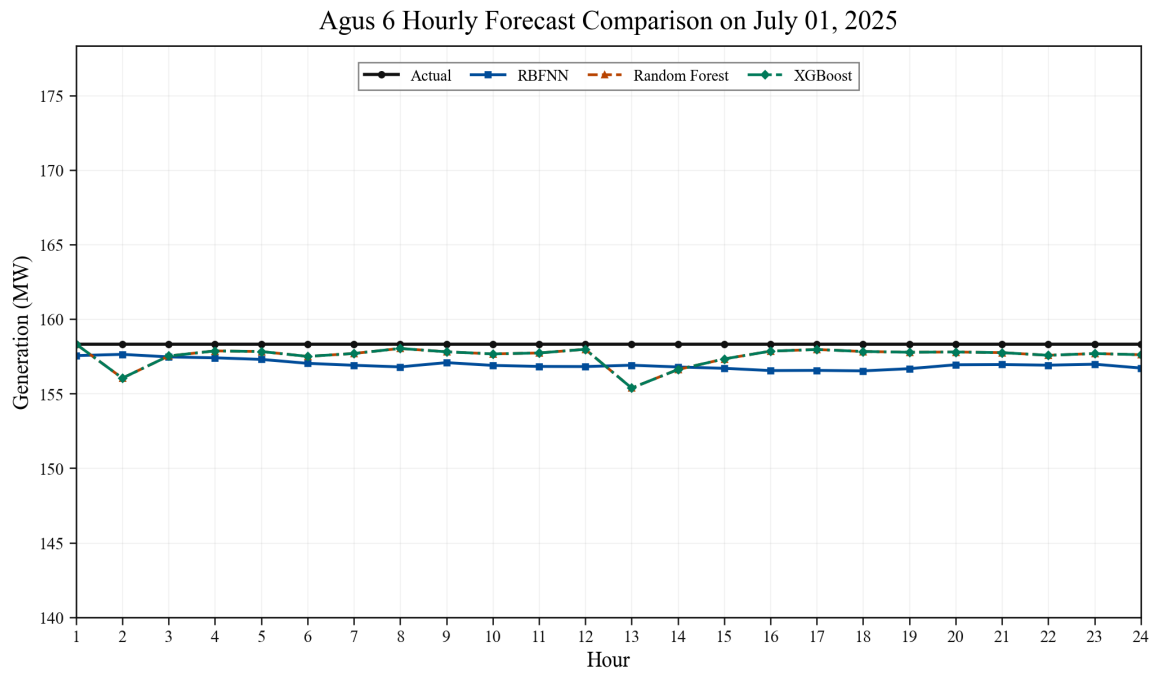


Figure 4.23. Actual and forecast generation for Agus 6 using the RBFNN model.

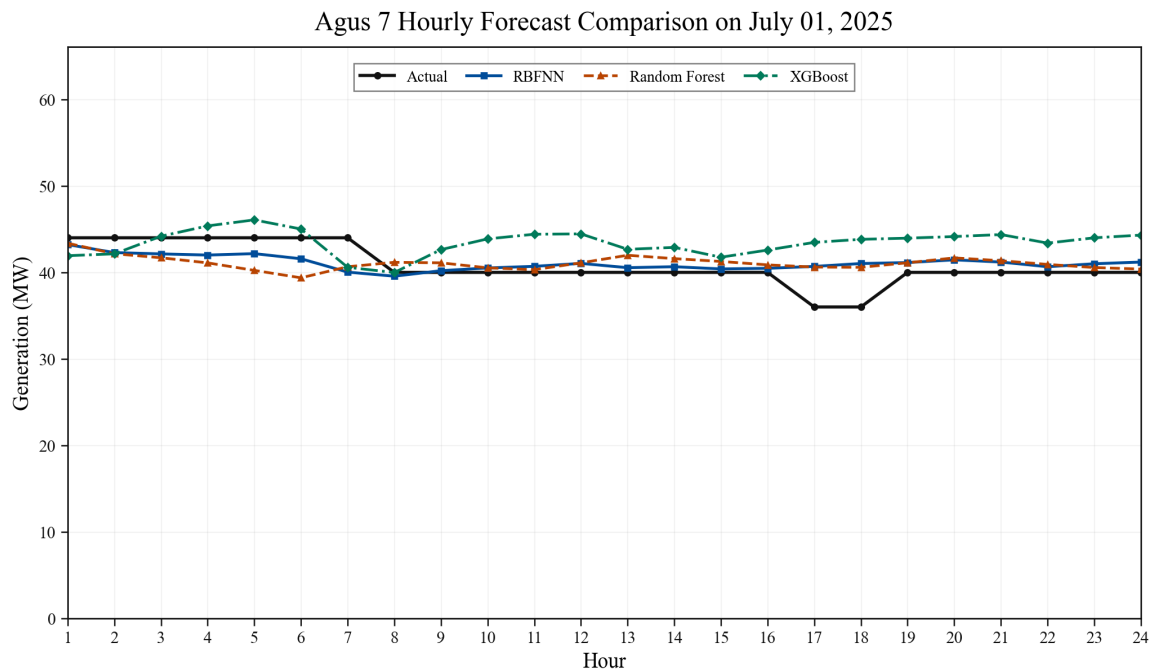


Figure 4.24. Actual and forecast generation for Agus 7 using the RBFNN model.

Figures 4.19 to 4.24 present sample actual-versus-forecast comparisons for the RBFNN model. These figures provide a visual check of how the model behaves across individual plant profiles. The actual values used for comparison were based on the

Daily Operation Report (DOR) data of NPC-MinGen rather than live real-time SCADA measurements, due to limitations in the acquired dataset. Therefore, the plotted actual data represent recorded operational values available from the DOR, while the forecast values represent the model-generated day-ahead estimates. The results show that the RBFNN was generally able to follow the main generation level and daily shape of the recorded data, although deviations occurred during abrupt changes in generation.

The sample forecasts are useful because numerical metrics alone cannot show whether the model follows the timing of generation changes. In the plots, the RBFNN forecast curves generally tracked the actual curves during normal operating periods, which indicates that the model captured recurring temporal and cascade-related patterns. Larger deviations were most visible during sudden changes, where the next-day operating condition may have been affected by dispatch decisions, unit status changes, or hydrologic conditions that were not available as same-day inputs.

In the forecasting task, the user can edit the planned outage or unit availability input before generating the day-ahead forecast. This allows the forecast to account for expected unit availability during the next 24 hours. However, the current prototype does not yet include manual editing of future gate openings, spillway conditions, or other detailed hydraulic control settings. These variables can also affect generation because gate operation influences water release, turbine flow, and downstream cascade behavior. Therefore, some forecast deviations may still occur when the actual gate operation during the forecast day differs from the historical or lagged values available to the model.

These visual results support the quantitative findings. The RBFNN did not perfectly reproduce every hourly change, but it produced forecasts that were sufficiently close to the recorded generation pattern to be useful as an offline decision-support reference for day-ahead capacity nomination and planning studies.

4.6. Day-Ahead Forecast

This section presents the prototype day-ahead forecast output of the developed forecasting framework using the selected RBFNN model. The system generates hourly fore-

casts for the Agus cascade based on the most recent available data while incorporating planned outage schedules at the unit level. Unlike the previous sections, which focused on model evaluation and benchmarking, this section demonstrates how the forecasting system may be used in a prototype decision-support setting.

To illustrate the proposed decision-support workflow, the day-ahead forecasting process is presented in terms of input configuration, model output, and system-level behavior.

4.6.1. Forecast Input: Planned Outage Schedule

The day-ahead forecasting process incorporated planned outage information to represent the expected availability of generating units during the forecast horizon. In the prepared outage input, each unit was represented using a binary status value, where 1 indicated that the unit was available and 0 indicated that the unit was scheduled for outage. This input allowed the forecasting model to consider the expected operational availability of each plant before producing the 24-hour forecast.

The planned outage schedule was treated as a forecast-time input because it represents information that may be known before the operating day. This differs from actual same-day outage status, which was not used as an input during validation and testing. Therefore, the use of planned outage information helped make the forecast more consistent with day-ahead planning conditions while avoiding the use of future actual operating data.

A sample of the planned outage input format is provided in Appendix E, Table E.1. The appendix figure shows how unit-level availability values were arranged before being used by the forecasting workflow.

4.6.2. Day-Ahead Forecast Output

After the planned outage and historical input data were processed, the RBFNN model generated hourly generation forecasts for the next 24-hour day-ahead horizon. The forecast output included unit-level generation, plant-level total generation, and total cascade generation. This format allowed the forecast to be reviewed both at the individual plant level and at the overall Agus cascade level.

The output table was designed to follow an operational reporting format so that the forecast values could be easily inspected, exported, and compared with planning records. Since the full forecast table contains many unit-level and plant-level columns, the detailed sample output is presented in Appendix E, Table E.2 instead of the main results section. This keeps the main chapter focused on interpreting the forecasting results, while still documenting the complete output format for reproducibility and review.

The day-ahead forecast output should be interpreted as a planning estimate rather than a final dispatch instruction. Actual generation may still differ from the forecast because of dispatch decisions, changes in hydrologic conditions, forced outages, gate operations, and other same-day plant constraints that may not be fully known before the forecast day.

4.6.3. Total Cascade Forecast

To evaluate the system-level behavior, the total cascade generation forecast is presented in Figure 4.25.

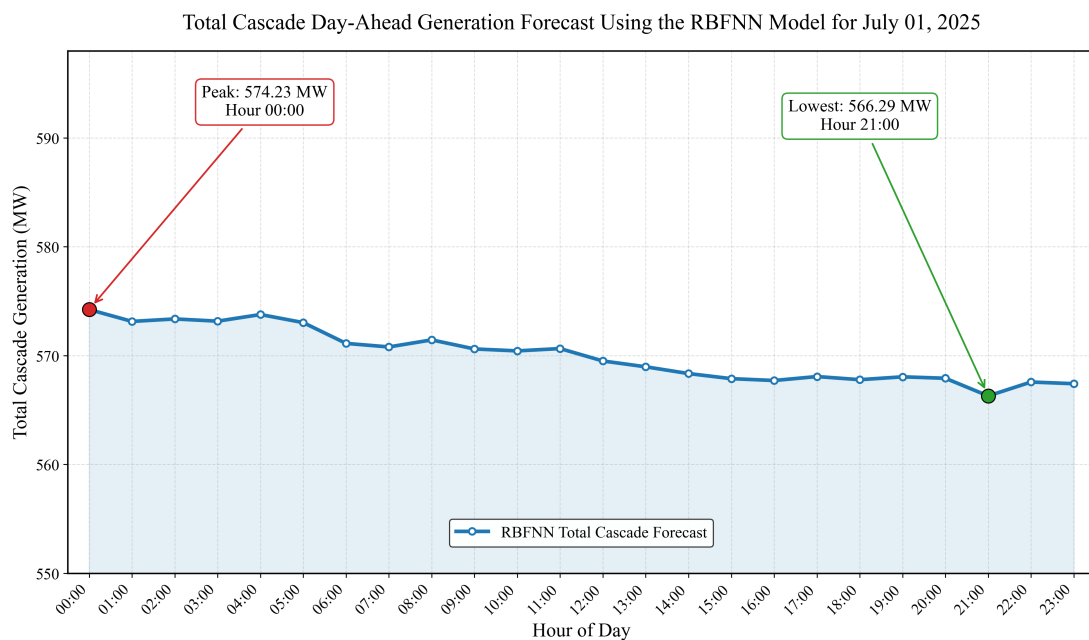


Figure 4.25. Total cascade day-ahead generation forecast using the RBFNN model for July 01, 2025.

Figure 4.25 shows the total cascade day-ahead generation forecast obtained by summing the plant-level forecasts. This total forecast provides an overall estimate of the ex-

pected generation contribution of the Agus cascade for the forecast day. Since the Agus plants are hydraulically connected, the total cascade forecast is important for understanding the combined behavior of the system rather than evaluating each plant in isolation.

The total cascade forecast also helps show whether the individual plant forecasts produce a reasonable overall generation profile. A smooth and physically plausible total forecast indicates that the model output is consistent at the cascade level. Nevertheless, deviations may still occur if actual operation differs from planned outage assumptions or if same-day dispatch and hydrologic conditions change after the forecast is generated.

4.7. Graphical User Interface (GUI) Implementation

To demonstrate the usability and potential decision-support application of the proposed forecasting framework, a Graphical User Interface (GUI) dashboard was developed for the NPC Agus Cascade Hydropower Generation Forecasting System. The dashboard serves as a prototype interface that allows users to visualize historical data, manage planned outage inputs, and review forecast outputs in a more accessible and user-friendly environment.

The developed GUI integrates the major components of the forecasting workflow, including data management, planned outage scheduling, RBFNN-based forecasting, and system monitoring. The interface was designed to simplify user interaction with the forecasting system by minimizing the need for direct code execution and manual inspection of raw output files.

The dashboard also provides visualization tools for analyzing hourly and daily generation forecasts. Line plots were used to display hourly generation behavior of each Agus plant, while bar charts were used to compare total daily generation across the cascade system. These visualizations may help users review forecast trends, generation variations, and the possible effects of planned outages on generation output.

In addition, the GUI was designed to support prototype-level monitoring by displaying the availability status of important datasets and forecast outputs, including raw Excel data, cleaned datasets, outage plans, and generated RBFNN forecasts. This provides users

with immediate confirmation that the required files and forecasting outputs are successfully updated and available within the system.

The developed GUI demonstrates the prototype implementation of the proposed forecasting framework by integrating forecasting, outage input management, visualization, and monitoring functions into a single interface. Figure 4.26 shows the developed NPC Agus Cascade Hydropower Generation Forecasting Dashboard.

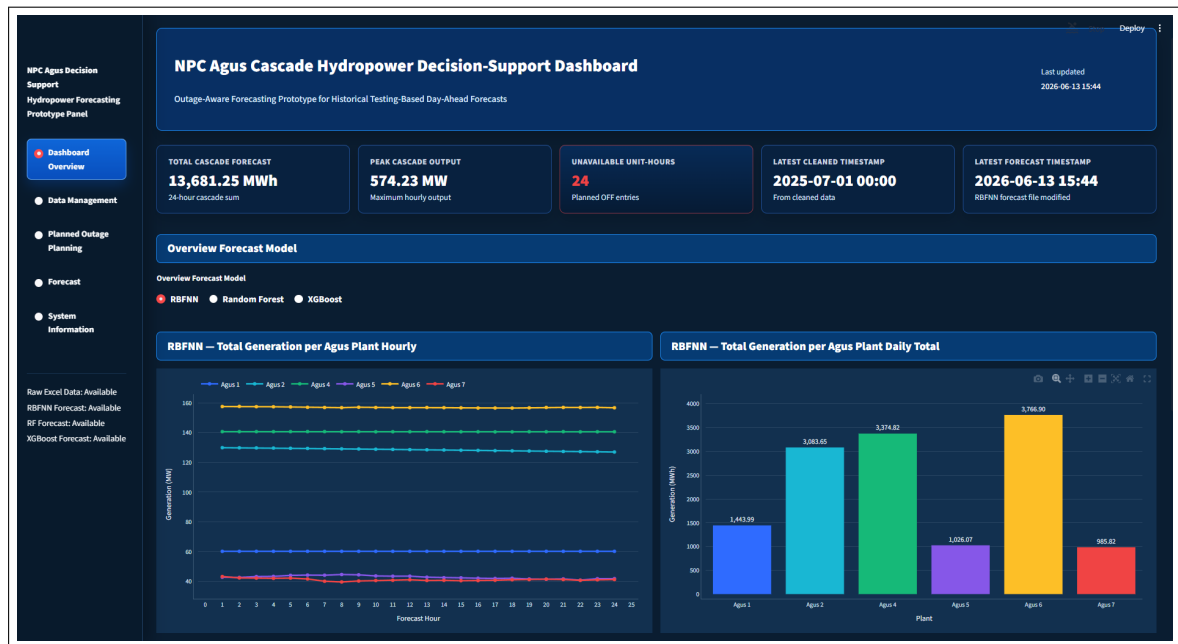


Figure 4.26. NPC Agus Cascade Hydropower Generation Forecasting Dashboard

4.8. Discussion of Results

This section synthesizes the main findings of the study based on the data preparation results, RBFNN validation and testing performance, benchmark comparison, day-ahead forecast output, and graphical user interface implementation. The discussion focuses on the behavior of the forecasting model under a leakage-free rolling 24-hour day-ahead evaluation and its potential use as a decision-support tool for cascaded hydropower generation planning.

4.8.1. Model Learning Behavior and Generalization

The RBFNN model demonstrated useful learning behavior across the Agus cascade by using plant-specific feature sets, residual forecasting, and validation-only calibration. The residual formulation allowed the model to predict the next-hour change in generation rather than directly predicting the full generation level. This helped maintain continuity because each forecast was anchored to the most recent available generation value.

The validation results showed that the model was able to learn meaningful nonlinear relationships from historical generation, unit availability, temporal patterns, hydrologic context, and upstream cascade behavior. However, validation performance was not equally strong for all plants. Agus 6 showed weaker validation R^2 compared with the other plants, indicating that the model had difficulty explaining the full variability of that validation period. This does not automatically indicate failed training because Agus 6 has higher capacity, more units, stronger ramping behavior, and greater exposure to operational changes.

The difference between validation and testing performance also shows the importance of chronological evaluation. Since validation and testing represent different time periods, each split may contain different outage patterns, dispatch behavior, and hydrologic conditions. Therefore, the results should be interpreted using MAPE, MAE, RMSE, R^2 , and actual-versus-forecast plots together.

4.8.2. Forecasting Accuracy and Reliability

The RBFNN achieved an average testing operational MAPE of 5.7949%, MAE of 3.4687 MW, RMSE of 8.5306 MW, and R^2 of 0.6705. These values indicate that the model produced useful day-ahead forecasts under offline testing conditions. The plant-level testing results also showed that the model maintained moderate percentage errors across all six plants.

The actual-versus-forecast plots showed that the RBFNN generally followed the main level and timing of generation changes. Forecast errors were more visible during sharp changes in plant output, which may be caused by dispatch instructions, unit availability changes, ramping events, water constraints, or other same-day operational decisions. Since

these factors may not be fully known before the forecast day, some error is expected in a realistic day-ahead forecasting setup.

The results therefore support the use of the RBFNN as a forecasting reference for planning. However, the model should not be described as fully deployable or ready for unsupervised real-world operation. Its reliability is supported only under historical offline testing, and further validation under actual plant operation is required before operational deployment.

4.8.3. Comparison with Benchmark Models

The benchmark comparison showed that the RBFNN achieved the best overall testing performance among the evaluated models under the implemented rolling 24-hour day-ahead setup. On average, the RBFNN obtained lower operational MAPE, MAE, and RMSE than Random Forest and XGBoost. These findings suggest that the RBFNN framework was effective in representing the generation patterns contained in the Agus dataset under the forecasting conditions examined in this study.

Random Forest and XGBoost remained useful benchmark models because they are established nonlinear regression methods. However, their overall errors were higher under the recursive day-ahead evaluation. One possible explanation is that small prediction errors accumulated across successive forecast hours because predicted values were reused to construct the lagged and rolling features for the succeeding forecast hours. The RBFNN's localized radial-basis representation and validation-selected calibration procedures may have contributed to the relative stability of its forecasts during recursive prediction.

The comparison supports the selection of the RBFNN as the primary forecasting model in this study. However, the result is specific to the Agus dataset, the implemented feature set, the validation-only calibration procedure, and the rolling 24-hour day-ahead evaluation. It should not be interpreted as evidence that Random Forest and XGBoost are unsuitable for other hydropower forecasting applications or that the RBFNN is universally superior to tree-based models.

4.8.4. Potential Operational Implications of the Forecasting Framework

The forecasting framework demonstrated potential value as a decision-support tool for day-ahead capacity nomination and short-term generation planning. By producing hourly forecasts at both plant and cascade levels, the model may help users review expected generation trends before the operating day. The inclusion of planned outage inputs also allows the forecast output to reflect expected unit availability, provided that the outage schedule is prepared before the forecast day and does not use actual same-day outage information.

The GUI dashboard further illustrates how the forecasting workflow can be organized into a more accessible prototype system. Through the dashboard, users can review input data, manage planned outage schedules, generate forecast outputs, and visualize generation trends without directly interacting with the computational workflow. This makes the framework easier to demonstrate and evaluate as a planning-support prototype.

However, these operational implications should be interpreted with caution. The framework was evaluated using historical offline data and was not tested under live plant operation. Actual hydropower operation involves additional factors such as real-time dispatch instructions, forced outages, operator interventions, water management constraints, communication delays, and market-related decisions. These conditions may affect actual generation but were not fully represented in the offline testing environment. Therefore, the framework should be presented as a prototype decision-support system with potential planning value, not as a fully deployed or operationally validated forecasting system.

4.8.5. Overall Assessment

Overall, the results indicate that the RBFNN-based forecasting framework produced useful short-term generation estimates for the Agus cascade under the strict rolling 24-hour day-ahead backtest. The RBFNN achieved the most favorable aggregate performance among the three evaluated models under the implemented testing procedure, although performance varied across plants and metrics.

The results also emphasize the importance of residual forecasting, validation-only

calibration, and leakage-free evaluation in improving the reproducibility and realism of the forecasting procedure. However, the framework was evaluated using historical offline data only. The findings provide evidence of forecasting potential, but they do not establish full operational readiness.

CHAPTER 5

CONCLUSION AND RECOMMENDATIONS

5.1. Conclusion

This study developed an AI-based short-term generation forecasting framework for the Agus 1, Agus 2, Agus 4, Agus 5, Agus 6, and Agus 7 cascaded hydropower plants using plant-specific Radial Basis Function Neural Network models. The framework incorporated historical generation, temporal indicators, lagged and rolling variables, unit availability, hydrologic context, and upstream cascade features to produce hourly generation forecasts for the succeeding operating day. A residual forecasting formulation was used in which each model predicted the change in generation relative to the most recently available plant output.

The historical dataset was cleaned, standardized, and organized into a consistent hourly time-series structure. Rainfall and Lake Lanao outflow were retained as daily-derived hourly-equivalent hydrologic context variables rather than being treated as direct hourly measurements. The models were developed using a chronological data split, with the training set used for parameter estimation, the validation set used for model selection and calibration, and the testing set reserved for final evaluation.

Forecasting performance was assessed through a rolling 24-hour day-ahead back-test. For each forecast day, the complete 24-hour horizon was generated recursively using only information available before the beginning of that day. Actual forecast-day generation and other same-day observations were excluded from the forecasting inputs and were used only after prediction for performance evaluation.

The RBFNN achieved an average validation operational MAPE of 6.1329%, MAE of 4.3216 MW, RMSE of 8.4214 MW, and R^2 of 0.4650. During testing, it achieved an average operational MAPE of 5.7949%, MAE of 3.4687 MW, RMSE of 8.5306 MW, and R^2 of 0.6705. These results indicate that the framework produced useful day-ahead generation estimates under historical offline evaluation. Performance nevertheless varied among the individual plants because of differences in generation variability, installed capacity, opera-

tional conditions, and abrupt generation changes.

Under the implemented Agus dataset, feature construction, model-selection procedure, and rolling day-ahead evaluation, the RBFNN achieved the most favorable aggregate testing performance among the evaluated models. Its average operational MAPE, MAE, and RMSE were lower than those of Random Forest and XGBoost. This finding supports the selection of the RBFNN as the primary model for the present study but does not establish its universal superiority over other forecasting methods.

A prototype graphical user interface was also developed to demonstrate how the forecasting workflow could be organized for data review, planned-outage input, forecast generation, and visualization of plant-level and cascade-level outputs. The interface demonstrated the potential usability of the framework but did not constitute deployment in an actual plant operating environment.

Overall, the study provides prototype-level evidence that an RBFNN-based forecasting framework may support the review of day-ahead generation trends and capacity nomination plans for the Agus cascade. However, the framework was validated only using historical offline data and was not tested under live dispatch, real-time data acquisition, unanticipated forced-outage conditions, or actual operator decision-making. It should therefore be regarded as a potential decision-support tool that requires further validation using live plant data, operator-reviewed inputs, real-time system integration, and continuous performance monitoring before operational use.

5.2. Recommendations

Based on the findings and limitations of the study, the following recommendations are proposed:

1. **Validate the forecasting framework under actual plant operation.** The RBFNN forecasting framework should be tested using actual operational workflows, operator-reviewed inputs, and real plant conditions. This would help determine whether the model can maintain reliable performance beyond historical offline testing.

2. **Improve the availability and quality of hydrologic input data.** Future work should improve the collection and updating of hydrologic variables such as rainfall, Lake Lanao outflow, forebay elevation, gate opening, and spillway-related measurements. More complete and higher-resolution hydrologic data may improve the model's ability to capture water availability and cascade behavior.
3. **Include future gate-opening and hydraulic control inputs.** The current prototype allows planned outage or unit availability inputs to be edited before forecast generation. However, future versions should also allow operators to specify expected gate openings, spillway conditions, and other hydraulic control actions, since these variables directly affect water release, turbine flow, and downstream generation.
4. **Extend the evaluation period and operating scenarios.** The model should be evaluated using a longer dataset that includes more seasonal variation, outage events, dispatch changes, low-water conditions, high-inflow periods, and abnormal operating cases. This would provide a more complete assessment of the model's robustness.
5. **Enhance the dashboard as a decision-support system.** The graphical user interface may be improved by adding database integration, automatic data updating, user access control, forecast confidence intervals, error monitoring, and operator feedback logging. These improvements would make the system more practical for planning review.
6. **Compare the RBFNN with additional forecasting models.** Future studies may compare the RBFNN with other advanced models such as LSTM, GRU, hybrid physical-data-driven models, and ensemble forecasting approaches. These comparisons should use the same leakage-free rolling 24-hour day-ahead evaluation procedure to ensure fairness.
7. **Apply continuous monitoring and recalibration.** If the framework is considered for operational use, its forecast accuracy should be monitored over time. The model should be retrained or recalibrated when major changes occur in plant operation, unit availability, hydrologic behavior, or data quality.

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Appendices

APPENDIX A

DATA CLEANING AND PREPROCESSING ARTIFACTS

A.1. Data Cleaning Rules

Table A.1. Data Cleaning and Preprocessing Rules Applied to the Hydropower Dataset

Cleaning Step	Rule Applied	Purpose
Near-zero generation treatment	Generation values at or below 0.05 MW were set to zero.	Removes insignificant near-zero generation readings and prevents them from being interpreted as meaningful plant output.
Short missing-gap interpolation	Linear interpolation was applied to short missing gaps using a maximum fill limit of two consecutive records.	Completes short data gaps while leaving unresolved values for subsequent imputation.
Short zero-run interpolation	Zero-generation runs consisting of one or two consecutive records were temporarily treated as missing and interpolated.	Corrects isolated zero readings that may have resulted from temporary recording interruptions.

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Cleaning Step	Rule Applied	Purpose
Long zero-run preservation	Zero-generation runs of three consecutive records or more were retained as actual zero values.	Preserves probable unit shutdown, outage, or sustained zero-generation periods.
Spike smoothing	Isolated spikes were identified using a local-neighbor comparison with a configured factor of 3.5 and were replaced using the neighboring-value average.	Reduces abnormal isolated spikes while retaining the general behavior of the time series.
Negative-value correction	Negative spillway-gate and total-gate values were replaced with zero. Negative rainfall values were also replaced with zero.	Prevents physically invalid negative operating and hydrologic measurements from entering the cleaned dataset.
Distance-weighted KNN imputation	Remaining missing numeric values were imputed using a distance-weighted K-Nearest Neighbors method with five nearest neighbors.	Completes unresolved missing values using observations with similar multivariable conditions.

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Cleaning Step	Rule Applied	Purpose
Outage-status standardization	Unit-status values were standardized into binary form, where 1 represented an available or operating unit and 0 represented an unavailable unit.	Provides a consistent representation of unit availability for model training and forecasting.
Hourly resampling	Numeric observations were averaged into hourly intervals. After resampling, outage-status values were restored to binary form using a threshold of 0.5.	Produces the unified hourly dataset used by the forecasting models while maintaining valid binary outage information.
Plant-total recalculation	Plant-level total generation was recalculated from the resampled unit-generation values.	Ensures that each plant-total generation value is equal to the sum of its corresponding unit outputs.
Daily-derived hydrologic variables	Rainfall and Lake Lanao outflow were retained as daily-derived hourly-equivalent hydrologic context variables and were not divided by 24 again.	Preserves the original hydrologic information without treating repeated daily values as independent hourly sensor measurements.

APPENDIX B

COMPLETE FORECASTING FEATURE LIST

B.1. Feature Groups

Table B.1. Summary of Feature Groups Used in the Forecasting Models

Feature Group	Implemented Features	Purpose
Calendar and temporal features	Hour sine and cosine, day-of-week sine and cosine, month sine and cosine, weekend indicator, target-hour sine and cosine, target-day sine and cosine, and target-weekend indicator.	Represent daily, weekly, monthly, and target-time periodic behavior.
Plant-generation lag features	Plant-total generation lags at 1, 2, 3, 6, 12, 24, 48, 72, and 168 hours.	Capture short-term dependence, daily recurrence, and weekly recurrence.
Rolling-generation features	Rolling mean, standard deviation, minimum, and maximum over 3, 6, 12, 24, 48, and 168 hours.	Represent recent generation level, variability, and operating range.

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Feature Group	Implemented Features	Purpose
Generation-change features	Generation differences over 1, 3, and 24 hours.	Describe recent upward or downward generation movement.
Same-target-hour features	Generation associated with the same target hour one day and one week earlier.	Provide daily and weekly profile references without using forecast-day actual values.
Unit-generation features	Current and lagged unit-generation values and current and lagged unit-generation shares.	Represent unit loading and each unit's contribution to plant output.
Outage and availability features	Binary unit-status values, number of running units, and plant-availability indicator.	Represent unit-level operating availability known before prediction.
Operational and hydraulic features	Current and lagged gate-opening, forebay-elevation, and spillway-related variables.	Describe recent operating and hydraulic conditions.
Daily-derived hydrologic features	Lagged rainfall and Lake Lanao outflow values at 1, 3, 6, 12, and 24 hours.	Provide coarse hydrologic context without using actual forecast-day values.

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Feature Group	Implemented Features	Purpose
Cascade features	Current upstream-plant generation and upstream-generation lags at 1, 2, 3, 6, 12, 24, 48, 72, and 168 hours.	Represent upstream-to-downstream dependence in the Agus cascade.

APPENDIX C

HYPERPARAMETER CONFIGURATION

Table C.1. RBFNN Model Configuration and Calibration Parameters per Agus Plant

Plant	Features	Cap.	MAPE Threshold	Centers	Learning Rate	Gamma	Batch	Patience	Shrinkage	Bin	Hour	Base Profile Blend	Day-Ahead Profile Blend
Agus 1	203	80.0	1.000	180	0.0003	0.50	16	16	1.00	Yes	No	No	No
Agus 2	234	180.0	1.800	120	0.0007	0.50	16	16	1.15	Yes	Yes	No	No
Agus 4	234	158.1	1.581	180	0.0003	0.50	16	16	1.15	Yes	Yes	No	No
Agus 5	213	55.0	1.000	80	0.0005	0.50	16	16	0.10	No	Yes	Yes	No
Agus 6	276	219.0	2.190	60	0.0005	0.50	16	16	1.15	Yes	Yes	No	No
Agus 7	213	54.0	1.000	120	0.0007	0.50	16	16	1.00	Yes	Yes	No	Yes

Table C.2. RBFNN Calibration and Training Configuration per Agus Plant

Plant	Bias Correction	Calibration Applied	Optimizer	Loss
Agus 1	-0.00543 MW	Bin calibration	Adam	MSE
Agus 2	0.01148 MW	Bin calibration; hourly residual correction	Adam	MSE
Agus 4	0.00320 MW	Bin calibration; hourly residual correction	Adam	MSE
Agus 5	-0.02640 MW	Hourly residual correction; same-hour-yesterday profile blending	Adam	MSE
Agus 6	-0.01242 MW	Bin calibration; hourly residual correction	Adam	MSE
Agus 7	0.00714 MW	Bin calibration; hourly residual correction; validation-only day-ahead profile blending	Adam	MSE

Table C.3. Calibration and Post-Processing Rule Summary

Calibration/Post-Processing Step	Formula/Rule	Candidate Values	Derived From	Testing Used?
Shrinkage	$\hat{y}_{t+1}^{\text{raw}} = y_t + s\hat{\Delta}y_{t+1}$	$s \in \{0.05, 0.10, 0.20, 0.35, 0.50, 0.75, 1.00, 1.15\}$	Validation data only	No
Bias correction	$b = \text{clip}(b_{\text{raw}}, -\tau_p, \tau_p)$, where $b_{\text{raw}} = \text{median}(y - \hat{y}^{\text{raw}})$ and $\tau_p = \max(1, 0.01C_p)$.	Validation-derived median residual, bounded by the plant-specific operational threshold.	Validation data only	No
Bin calibration	$\hat{y}^{\text{bin}} = \hat{y} + \alpha c_k$, where c_k is the median residual correction for bin k .	Quantile-bin candidates and correction-scale values from 0.50 to 2.50.	Validation data only	No
Hourly residual correction	$\hat{y}^{\text{hour}} = \hat{y} + \beta q_h$, where q_h is the median residual correction for target hour h .	$\beta \in \{0, 0.25, 0.50, 0.75, 1.00, 1.25\}$	Validation data only	No

Continued on the next page

Table C.3 continued from the previous page

Calibration/Post-Processing Step	Formula/Rule	Candidate Values	Derived From	Testing Used?
Initial validation profile blend	$\hat{y}^{\text{blend}} = (1 - w)\hat{y} + wy_{t-24}$	$w \in \{0, 0.10, 0.20, 0.30, 0.40, 0.55, 0.70, 0.85\}$	Validation data, with acceptance checked through rolling day-ahead validation.	No
Rolling day-ahead profile blend	$\hat{y}^{\text{DA}} = (1 - w_{\text{DA}})\hat{y} + w_{\text{DA}}y_{t-24}$	$w_{\text{DA}} = 0.05, 0.10, \dots, 0.90$. Retained only when operational MAPE and RMSE improved and R^2 did not decrease.	Rolling 24-hour day-ahead validation predictions only.	No
Ramp limiting	$\hat{y}_{t+1}^{\text{ramp}} = \text{clip}(\hat{y}_{t+1}, y_t - r_p, y_t + r_p)$, where $r_p = \max[1, Q_{0.98}(\Delta y_{\text{train}})]$.	Fixed plant-specific limit calculated from the training-generation series.	Training data only	No
Capacity clipping	$0 \leq \hat{y}_{p,t}^{\text{clip}} \leq 1.05C_p$	Fixed upper limit of 105% of installed plant capacity.	Installed plant capacity.	No

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Table C.3 continued from the previous page

Calibration/Post-Processing Step	Formula/Rule	Candidate Values	Derived From	Testing Used?
Planned-outage adjustment	$\hat{y}_{p,t}^{\text{out}} = \text{clip} \left(\hat{y}_{p,t}^{\text{ramp}}, 0, C_{p,t}^{\text{plan}} \right)$	Fixed rule based on the available capacity of units scheduled as operating.	Planned outage schedule or latest known pre-forecast status.	No
Unit allocation	The adjusted plant-total forecast was distributed among available units using current unit share, recent median share, and same-hour historical share. Installed-capacity shares were used as the fallback.	Fixed weighted allocation rule subject to individual unit-capacity limits.	Historical information available before forecasting and the planned outage schedule.	No

Table C.4. Reproducibility Artifacts

Artifact	File/Folder	Purpose
Cleaned data	outputs/cleaned_data/cleaned_hourly_data.xlsx	Model input
RBFNN model	models/rbfnn/rbfnn_<plant>.keras	Trained plant model
Input scaler	models/rbfnn/x_scaler_<plant>.pkl	Feature scaling
Target scaler	models/rbfnn/y_scaler_<plant>.pkl	Residual scaling
Plant metadata	models/rbfnn/meta_<plant>.json	Selected features and calibration
Calibration report	metadata/rbfnn/rbfnn_calibration_report.xlsx	Calibration summary
Selection audit	metadata/overall_metrics/model_selection_audit.json	Validation-only selection record
Leakage audit	metadata/leakage_audit.json	Feature leakage check
Day-ahead predictions	metadata/day_ahead_backtest/	Strict validation/testing outputs

APPENDIX D

PROTOTYPE DASHBOARD REFERENCE

This appendix provides a step-by-step operating guide for the Streamlit-based prototype developed for the Agus Cascade Hydropower Generation Forecasting System. The dashboard runs locally on the user's computer and provides a graphical interface for data management, planned outage scheduling, day-ahead forecasting, forecast comparison, output review, and system-status monitoring.

The dashboard supports the Radial Basis Function Neural Network (RBFNN), Random Forest, and XGBoost forecasting models. The RBFNN serves as the primary forecasting model of the study, while Random Forest and XGBoost serve as benchmark models. All three models can be operated through the same interface after the required data and trained model files have been prepared.

D.1. Required Project Structure

The dashboard must be launched from the main Thesis Forecasting project folder. The important directories within the project are as follows:

- `data/` contains the original Excel data workbook.
- `scripts/` contains the dashboard, data-cleaning, model-training, forecasting, and thesis-output scripts.
- `models/` contains the trained RBFNN, Random Forest, and XGBoost model artifacts.
- `outputs/` contains the cleaned data, planned outage input, and generated day-ahead forecast files.
- `metadata/` contains validation and testing metrics, day-ahead backtest results, prediction files, and audit records.
- `benchmark/` contains supporting files associated with the Random Forest and XGBoost benchmark models.
- `thesis_figures/` contains the figures generated for the results and discussion chapter.

The main dashboard application is located at

`ThesisForecasting/scripts/streamlit_app.py`

while the Windows launcher is located at

`ThesisForecasting/RUN_DASHBOARD.bat`.

D.2. Required Input Files

Before launching the dashboard, the user must confirm that the required data files are available. The original data workbook is stored at

`ThesisForecasting/data/DATA(JAN2024-JUNE2025).xlsx`.

The data-cleaning process produces the following cleaned-data files:

`ThesisForecasting/outputs/cleaned_data/cleaned_hourly_data.xlsx`

and

`ThesisForecasting/outputs/cleaned_data/cleaned_hourly_data.parquet`.

The planned unit-availability input is stored at

`ThesisForecasting/outputs/outages_planning/Planned_Outages_Input.xlsx`.

The planned outage workbook contains the unit status for each of the 24 forecast hours. A value of 1 indicates that a generating unit is available or operating, while a value of 0 indicates that the unit is unavailable or scheduled for a planned outage.

D.3. Launching the Dashboard

The recommended method for launching the dashboard on a Windows computer is through the provided batch file.

1. Open the main Thesis Forecasting project folder.

2. Locate `RUN_DASHBOARD.bat` and double-click the file.
3. Wait while the terminal initializes the required environment and starts the Streamlit application.
4. When the terminal displays the local Streamlit address, open a web browser and proceed to

<http://localhost:8501>.

5. Keep the terminal window open while the dashboard is being used. Closing the terminal window terminates the Streamlit application.

The dashboard may also be launched manually through Visual Studio Code or another terminal application.

1. Open Visual Studio Code.
2. Open the `Thesis Forecasting` folder as the active project folder.
3. Open a terminal within Visual Studio Code.
4. Execute the following command:

```
python -m streamlit run scripts/streamlit_app.py
```

5. Open the local browser address displayed in the terminal.

D.4. Dashboard Navigation

The dashboard sidebar contains five main pages:

1. *Dashboard Overview*
2. *Data Management*
3. *Planned Outage Planning*
4. *Forecast*
5. *System Information*

The user may move between pages using the navigation controls displayed in the dashboard sidebar. Each page performs a separate function within the forecasting workflow.

D.5. Dashboard Overview

The *Dashboard Overview* page provides a summary of the current condition of the forecasting system. It displays the total cascade forecast, peak cascade generation, unavailable unit-hours, latest cleaned-data timestamp, latest forecast timestamp, hourly generation charts, daily generation summaries, recent system activity, and the availability of the forecast files for each model.

To review the system overview:

1. Open the *Dashboard Overview* page.
2. Select RBFNN, Random Forest, or XGBoost from the *Overview Forecast Model* control.
3. Confirm that the raw data, cleaned data, planned outage file, and selected forecast file are available.
4. Review the hourly total-generation chart for the individual Agus plants.
5. Review the daily total-generation summary and recent file activity.
6. Use the cumulative distribution function section to compare the absolute testing errors of RBFNN, Random Forest, and XGBoost.

The cumulative distribution function displayed on this page is based on the held-out rolling 24-hour day-ahead testing predictions. It does not use the current operational day-ahead forecast workbook. A forecast file must already exist before its corresponding forecast charts can be displayed.

D.6. Data Management

The *Data Management* page allows the user to upload or replace the original data workbook, reload data files from disk, and execute the data-cleaning process.

The page contains the following main controls:

- *Upload Excel File*, which replaces the raw workbook used by the cleaning procedure;

- *Refresh Data From Disk*, which clears the cached dashboard data and reloads the available files; and
- *Run Data Cleaning*, which executes `cell1_clean_data.py`.

To update and clean the source data:

1. Open the *Data Management* page.
2. Upload a new Excel workbook, or manually place the updated file at

`ThesisForecasting/data/DATA(JAN2024-JUNE2025).xlsx`.

3. Click *Refresh Data From Disk* when a file has been modified outside the dashboard.
4. Click *Run Data Cleaning* when the raw data have been updated or when the cleaned files are unavailable.
5. Wait until the data-cleaning process reports successful completion.
6. Confirm that the cleaned-data status, number of records, number of variables, and latest timestamp have been updated.

The cleaning process creates or updates the Excel and Parquet cleaned-data files, the data-cleaning metadata file, and the 24-hour planned outage template. The planned outage workbook should be closed in Microsoft Excel before this process is executed to prevent file-locking errors.

D.7. Planned Outage Planning

The *Planned Outage Planning* page allows the user to define the expected unit availability for the next 24-hour forecast horizon. The planned outage table uses the following binary coding:

Value	Meaning
1	Unit available or operating
0	Unit unavailable or scheduled for outage

To prepare the planned outage schedule:

1. Open the *Planned Outage Planning* page.
2. Review the displayed unavailable unit-hours, available unit-hours, affected plants, and outage-plan modification time.
3. Edit the unit-status entries for each forecast hour.
4. Enter only 1 or 0 in the unit-status columns.
5. Click *Save Outage Plan*.
6. Confirm that the dashboard reports that the outage plan was saved successfully.

The saved table is written to

`outputs/outages_planning/Planned_Outages_Input.xlsx`.

After saving the outage plan, the user may click *Forecast Day-Ahead*. The dashboard then validates the availability of the cleaned data, outage plan, and trained models before running the forecast-only procedures for RBFNN, Random Forest, and XGBoost.

The *Forecast Day-Ahead* option does not retrain the models. The *Retrain RBFNN Model* and *Retrain Benchmark Models* controls should only be used when new cleaned data must be learned or when the required model files are unavailable.

D.8. Forecast Generation and Review

The *Forecast* page allows the user to run, visualize, compare, and download the 24-hour generation forecasts. The user may select RBFNN, Random Forest, or XGBoost from the model selector.

To generate and review a forecast:

1. Open the *Forecast* page.
2. Select the desired forecasting model.
3. Click *Run Selected Model Forecast*.
4. Wait until the dashboard displays the successful completion message.

5. Review the total cascade forecast, peak cascade output, affected plants, and forecast-generation timestamp.
6. Select one of the available forecast views: *Total Cascade*, *All Plants*, *Agus 1*, *Agus 2*, *Agus 4*, *Agus 5*, *Agus 6*, or *Agus 7*.
7. Review the forecast graph, summary values, and hourly forecast table.
8. Click *Download Forecast Excel* to export the selected model forecast.

The operational forecast workbooks are stored in the following locations:

- RBFNN: `outputs/rbfnn_forecast/Day_Ahead_24H_RBFNN_Forecast.xlsx`
- Random Forest: `outputs/random_forest_forecast/Day_Ahead_24H_RANDOM_FOREST.xlsx`
- XGBoost: `outputs/xgboost_forecast/Day_Ahead_24H_XGB00ST.xlsx`

The cascade-comparison section allows the user to select the visible models, choose between lines with markers and lines only, control the vertical-axis range, and display the hourly spread among the selected model forecasts.

D.9. System Information

The *System Information* page summarizes the forecasting workflow, runtime environment, file availability, supported models, forecast horizon, and most recent system check.

This page should be used when diagnosing missing inputs, absent model artifacts, unavailable forecast files, or problems related to Python and Streamlit. The displayed file-status information allows the user to identify the stage of the workflow that requires attention before a new forecast can be generated.

D.10. Normal Daily Operating Workflow

The following procedure is recommended when trained model files already exist and only a new day-ahead forecast is required:

1. Start the dashboard.
2. Open the *Data Management* page and review the raw and cleaned-data status.
3. Run the data-cleaning process only when the original workbook has been updated.
4. Open the *Planned Outage Planning* page.
5. Edit the 24-hour unit-availability schedule using 1 for available units and 0 for unavailable units.
6. Click *Save Outage Plan*.
7. Click *Forecast Day-Ahead*.
8. Wait until the forecast-only procedures for the three models have finished.
9. Open the *Forecast* page.
10. Select the model to be reviewed.
11. Review the charts, key performance indicators, and forecast table.
12. Download the required Excel forecast output.

D.11. First-Run or Full-Update Workflow

The following procedure should be used when the forecasting system is prepared for the first time or when new source data must be included in model training:

1. Place the original workbook in the data folder.
2. Start the dashboard.
3. Open the *Data Management* page and click *Run Data Cleaning*.
4. Open the *Planned Outage Planning* page.
5. Review or edit the planned outage schedule and save the changes.
6. Click *Retrain RBFNN Model* and wait until the training procedure is completed.
7. Click *Retrain Benchmark Models* and wait until the Random Forest and XGBoost training procedures are completed.

8. Click *Forecast Day-Ahead*, or open the *Forecast* page and execute the models individually.
9. Review and download the generated forecast outputs.

Model retraining requires more computation time than forecast-only operation because the plant-specific models, calibration parameters, and saved artifacts must be regenerated.

D.12. Troubleshooting

Dashboard Does Not Open

1. Confirm that Python and Streamlit are installed.
2. Open a terminal in the Thesis Forecasting folder.
3. Execute

```
python -m streamlit run scripts/streamlit_app.py.
```

Raw Data File Is Missing

1. Confirm that the workbook is stored at

```
data/DATA(JAN2024-JUNE2025).xlsx.
```

2. Upload the workbook through the dashboard or copy it manually to the required folder.
3. Click *Refresh Data From Disk*.

Cleaned Data Are Missing

1. Open the *Data Management* page.
2. Click *Run Data Cleaning*.
3. Wait until the cleaning procedure is completed.

Forecast-Only Operation Fails

1. Confirm that the cleaned-data files are available.
2. Confirm that the planned outage workbook exists.
3. Confirm that the required saved models are present in the `models` folder.
4. Retrain the corresponding model when its saved artifacts are missing.

Planned Outage Workbook Cannot Be Saved

1. Close `Planned_Outages_Input.xlsx` in Microsoft Excel.
2. Return to the dashboard.
3. Click *Save Outage Plan* again.

Forecast Graph Is Not Updated

1. Click *Refresh Forecast Files*.
2. Review the displayed forecast-generation timestamp.
3. Run the selected model forecast again when necessary.

APPENDIX E

GRAPHICAL USER INTERFACE DASHBOARD REFERENCE

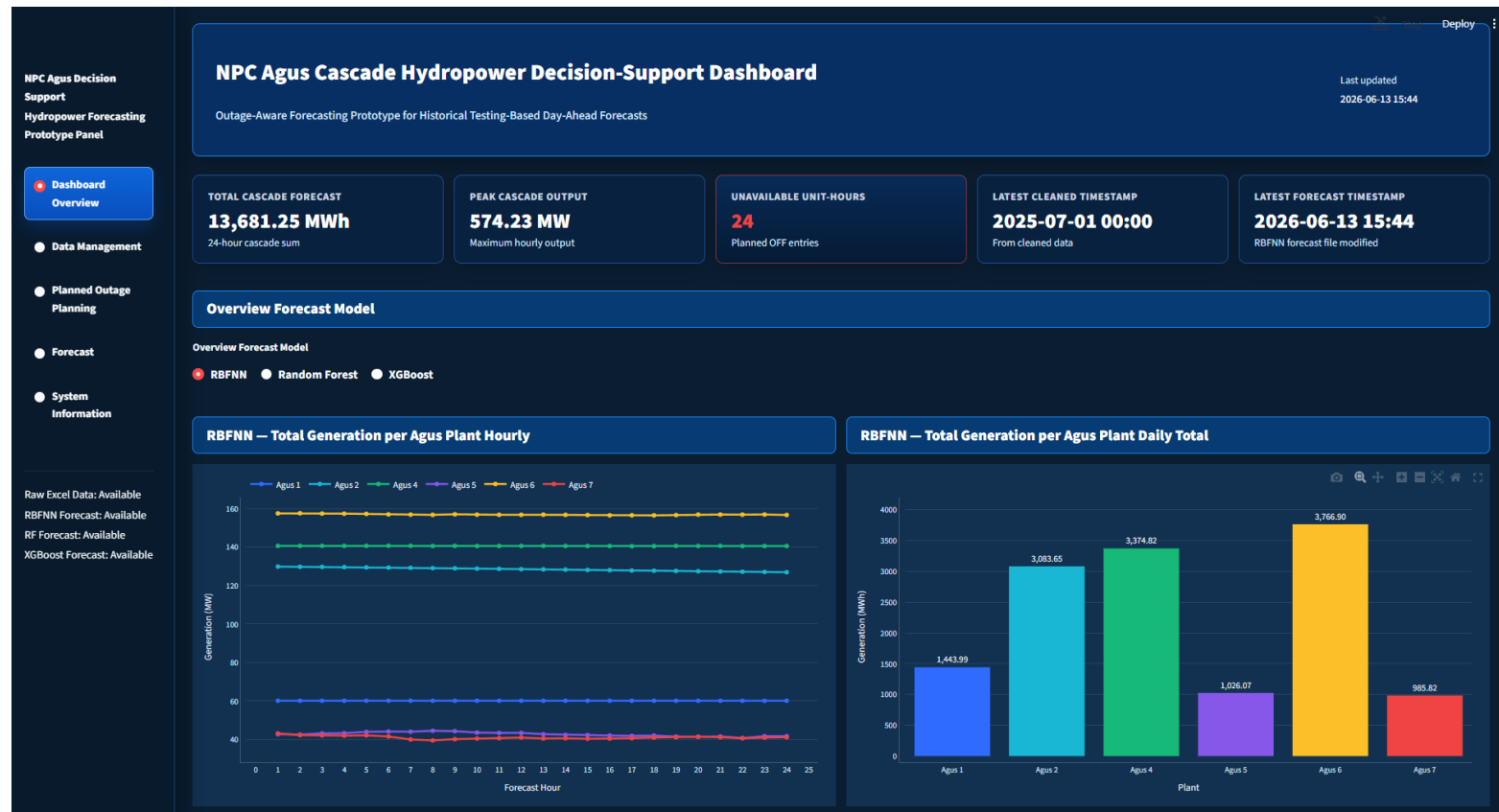


Figure E.1. Dashboard home and system-summary page.

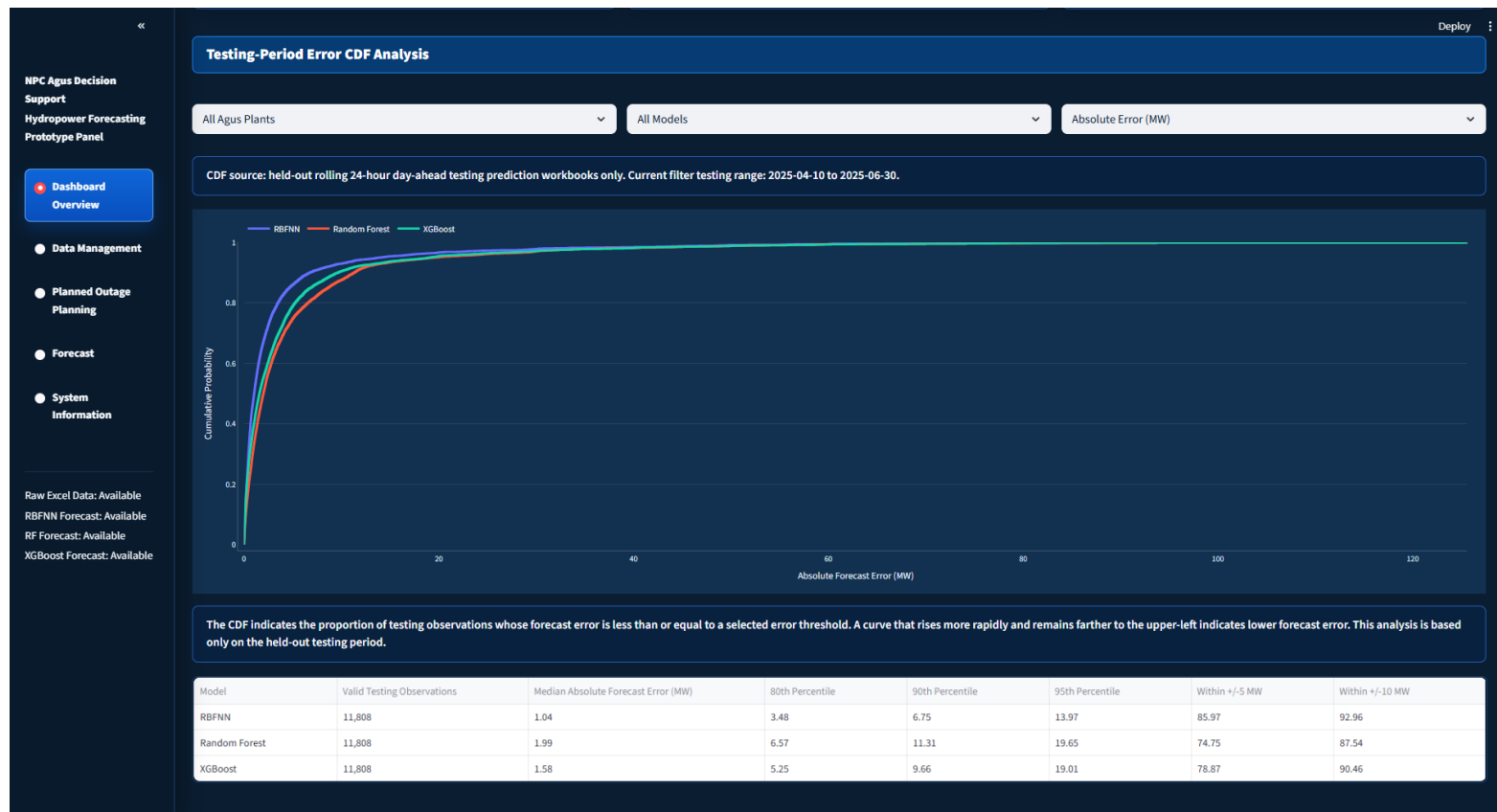


Figure E.2. The cumulative distribution function analysis for errors.

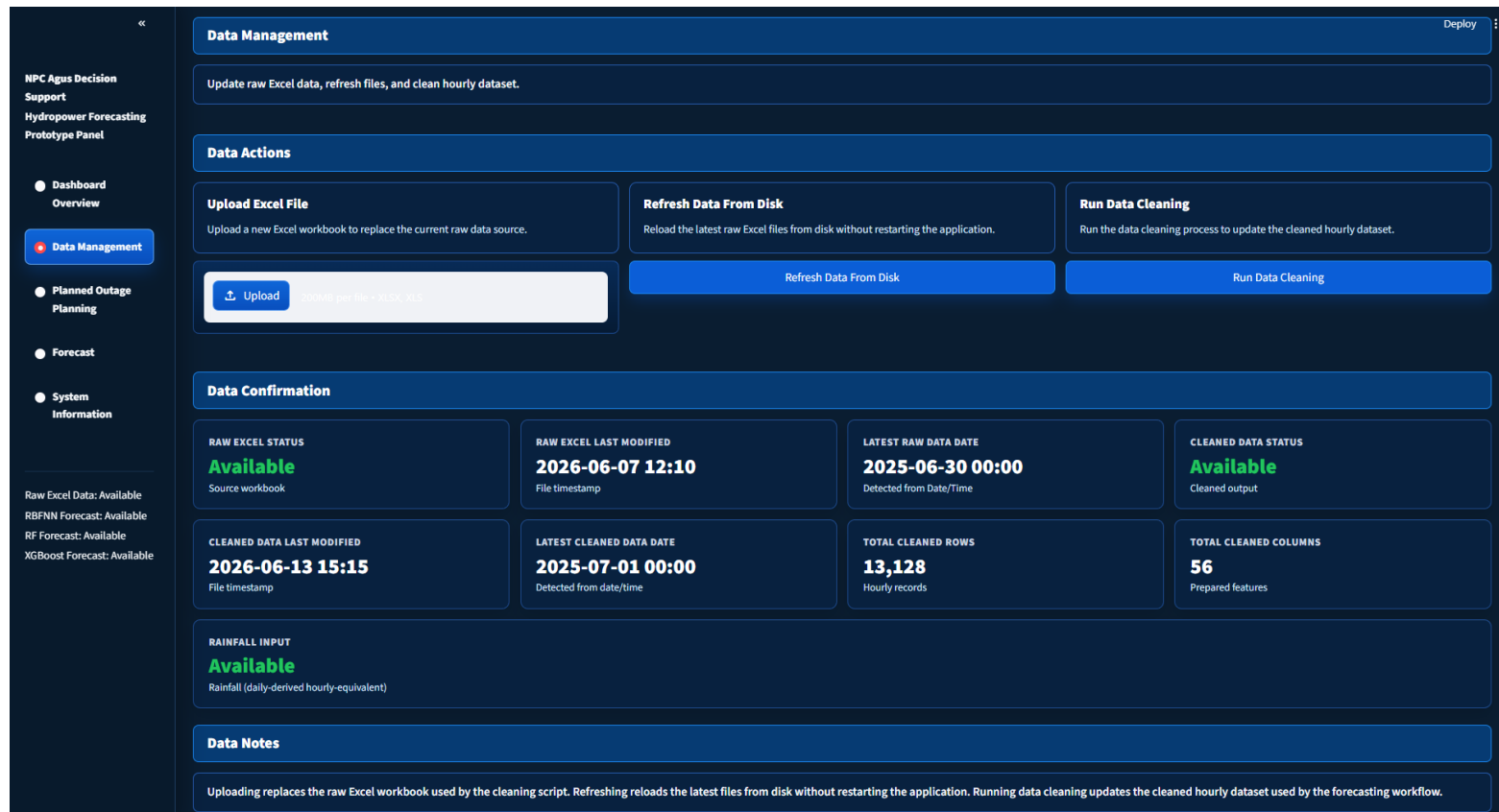


Figure E.3. Data-management and input-review page.



Figure E.5. Day-ahead forecasting results page.

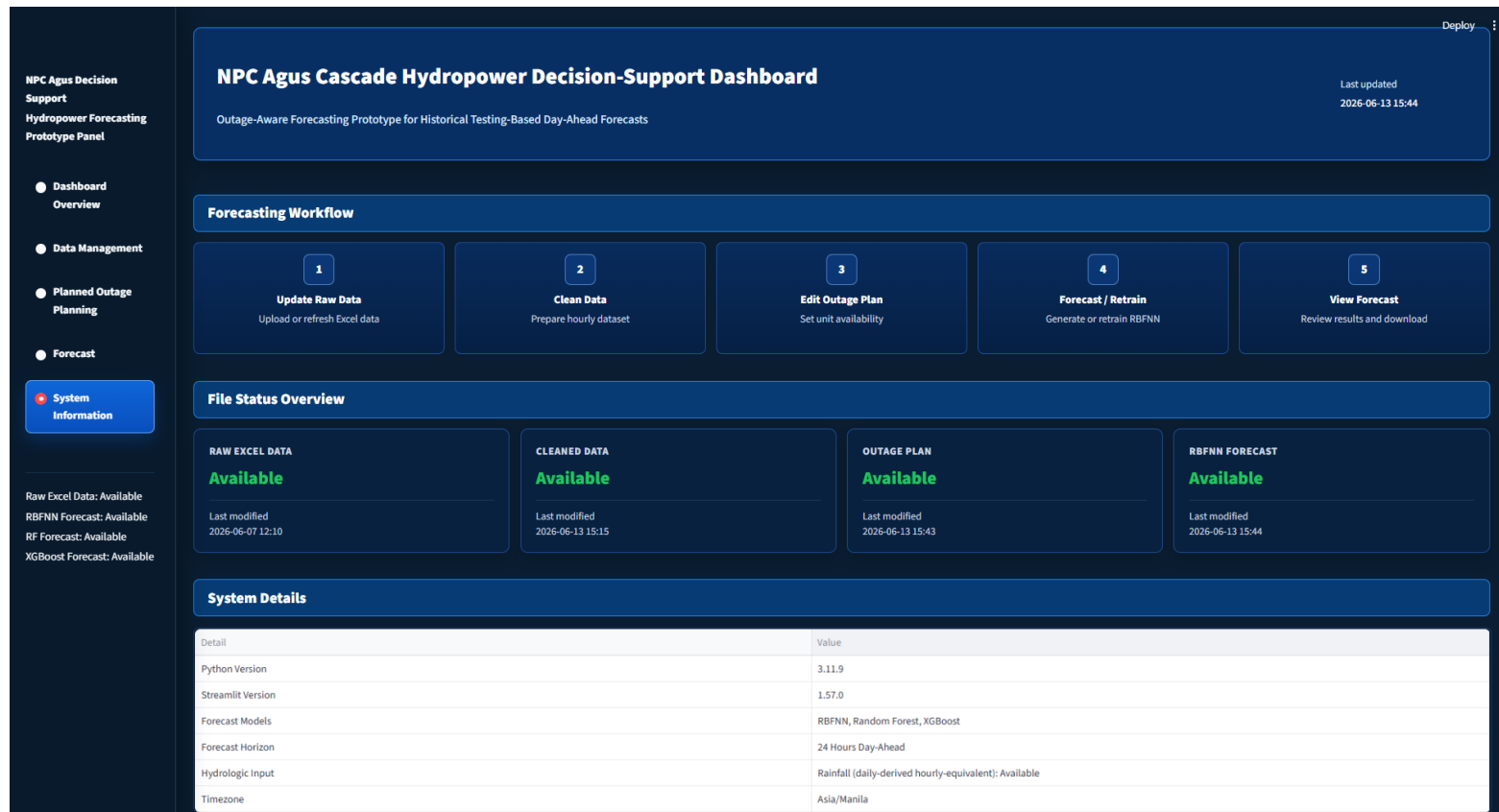


Figure E.6. System-operation guide and user-instruction page.

Table E.1. Sample planned outage input table used for day-ahead forecasting in the GUI.

Date	Hour	Agus 1	Agus 1	Agus 2	Agus 2	Agus 2	Agus 4	Agus 4	Agus 4	Agus 5	Agus 5	Agus 6	Agus 6	Agus 6	Agus 6	Agus 6	Agus 7	Agus 7
		Unit 1	Unit 2	Unit 1	Unit 2	Unit 3	Unit 1	Unit 2	Unit 3	Unit 1	Unit 2	Unit 1	Unit 2	Unit 3	Unit 4	Unit 5	Unit 1	Unit 2
2025-07-01	0:00	1	1	1	1	1	1	1	1	1	1	1	0	1	1	1	1	1
2025-07-01	1:00	1	1	1	1	1	1	1	1	1	1	1	0	1	1	1	1	1
2025-07-01	2:00	1	1	1	1	1	1	1	1	1	1	1	0	1	1	1	1	1
2025-07-01	3:00	1	1	1	1	1	1	1	1	1	1	1	0	1	1	1	1	1
2025-07-01	4:00	1	1	1	1	1	1	1	1	1	1	1	0	1	1	1	1	1
2025-07-01	5:00	1	1	1	1	1	1	1	1	1	1	1	0	1	1	1	1	1
2025-07-01	6:00	1	1	1	1	1	1	1	1	1	1	1	0	1	1	1	1	1
2025-07-01	7:00	1	1	1	1	1	1	1	1	1	1	1	0	1	1	1	1	1
2025-07-01	8:00	1	1	1	1	1	1	1	1	1	1	1	0	1	1	1	1	1
2025-07-01	9:00	1	1	1	1	1	1	1	1	1	1	1	0	1	1	1	1	1
2025-07-01	10:00	1	1	1	1	1	1	1	1	1	1	1	0	1	1	1	1	1
2025-07-01	11:00	1	1	1	1	1	1	1	1	1	1	1	0	1	1	1	1	1
2025-07-01	12:00	1	1	1	1	1	1	1	1	1	1	1	0	1	1	1	1	1
2025-07-01	13:00	1	1	1	1	1	1	1	1	1	1	1	0	1	1	1	1	1
2025-07-01	14:00	1	1	1	1	1	1	1	1	1	1	1	0	1	1	1	1	1
2025-07-01	15:00	1	1	1	1	1	1	1	1	1	1	1	0	1	1	1	1	1
2025-07-01	16:00	1	1	1	1	1	1	1	1	1	1	1	0	1	1	1	1	1
2025-07-01	17:00	1	1	1	1	1	1	1	1	1	1	1	0	1	1	1	1	1
2025-07-01	18:00	1	1	1	1	1	1	1	1	1	1	1	0	1	1	1	1	1
2025-07-01	19:00	1	1	1	1	1	1	1	1	1	1	1	0	1	1	1	1	1
2025-07-01	20:00	1	1	1	1	1	1	1	1	1	1	1	0	1	1	1	1	1
2025-07-01	21:00	1	1	1	1	1	1	1	1	1	1	1	0	1	1	1	1	1
2025-07-01	22:00	1	1	1	1	1	1	1	1	1	1	1	0	1	1	1	1	1
2025-07-01	23:00	1	1	1	1	1	1	1	1	1	1	1	0	1	1	1	1	1

Note. A value of 1 indicates an available generating unit, while a value of 0 indicates a scheduled outage. In this sample, Agus 6 Unit 2 is marked unavailable during the 24-hour forecast horizon.

Table E.2. Day-ahead hourly generation forecast output for July 01, 2025.

Date	Hour	Agus 1	Agus 1	Agus 2	Agus 2	Agus 2	Agus 4	Agus 4	Agus 4	Agus 5	Agus 5	Agus 6
		Unit 1	Unit 2	Unit 1	Unit 2	Unit 3	Unit 1	Unit 2	Unit 3	Unit 1	Unit 2	Unit 1
2025-07-01	00:00	30.62	29.55	40.23	38.62	51.03	47.52	45.08	48.07	21.47	21.28	34.50
2025-07-01	1:00	30.56	29.61	40.27	38.07	51.44	47.41	45.16	48.09	21.31	21.28	34.50
2025-07-01	2:00	30.54	29.63	40.40	37.75	51.58	47.35	45.20	48.11	21.57	21.63	34.50
2025-07-01	3:00	30.53	29.64	40.43	37.59	51.60	47.32	45.22	48.11	21.61	21.71	34.50
2025-07-01	4:00	30.54	29.62	40.42	37.47	51.59	47.32	45.24	48.12	21.92	22.05	34.50
2025-07-01	5:00	30.51	29.66	40.30	37.45	51.62	47.32	45.26	48.13	22.02	22.16	34.50
2025-07-01	6:00	30.51	29.66	40.17	37.42	51.65	47.31	45.26	48.13	22.01	22.08	34.50
2025-07-01	7:00	30.52	29.65	40.06	37.40	51.64	47.27	45.23	48.10	22.26	22.31	34.50
2025-07-01	8:00	30.50	29.66	40.01	37.36	51.63	47.29	45.24	48.12	22.18	22.16	34.50
2025-07-01	9:00	30.49	29.68	39.96	37.30	51.59	47.27	45.21	48.09	21.84	21.79	34.50
2025-07-01	10:00	30.48	29.68	39.92	37.26	51.54	47.28	45.22	48.09	21.75	21.68	34.50
2025-07-01	11:00	30.47	29.70	39.87	37.22	51.46	47.29	45.22	48.09	21.77	21.69	34.50
2025-07-01	12:00	30.48	29.69	39.85	37.18	51.43	47.32	45.24	48.10	21.42	21.35	34.50
2025-07-01	13:00	30.47	29.69	39.81	37.13	51.38	47.29	45.23	48.08	21.25	21.18	34.50
2025-07-01	14:00	30.47	29.69	39.76	37.08	51.32	47.28	45.22	48.08	21.21	21.14	34.50
2025-07-01	15:00	30.46	29.70	39.71	37.01	51.26	47.28	45.22	48.08	21.07	21.00	34.50
2025-07-01	16:00	30.48	29.69	39.67	37.01	51.19	47.26	45.20	48.05	20.99	20.93	34.50
2025-07-01	17:00	30.48	29.68	39.66	36.98	51.12	47.26	45.20	48.07	21.07	21.00	34.50
2025-07-01	18:00	30.46	29.70	39.54	37.08	51.01	47.29	45.24	48.10	20.81	20.74	34.50
2025-07-01	19:00	30.46	29.70	39.40	37.22	50.85	47.28	45.23	48.09	20.75	20.68	34.50
2025-07-01	20:00	30.48	29.68	39.31	37.27	50.75	47.29	45.25	48.12	20.85	20.77	34.50
2025-07-01	21:00	30.48	29.68	39.26	37.29	50.68	47.25	45.21	48.08	20.39	20.41	34.50
2025-07-01	22:00	30.49	29.68	39.21	37.27	50.62	47.26	45.22	48.09	20.84	20.91	34.50
2025-07-01	23:00	30.48	29.68	39.16	37.24	50.57	47.26	45.23	48.10	20.85	20.94	34.50

Table E.2. Day-ahead hourly generation forecast output for July 01, 2025 (continued).

Date	Hour	Agus 6	Agus 6	Agus 6	Agus 6	Agus 7	Agus 7	Total Gen	Total Gen	Total Gen	Total Gen	Total Gen	Total Gen	Total Cascade
		Unit 2	Unit 3	Unit 4	Unit 5	Unit 1	Unit 2	Agus 1	Agus 2	Agus 4	Agus 5	Agus 6	Agus 7	Generation
2025-07-01	00:00	0.00	47.12	28.15	47.79	21.74	21.48	60.17	129.88	140.67	42.75	157.55	43.22	574.23
2025-07-01	1:00	0.00	45.88	27.26	50.00	21.30	20.99	60.17	129.79	140.67	42.59	157.64	42.29	573.14
2025-07-01	2:00	0.00	46.87	26.10	50.00	21.24	20.89	60.17	129.73	140.67	43.20	157.46	42.14	573.37
2025-07-01	3:00	0.00	47.30	25.60	50.00	21.18	20.82	60.17	129.62	140.65	43.32	157.40	42.00	573.16
2025-07-01	4:00	0.00	47.96	24.84	50.00	21.27	20.90	60.17	129.48	140.68	43.98	157.30	42.17	573.77
2025-07-01	5:00	0.00	48.39	24.14	50.00	20.97	20.60	60.17	129.37	140.71	44.18	157.03	41.57	573.03
2025-07-01	6:00	0.00	48.84	23.56	50.00	20.20	19.84	60.17	129.24	140.69	44.08	156.90	40.03	571.12
2025-07-01	7:00	0.00	49.91	23.34	49.04	19.94	19.61	60.17	129.10	140.60	44.57	156.79	39.56	570.79
2025-07-01	8:00	0.00	47.93	24.65	50.00	20.32	19.87	60.17	129.00	140.65	44.34	157.08	40.19	571.44
2025-07-01	9:00	0.00	47.81	24.59	50.00	20.48	20.03	60.17	128.84	140.57	43.63	156.89	40.51	570.61
2025-07-01	10:00	0.00	47.76	24.56	50.00	20.64	20.06	60.17	128.71	140.59	43.44	156.83	40.70	570.43
2025-07-01	11:00	0.00	47.76	24.56	50.00	20.86	20.18	60.17	128.56	140.60	43.46	156.82	41.04	570.64
2025-07-01	12:00	0.00	47.82	24.59	50.00	20.61	19.93	60.17	128.45	140.67	42.77	156.91	40.54	569.50
2025-07-01	13:00	0.00	47.74	24.55	50.00	20.70	19.96	60.17	128.33	140.60	42.43	156.79	40.66	568.97
2025-07-01	14:00	0.00	47.68	24.52	50.00	20.53	19.87	60.17	128.16	140.58	42.35	156.70	40.40	568.35
2025-07-01	15:00	0.00	46.47	25.58	50.00	20.51	19.95	60.16	128.03	140.59	42.08	156.55	40.47	567.88
2025-07-01	16:00	0.00	43.00	29.06	50.00	20.63	20.06	60.16	127.87	140.51	41.92	156.56	40.69	567.71
2025-07-01	17:00	0.00	41.75	30.27	50.00	20.82	20.20	60.16	127.76	140.53	42.07	156.53	41.02	568.07
2025-07-01	18:00	0.00	42.36	29.82	50.00	20.89	20.25	60.16	127.63	140.63	41.55	156.67	41.13	567.79
2025-07-01	19:00	0.00	42.94	29.49	50.00	21.05	20.40	60.16	127.47	140.60	41.43	156.94	41.45	568.05
2025-07-01	20:00	0.00	43.32	29.14	50.00	20.93	20.26	60.16	127.33	140.66	41.62	156.95	41.19	567.91
2025-07-01	21:00	0.00	43.59	28.82	50.00	20.60	20.06	60.16	127.23	140.53	40.80	156.91	40.65	566.29
2025-07-01	22:00	0.00	43.88	28.60	50.00	20.75	20.25	60.16	127.11	140.58	41.75	156.97	41.00	567.57
2025-07-01	23:00	0.00	43.95	28.27	50.00	20.83	20.37	60.16	126.97	140.59	41.79	156.72	41.19	567.42

APPENDIX E

SOURCE CODE REPOSITORY

The complete implementation of the thesis forecasting pipeline, including the data preprocessing scripts, model training workflows, benchmark models, saved outputs, dashboard files, configuration files, and supporting documentation, is maintained in a version-controlled repository hosted on GitHub.

Table F.1. Repository Reference

Field	Details
URL	https://github.com/allanaspejr-ship-it/THESIS/tree/new-branch-panel_revision
Branch	new-branch-panel_revision

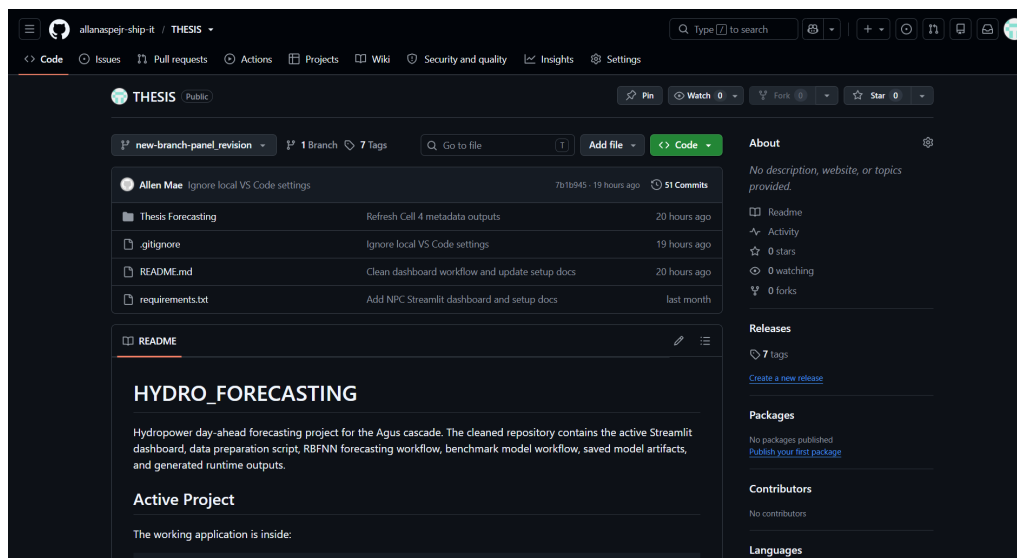


Figure F.1. GitHub repository root directory showing the thesis forecasting project files, including the cleaned dataset workflow, model scripts, benchmark implementation, dashboard components, saved outputs, and documentation files.

APPENDIX G

PSEUDOCODE

This appendix presents algorithm-style pseudocode for the core procedures used in the study. The prose methodology in Chapter 3 remains the authoritative description, while the algorithms below provide concise implementation supplements.

Algorithm 1 Agus Cascade Day-Ahead Hydropower Forecasting Study

Require: Raw Excel workbook containing Agus generation, outage/status, hydrologic, date, and time variables

Ensure: Cleaned hourly data, planned-outage template, trained model artifacts, 24-hour forecasts, metrics, figures, and metadata

- 1: Run Cell 1 to clean and resample the raw data.
 - 2: Run Cell 2 to train or load the RBFNN and generate forecasts.
 - 3: Run Cell 3 to train or load Random Forest and XGBoost benchmark models and generate forecasts.
 - 4: Run Cell 4 to generate thesis figures and organized result files.
 - 5: Use the Streamlit dashboard for operational execution and review.
-

Algorithm 2 Clean Raw Agus Hydropower Dataset

Require: Raw workbook data/ DATA(JAN2024- JUNE2025) .xlsx

Ensure: Cleaned Excel and Parquet data, cleaning metadata, and Planned_Outages_Input.xlsx

- 1: Set the plant list to Agus 1, Agus 2, Agus 4, Agus 5, Agus 6, and Agus 7.
- 2: Set the short zero-run threshold to 2 records.
- 3: Set the long zero-run threshold to 3 records.
- 4: Set the short missing-gap threshold to 2 records.
- 5: Set the spike factor to 3.5 and KNN neighbors to 5.
- 6: Read the raw workbook without assuming a single header row.
- 7: Forward-fill the upper source-header rows.
- 8: Sanitize the header text into lowercase underscore-separated column names.
- 9: Combine the three header levels into one machine-readable name.
- 10: Make duplicate column names unique.
- 11: Standardize the plant, unit, generation, outage, gate, elevation, outflow, and rainfall columns.
- 12: **if** rainfall columns exist **then**
- 13: Convert the rainfall candidates to numeric values.
- 14: Consolidate the candidates into one rainfall column.
- 15: **else**
- 16: Continue without adding a rainfall column.
- 17: **end if**
- 18: Convert the date field into valid date values.
- 19: Convert the time field into numeric hour values.
- 20: Compute $datetime = date + (time - 1)$ hours.

```

21: Sort all records according to datetime.
22: Identify the generation, outage/status, hydrologic, and operational-context columns.
23: for each numeric cleaning columns do
24:     Convert the values to numeric form.
25:     Interpolate short missing gaps.
26:     Treat short zero runs as missing observations.
27:     Interpolate the resulting short zero runs.
28:     Preserve long zero runs as actual shutdowns.
29:     Smooth isolated spikes using neighboring values.
30: end for
31: Select the remaining numeric columns.
32: Apply KNN imputation using five nearest neighbors.
33: Fill the remaining missing numeric values.
34: for each plants  $p$  do
35:     Sum the cleaned unit-generation columns of  $p$ .
36:     Save the result as total_gen_<plant>.
37: end for
38: for each unit-generation columns do
39:     if unit generation is greater than zero then
40:         Set the unit status to 1.
41:     else
42:         Set the unit status to 0.
43:     end if
44: end for
45: Clip all outage/status fields to the interval  $[0, 1]$ .
46: Round the outage/status fields to binary values.
47: Set datetime as the temporary time index.
48: Resample the records into hourly intervals.
49: Average the numeric values within each hour.
50: Restore outage/status fields as binary values.
51: Retain only the columns required by downstream training and forecasting.
52: Rename internal fields to their final public-output names.
53: Retain usable date and time fields in the cleaned output.
54: Save the cleaned dataset to Excel and Parquet.
55: Save the latest timestamp, row count, column names, and rainfall metadata.
56: Obtain the latest available historical row.
57: Create the next 24 forecast-hour records.
58: for each outage/status columns do
59:     Copy the latest known status into all 24 hours.
60:     if no valid historical status is available then
61:         Use 1 as the default available status.
62:     end if
63: end for
64: Save and format outputs/outages_planning/Planned_Outages_Input.xlsx.

```

Algorithm 3 Build Forecasting Features

Require: Cleaned hourly hydropower dataset

Ensure: Plant-level modeling frame containing lagged, rolling, outage, hydrologic, calendar, and cascade features

- 1: Rebuild datetime from the date and time fields when needed.
- 2: Add hour-of-day indicators.

- 3: Add day-of-week indicators.
 - 4: Add month and day-of-year indicators.
 - 5: Add target-hour and target-day calendar indicators.
 - 6: **for each** plants p **do**
 - 7: Define the target-generation column as `total_gen_p`.
 - 8: Add generation lags at 1, 2, 3, 6, 12, 24, 48, 72, and 168 hours.
 - 9: Add rolling statistics for windows of 3, 6, 12, 24, 48, and 168 hours.
 - 10: Add the current generation value y_t .
 - 11: Add the next-hour generation value y_{t+1} .
 - 12: Compute the residual target

$$\Delta y_{t+1} = y_{t+1} - y_t.$$
 - 13: Add unit-generation lag features.
 - 14: Add unit-generation share features.
 - 15: Add unit outage/status features.
 - 16: Add available-unit and plant-availability context.
 - 17: **if** p has an upstream plant **then**
 - 18: Add upstream-generation and cascade lag features.
 - 19: **end if**
 - 20: Add lagged rainfall when rainfall is available.
 - 21: Add lagged Lake Lanao outflow when available.
 - 22: Add lagged gate, elevation, and spillway context.
 - 23: Remove rows that cannot support the required lag or next-hour target.
 - 24: **end for**
-
-

Algorithm 4 Split Time Series Without Shuffling

- Require:** Plant-level feature dataset sorted according to time
Ensure: Chronological training, validation, and testing datasets
- 1: Sort all usable records according to datetime.
 - 2: Let n be the total number of usable records.
 - 3: Compute the training endpoint:

$$i_{\text{train}} = \lfloor 0.70n \rfloor.$$
 - 4: Compute the validation endpoint:

$$i_{\text{validation}} = \lfloor 0.85n \rfloor.$$
 - 5: Assign records 1 through i_{train} to the training set.
 - 6: Assign records $i_{\text{train}} + 1$ through $i_{\text{validation}}$ to the validation set.
 - 7: Assign the remaining records to the testing set.
 - 8: Do not shuffle any records.
 - 9: Fit scalers and model parameters using training data only.
 - 10: Use validation data for tuning, calibration, and model selection.
 - 11: Reserve testing data exclusively for final evaluation.
-
-

Algorithm 5 Train Cascade-Aware, Outage-Aware RBFNN

Require: Cleaned hourly dataset and planned-outage template

Ensure: Saved plant-specific RBFNN models, scalers, metadata, forecasts, and validation/testing results

- 1: Load the cleaned hourly data.
- 2: Load the planned-outage input.
- 3: Rebuild runtime-compatible datetime fields.
- 4: Construct the complete forecasting feature frame.
- 5: **for each** plants p in cascade order **do**
- 6: Select the plant-specific feature columns.
- 7: Run the leakage audit on the selected features.
- 8: Split the plant frame chronologically into training, validation, and testing sets.
- 9: Fit the input scaler using training features only.
- 10: Fit the residual-target scaler using training targets only.
- 11: Generate candidate configurations for:
 - number of RBF centers,
 - learning rate,
 - Gaussian-width initialization,
 - batch size, and
 - early-stopping patience.
- 12: **for each** candidate configurations c **do**
- 13: Build an RBFNN with an input layer, Gaussian RBF hidden layer, and linear output layer.
- 14: Initialize the RBF centers using K-means clustering on scaled training features.
- 15: Train the RBFNN using the training split.
- 16: Monitor validation loss for early stopping.
- 17: Restore the weights associated with the best validation loss.
- 18: Predict validation residual generation changes.
- 19: Inverse-transform the predicted residuals.
- 20: Reconstruct generation levels around persistence.
- 21: Tune shrinkage using validation data only.
- 22: Compute the median validation bias correction.
- 23: Clip the bias according to the plant-specific operational threshold.
- 24: Compute operational MAPE, MAE, RMSE, and R^2 .
- 25: Compute the validation-aware candidate score.
- 26: Store the candidate score and configuration metadata.
- 27: **end for**
- 28: Retain the strongest validation finalists.
- 29: **for each** finalist candidates **do**
- 30: Run a recursive rolling 24-hour validation day-ahead backtest.
- 31: Score the finalist using operational MAPE, RMSE, R^2 , and a negative- R^2 penalty.
- 32: **end for**
- 33: Select the best finalist for plant p .
- 34: Tune validation-only value-bin calibration when it improves the selection score.
- 35: Tune validation-only hourly residual correction when it improves the selection score.
- 36: Tune same-hour-yesterday profile blending when it improves the selection score.
- 37: Apply validation-derived ramp limiting where applicable.
- 38: Evaluate the final selected model on validation data.
- 39: Evaluate the final selected model on testing data.
- 40: Save the Keras RBFNN model.
- 41: Save the input and output scalers.
- 42: Save the selected feature list.
- 43: Save plant metadata and calibration information.
- 44: Save the training and validation loss history.
- 45: Save validation and testing metrics.
- 46: **end for**
- 47: Save the feature-leakage audit.
- 48: Save the model-selection audit.
- 49: Generate the operational 24-hour RBFNN forecast.
- 50: Save the RBFNN forecast to Excel and CSV.

- 51: Run leakage-safe rolling validation and testing backtests.
- 52: Save the day-ahead predictions and performance metrics.

Algorithm 6 Generate RBFNN 24-Hour Forecast From Saved Models

- Require:** Cleaned hourly data, planned-outage workbook, and saved RBFNN artifacts
Ensure: `Day_Ahead_24H_RBFNN_Forecast.xlsx` and its corresponding CSV file
- 1: Verify that cleaned data exist.
 - 2: Verify that the planned-outage workbook exists.
 - 3: Verify that models, scalers, and metadata exist for every plant.
 - 4: Load the cleaned historical dataset.
 - 5: Load the 24-hour planned-outage table.
 - 6: Load the saved RBFNN bundle for each plant.
 - 7: Initialize an empty 24-hour forecast table.
 - 8: **for** $h = 1$ to 24 **do**
 - 9: **for each** plants p in cascade order **do**
 - 10: Build the next feature row using the latest temporary history.
 - 11: Apply the saved input scaler.
 - 12: Predict the residual $\widehat{\Delta y}_{t+1}$ using the saved RBFNN.
 - 13: Inverse-transform the predicted residual.
 - 14: Add the residual to the current generation anchor.
 - 15: Apply the saved shrinkage factor.
 - 16: Apply the saved bias correction.
 - 17: Apply saved bin and hourly corrections.
 - 18: Apply profile blending and ramp limiting when configured.
 - 19: Read planned unit availability for forecast hour h .
 - 20: Limit the plant forecast according to available plant capacity.
 - 21: Allocate the plant forecast among available units.
 - 22: Enforce the individual unit capacity limits.
 - 23: Save the unit-level and plant-total forecasts.
 - 24: Append the predicted plant total to recursive history.
 - 25: **end for**
 - 26: Compute total cascade generation as the sum of the plant totals.
 - 27: **end for**
 - 28: Verify that planned-outage units have zero generation.
 - 29: Verify that unit forecasts do not exceed unit capacities.
 - 30: Verify that every plant total equals its unit sum.
 - 31: Verify that cascade generation equals the sum of plant totals.
 - 32: Format the forecast columns and hour labels.
 - 33: Save the forecast to Excel and CSV.

Algorithm 7 Train and Forecast Random Forest and XGBoost Benchmarks

- Require:** Cleaned hourly dataset and planned-outage template
Ensure: Saved benchmark artifacts, metrics, predictions, and operational forecast workbooks
- 1: Load the cleaned hourly dataset.

```

2: Load the planned-outage input.
3: Construct the common forecasting feature structure.
4: for each plants  $p$  do
5:   Split the records chronologically into training, validation, and testing sets.
6:   Select the plant-specific feature columns.
7:   Run the feature-leakage audit.
8:   for each models  $m \in \{\text{Random Forest, XGBoost}\}$  do
9:     Generate candidate hyperparameter configurations.
10:    for each candidate configurations  $c$  do
11:      Train model  $m$  using the training split only.
12:      Predict validation residual generation changes.
13:      Convert residual predictions into generation-level forecasts.
14:      Tune shrinkage and bias using validation data only.
15:      Tune additional calibration using validation data only.
16:      Run rolling validation day-ahead scoring when required.
17:      Store the candidate score and metadata.
18:    end for
19:    Select the best candidate using the validation-aware score.
20:    Evaluate the selected model using validation data.
21:    Evaluate the selected model using testing data.
22:    Save the trained estimator.
23:    Save the exact feature order.
24:    Save validation-derived calibration data.
25:    Save validation/testing metrics and prediction rows.
26:  end for
27: end for
28: Save the benchmark-fairness audit.
29: Save the model-selection audit.
30: Generate operational 24-hour Random Forest and XGBoost forecasts.
31: Save the benchmark forecasts to Excel and CSV.
32: Run leakage-safe rolling benchmark backtests.
33: Save benchmark backtest predictions and metrics.

```

Algorithm 8 Generate Benchmark Forecasts From Saved Models

Require: Cleaned hourly data, planned-outage input, and saved benchmark model payloads
Ensure: Random Forest and/or XGBoost 24-hour Excel and CSV forecast files

```

1: Read the optional benchmark-model filter from the command-line arguments.
2: Load the cleaned hourly data.
3: Load the planned-outage table.
4: Load the selected Random Forest and/or XGBoost model payloads.
5: for each selected benchmark models  $m$  do
6:   Initialize an empty 24-hour forecast table.
7:   for  $h = 1$  to 24 do
8:     for each plants  $p$  in cascade order do
9:       Build the feature row using the latest available history.
10:      Align the features with the exact order saved with model  $m$ .
11:      Predict the residual generation change.
12:      Convert the residual into a plant-level forecast.
13:      Apply the saved validation-only corrections.
14:      Read planned unit status for forecast hour  $h$ .
15:      Allocate the plant forecast among available units.

```

```

16:         Enforce all individual unit capacity limits.
17:         Append the predicted plant total to recursive history.
18:     end for
19:     Compute total cascade generation.
20: end for
21:     Validate unit, plant, and cascade consistency.
22:     Save the benchmark forecast to Excel and CSV.
23: end for

```

Algorithm 9 Rolling Day-Ahead Validation and Testing Backtest

Require: Cleaned hourly data, saved model artifacts, and chronological split boundaries
Ensure: Day-ahead validation/testing predictions, metrics, and leakage-control metadata

```

1: Identify forecast days containing complete 24-hour actual records.
2: Assign valid days to validation and testing ranges according to the chronological split.
3: for each splits  $s \in \{\text{validation, testing}\}$  do
4:     Initialize an empty prediction table.
5:     for each forecast days  $d$  in split  $s$  do
6:         Create historical data using only records before 00:00 of day  $d$ .
7:         if history contains any record from forecast day  $d$  then
8:             Reject the case because it contains same-day information.
9:         end if
10:        Obtain unit outage/status information from the final historical record before day  $d$ .
11:        Construct a 24-hour outage plan using the last-known historical statuses.
12:        Do not use actual forecast-day unit statuses.
13:        Generate a recursive 24-hour forecast using the saved model artifacts.
14:        Load the actual generation of day  $d$  only after prediction is completed.
15:        for each plants and forecast hours do
16:            Save the forecast date and hour.
17:            Save the plant name and model name.
18:            Save actual and predicted generation.
19:        end for
20:    end for
21:    Compute operational MAPE for each plant.
22:    Compute MAE, RMSE, and  $R^2$  for each plant.
23:    Count the number of forecast days and hours.
24:    Count the MAPE-included and MAPE-excluded rows.
25:    Save the split prediction workbook.
26:    Save the split metrics workbook.
27: end for
28: Save the chronological split strategy.
29: Save the models included in the evaluation.
30: Save the outage-input rule.
31: Record same-day actual outage usage as false.
32: Record future-data usage as false.
33: Record testing use for training or tuning as false.

```

Algorithm 10 Compute Operational MAPE

Require: Actual values y_i , predicted values $\hat{y}_i$, and plant capacity C_p

Ensure: Operational MAPE and inclusion/exclusion counts

1: Compute the plant-specific threshold:

$$\tau_p = \max(1 \text{ MW}, 0.01C_p).$$

2: Initialize the included-row set $\mathcal{J} \leftarrow \emptyset$.

3: **for** $i = 1$ to n **do**

4: **if** $|y_i| \geq \tau_p$ **then**

5: Add row i to $\mathcal{J}$.

6: **else**

7: Exclude row i because percentage error is unstable near zero generation.

8: **end if**

9: **end for**

10: **if** $\mathcal{J} = \emptyset$ **then**

11: Return undefined operational MAPE.

12: Return the inclusion and exclusion counts.

13: **else**

14: Compute

$$\text{MAPE} = \frac{100}{|\mathcal{J}|} \sum_{i \in \mathcal{J}} \left| \frac{y_i - \hat{y}_i}{y_i} \right|.$$

15: Return operational MAPE.

16: Return the inclusion and exclusion counts.

17: **end if**

Algorithm 11 Allocate Plant Forecast to Available Units

Require: Plant-level forecast, planned unit status, historical unit shares, and unit capacities

Ensure: Capacity-consistent unit-level generation forecasts

1: Identify available units for which planned status equals 1.

2: Identify unavailable units for which planned status equals 0.

3: **for each** unavailable units j **do**

4: Set the unit forecast $\hat{y}_j \leftarrow 0$.

5: **end for**

6: Compute available plant capacity:

$$C_{\text{available}} = \sum_{j: u_j=1} C_j.$$

7: Limit the plant forecast to the interval $[0, C_{\text{available}}]$.

8: Estimate allocation weights using historical unit shares.

9: **if** historical shares are unavailable or insufficient **then**

10: Use unit-capacity shares as fallback weights.

11: **end if**

12: **for each** available units j **do**

13: Compute the initial allocation:

$$\hat{y}_j = \hat{y}_p w_j.$$

- 14: Clip $\hat{y}_j$ to the interval $[0, C_j]$.
 - 15: **end for**
 - 16: **if** $\sum_j \hat{y}_j \neq \hat{y}_p$ **then**
 - 17: Compute the remaining unallocated generation.
 - 18: Redistribute the remainder among available units with unused capacity.
 - 19: **end if**
 - 20: Set the final plant total equal to $\sum_j \hat{y}_j$.
-

Algorithm 12 Generate Thesis Figures and Organized Outputs

- Require:** Cleaned data, model forecasts, validation/testing metrics, and backtest outputs
Ensure: Thesis figures, organized metrics, comparison tables, and plot-source workbooks
- 1: Create the required thesis-figure directories.
 - 2: Create the required organized metadata directories.
 - 3: Read the cleaned hourly dataset.
 - 4: **for each** Agus plants **do**
 - 5: Generate the cleaned hourly generation-profile figure.
 - 6: **end for**
 - 7: Read the RBFNN testing predictions.
 - 8: Generate plant-level actual-versus-RBFNN testing figures.
 - 9: Generate the all-plant RBFNN testing panel.
 - 10: Read RBFNN, Random Forest, and XGBoost testing predictions.
 - 11: Generate actual-versus-forecast figures for each model.
 - 12: Generate all-model actual-versus-forecast comparisons.
 - 13: Generate the RBFNN actual-versus-predicted scatter plot.
 - 14: Generate the RBFNN residual-distribution plot.
 - 15: Generate the testing-MAPE-per-plant plot.
 - 16: Read the rolling day-ahead model metrics.
 - 17: Generate the testing operational-MAPE comparison.
 - 18: Generate the testing RMSE comparison.
 - 19: Read current operational day-ahead forecast workbooks.
 - 20: Generate the total-cascade forecast-comparison figure.
 - 21: Write average validation metrics.
 - 22: Write average testing metrics.
 - 23: Write validation/testing summaries and prediction tables.
 - 24: Write hydrologic-input metadata.
 - 25: Write overall and plant-level model comparisons.
 - 26: Write the best-model summary.
 - 27: Write the all-model day-ahead backtest comparison.
 - 28: Synchronize metric workbooks into the corresponding model-specific metadata folders.
-

Algorithm 13 Dashboard Operation

- Require:** User actions in the Streamlit dashboard and existing project files
Ensure: Updated cleaned data, outage plan, forecasts, charts, downloads, and status displays
- 1: Configure the Streamlit page title and wide layout.

```

2: Define project paths for data, outputs, models, metadata, and scripts.
3: Define model settings for RBFNN, Random Forest, and XGBoost.
4: Render the sidebar pages:
    Dashboard Overview,
    Data Management,
    Planned Outage Planning,
    Forecast, and
    System Information.
5: if Dashboard Overview is selected then
6:     Load the selected model forecast when available.
7:     Load cleaned data and the outage plan when available.
8:     Display KPI cards, charts, file status, and the testing-error CDF section.
9: end if
10: if Data Management is selected then
11:     Allow the user to upload a raw Excel workbook.
12:     if Refresh Data From Disk is clicked then
13:         Clear the cached dashboard data.
14:     end if
15:     if Run Data Cleaning is clicked then
16:         Execute scripts/cell1_clean_data.py.
17:         Display the script output or error message.
18:         Reload the cleaned-data status.
19:     end if
20: end if
21: if Planned Outage Planning is selected then
22:     Load Planned_Outages_Input.xlsx.
23:     Display the editable 24-hour outage/status table.
24:     if Save Outage Plan is clicked then
25:         Save the edited outage table to Excel.
26:     end if
27:     if Forecast Day-Ahead is clicked then
28:         Save the edited outage plan.
29:         Validate required data and model artifacts.
30:         Run the RBFNN forecast-only script.
31:         Run the benchmark forecast-only script.
32:         Clear cached dashboard data.
33:         Display the forecast-completion status.
34:     end if
35:     if Retrain RBFNN Model is clicked then
36:         Execute scripts/cell2_rbfnn.py --train.
37:     end if
38:     if Retrain Benchmark Models is clicked then
39:         Execute scripts/cell3_benchmark.py --train.
40:     end if
41: end if
42: if Forecast is selected then
43:     Allow the user to select a forecast model.
44:     if Run Selected Model Forecast is clicked then
45:         Execute the selected forecast-only script.
46:         Clear cached dashboard data after successful execution.
47:     end if
48:     if Refresh Forecast Files is clicked then
49:         Clear the cached dashboard data.
50:     end if
51:     Load the selected forecast workbook.
52:     Display KPI cards, charts, forecast tables, and the Excel download control.
53:     Load all available model forecasts.
54:     Render the total-cascade model-comparison chart.

```

```
55:     Render model-comparison summary tables.  
56: end if  
57: if System Information is selected then  
58:     Display the workflow order.  
59:     Display project-file availability.  
60:     Display runtime-environment information.  
61:     Display the latest system checks.  
62: end if
```

Curriculum Vitae

ASPE, ALLAN T. JR.

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PERSONAL INFORMATION

Date of Birth: May 27, 2004

Place of Birth: Libertad, Butuan City

Age: 22

Civil Status: Single

Gender: Male

Religion: Roman Catholic

CGPA: 1.36304

EDUCATIONAL ATTAINMENT

Tertiary Education

- School: Mindanao State University - Iligan Institute of Technology
B.S. Electrical Engineering
2022 - 2026

Secondary Education

- School: Cabadbaran City National High School
With Highest Honor - Batch's Valedictorian
2016 - 2022

Elementary Education

- School: Cabadbaran South Central Elementary School

TRAINING EXPERIENCE

- On-the-Job Training at MERALCO Networks, June-July 2025, Ortigas, Pasig City
- Robotics Training for the College of Engineering Students of MSU-IIT, 2025

EXPERIENCE

- Mindanao State University - Iligan Institute of Technology, 2024 - 2025
Institute of Integrated Electrical Engineering Students - Treasurer
- Designed and developed a voting machine using integrated circuits, 2024
- Built and tested an FM transmitter for wireless audio signal transmission, 2024

SKILLS

- Programming and Software Skills
C/C++, MATLAB, Arduino IDE, Code::Blocks, SciLab, NI Multisim
- Circuit Design and Analysis
Proficiency in designing and analyzing electrical and electronic circuits
- Office Productivity Applications
Microsoft Word, Microsoft Excel, and Microsoft PowerPoint
- Organizational and Research Skills
Experience in student-organization financial responsibility and academic research tasks
- Communication Skills
Effective use of technology and communication tools to improve productivity
- Languages
Cebuano, Filipino, and English

CONFERENCE/SEMINAR/WORKSHOP ATTENDED

- Robotics Training for the College of Engineering Students of MSU-IIT, 2025
- National Student Convention Math Wizard 2nd Placer, 2024
- National Student Convention Quiz Show Participant, 2024
- National Student Convention Math Wizard Participant, 2025
- National Student Convention Quiz Show Participant, 2025
- Regional Student Convention Math Wizard 2nd Placer, 2024
- Regional Student Convention Quiz Show 2nd Placer, 2024
- Regional Student Convention Math Wizard 1st Placer, 2025
- Regional Student Convention Quiz Show 2nd Placer, 2025

ACHIEVEMENTS

- Rizal's Lister (MSU-Iligan Institute of Technology, Iligan City)
 - 1st - 2nd Sem. SY: 2022-2023
 - 1st Sem. SY: 2024-2025
- Chancellor's Lister (MSU-Iligan Institute of Technology, Iligan City)
 - 2nd Sem. SY: 2024-2025
- Dean's Lister (MSU-Iligan Institute of Technology, Iligan City)
 - 1st-2nd Sem. SY: 2023-2024
 - 1st Sem. SY: 2025-2026